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Purpose

This work instruction serves as a standard for all configuration and/or testing steps required by the OTArmor™ build and intended for Pre-CWT use.

Scope & Applicability

This document is applicable to all M&C CS locations performing OTArmor™ Engineering for requisition projects. Newly acquired businesses must comply with plans defined by the Business Development team and the timetable requiring Quality Management System assimilation.

Terms, Definitions, Acronyms:

- **BOM** – Bill of Materials
- **PVP** – Patch Validation Program
- **Compliance Validator** - The compliance validation tool used to check TR&S compliance of CS Cyber BOM before ordering
- **DC** – Domain Controller
- **DLP** – Data Loss Prevention
- **DNS** – Domain Name System
- **EICAR** – European Institute for Computer Antivirus Research: Anti-Virus Test File
- **CWT** – Customer Witness Test
- **GPO** – Group Policy Object
- **HMI** – Human Machine Interface
- **iCWT** – Integrated Factory Acceptance Test: A validation test with multiple subsystems connected to each other through test interfaces and tested as a system. An iCWT may be performed within a GE M&C CS Facility or at an external location.
- **Internal Validation Test** - A validation test performed within a GE M&C CS facility conducted without participation from or acceptance by the customer and/or customer representative.
- **MAC** – Media Access Control
- **Mark VIe** – GE Proprietary turbine/DCS control
- **OTArmor™** – Proprietary Cyber Asset Protection Platform
- **NIDS** – Network Intrusion Detection System

- **NTFS** – New Technology File System
- **NTP** – Network Time Protocol
- **PDH** – Plant Data Highway
- **PIN Number** – A Personal Identification Number only known by the user who configured it.
- **Project** – The translation of customer requirements into a system that meets those requirements. The system is then shipped to the customer site for installation and commissioning.
- **RADIUS** – Remote Authentication Dial-in User Service
- **SIEM** – Security Information & Event Monitor
- **TCP** (network) – Transmission Control Protocol
- **TFTP** – Trivial File Transfer Protocol
- **Transactional Compliance Plan (TCP)** – The TCP is a process to develop, implement, and close out, a plan to ensure that TRS is identified and disposition over the course of an ITO/OTR project. The TCP is initiated at R1 during the ITO process and is completed at the end of the OTR process.
- **Two Factor Authentication** – An authentication method that requires the user to have two forms of validity, something they have (a Yubikey with a certificate) and something they know like a PIN number.
- **UDH** – Unit Data Highway
- **UTM** – Unified Threat Manager
- **VLAN** – Virtual Local Area Network
- **Xona** – Critical Systems Gateway (CSG)
- **Yubikey** – A cryptographic device that stores a certificate used to validate the identity of a user.
- **AM1** – Asset Management Server
- **CA1** – Certificate of Authority Server
- **CM1** – Configuration Management Server
- **DC1** – Primary Domain Controller Server
- **DC2** – Secondary Domain Controller Server
- **DR1** – Disaster Recovery Server
- **LM1** – Log Management Server
- **PM1** – Patch Management Server
- **SP1** – Site Protection Server
- **SP2** – Trend Micro Apex Central Server
- **SP3** – Trend Micro Apex One Server

Reference Documents

Note: Documents with a **yellow highlight** are referring to the legacy Nexus / SAP release numbers. The documents are now available in GEV / PLM.

The status is in progress and the work instruction will be updated once the numbers have been assigned.

- **313T0183** – Default Passwords Document – **Important:** This is a **Class III** document, DO NOT share via email or upload to a file share (Box, OneDrive, etc.).

- **313T7465P0420AA** – Control Server – Final Configuration
- **313T7465P0401AA** – DC1 Virtual – Server Configuration
- **313T7465P0401BA** – DC1 Virtual – Post Configuration
- **313T7465P0404BA** – DR1 Virtual – Post Configuration
- **GEH-6887** – OTArmor Log Management (Splunk) Maintenance Guide
- **GEH-6890** – OTArmor Backup & Recovery Maintenance Guide
- **GEH-6857** – Control Server HP Windows 10 Thin Client HMI System Support and Maintenance Guide
- **NCM10015** – OTArmor Network Monitor Maintenance Guide
- **427A7365** – Cisco Commands
- **427A4442** – Adding MAC Address Filtering
- **131M0028** – Cisco 2960X Upgrade Procedure
- **152M3647** – Fortinet Firewall Configuration
- **NCM10067** – Trend Micro Safe Lock TXOne Intelligent Manager Maintenance Guide
- **NCM10068** – Trend Micro Safe Lock TXOne Edition Maintenance Guide
- **NCM10069** – Trend Micro Apex One and Apex Central Maintenance Guide
- **151M6218** – OTArmor AM1 OTR/Field Services Configuration Guide
- **153M6895** – OTArmor Xona CSG Setup and Configuration
- **132M6452** – Controls Ports and Services
- **136M5793** – OTArmor FAT (CWT) Plan Template
- **136M5795** – OTArmor PreFAT (CWT) Checklist Template (Latest Release on Confluence)
- **136M5796** – OTArmor Firewall Policies Template
- **136M5798** – OTArmor Network Topology Template
- **DOC** – Startup-Shutdown VSAN Cluster.docx

IMPORTANT: Please execute the following steps in sequential order; do not jump between steps (unless unapplicable to the system).

1.0 **Network Verification – OTArmor**

1.1 **Review Design & Equipment**

Review network topology – created based on **REF – 136M5798** and verify the equipment (HMIs, OTArmor™ Servers, Firewalls, and Switches) is available for simulation testing:

- 1.1.1 If the equipment is available and connected, start testing by following the next step described in step 1.2.
- 1.1.2 If the equipment is not available, locate the essential test equipment as defined by the network topology and start testing described in step 1.2.

1.2 **HMI Connections**

Verify all HMIs are connected to the UDH and PDH simulating the network topology (4108), as closely as possible:

- 1.2.1 Login to the HMI -> Desktop -> right-click **Start** -> select **Command Prompt (Admin)** -> type: **ipconfig**
- 1.2.2 Verify the UDH and PDH IP addresses comply with the network topology.
 - 1.2.2.1 If not -> setup IP addresses for the UDH and PDH networks according to 4108.
 - 1.2.2.2 Desktop -> click **Start** -> **Control Panel** -> open **Network and Sharing Center** -> click **Change adapter settings**. Change setting to match the 4108.
- 1.2.3 Verify physical network connection for UDH and PDH (Primary and Secondary) are correctly connected to the Switch.
- 1.2.4 If a firewall is included, verify the default gateway has been configured for the PDH network and a static route has been added for UDH network (when needed).
- 1.2.5 Desktop -> click **Start** -> select **Computer** -> Right-click and select **Properties**.
 - 1.2.5.1 Verify the HMI name complies with the network topology.

Note: If using a test HMI, select the master HMI to simulate with.

1.3 **Switch Connections**

- 1.3.1 Verify the physical connections match the 4108 connections for the UDH and PDH for every device.

Note: If using test equipment, verify the configurations are clean.

- 1.3.2 Verify the Switch has a PDH IP address. Refer to the following (connect/view the switch configuration for IP verification). **PRO – 427A7365.**

1.4 **Controller Connections**

- 1.4.1 Verify the **Controller** is connected to the **UDH** switch.
- 1.4.2 Physically check the ethernet cable, verify it is connected to the **ENET1 port** on the controller and connected to the UDH port(s) on the switch.
- 1.4.3 If a firewall is included, make sure to update the TCW with the default gateway IP Address for PDH and UDH.

1.5 **Firewall Connections**

- 1.5.1 Verify the physical connections match the 4108 connections for UDH and PDH for every device.
- 1.5.2 Make sure the firewall has the rules applied to permit **all** for UDH and PDH.

1.6 **OTArmor Server Connections**

- 1.6.1 Verify Physical Servers are connected to **PDH/UDH**.
 - 1.6.1.1 Physically check the ethernet cables, verify they are connected to the correct ports on the server and connected to the Switch/Firewall.
- 1.6.2 If a firewall is included, verify the default gateway has been configured for the PDH network and a static route has been added for UDH network (when needed).

1.7 **Test Connections**

- 1.7.1 Ping starting from the HMIs to all **OTArmor™** servers (ex. DC1/DC2, CA1, DR1, SP1, PM1, LM1):
 - 1.7.1.1 Desktop -> right-click **Start** -> select **Command Prompt (Admin)** -> type **ping <enter ip address>** for the OTArmor™ servers.
 - 1.7.1.2 Verify successful reply from each device.
- 1.7.2 Ping starting from HMIs to the controllers:
 - 1.7.2.1 Desktop -> right-click **Start** -> select **Command Prompt (Admin)** -> type **ping <enter ip address>** for the Controller.
 - 1.7.2.2 Verify successful reply from each device.
- 1.7.3 Ping starting from **OTArmor™** Servers (CA1 and LM1) to the controllers:
 - 1.7.3.1 Desktop -> right-click **Start** -> select **Command Prompt (Admin)** -> type **ping <enter ip address>** for the Controller.
 - 1.7.3.2 Verify successful reply from each device.

1.8 Verify VMware Licensing

1.8.1 Verify the **Host(s)/Hypervisor(s)** are licensed.

1.8.1.1 **Contact the build team** for the license if needed -> the build team will need to license the host(s)/hypervisor(s).

1.8.2 Verify the **Cluster and vCenter** are licensed.

1.8.2.1 **Contact the build team** for the license if needed -> the build team will need to license the cluster/vcenter.

1.9 Validate Network Topology

Verify the network topology is consistent.

1.9.1 Verify the physical and virtual hardware complies with the network topology (4108).
Redline any change executed.

1.9.2 Review network topology (4108) for any potential asset changes and modify IP scheme accordingly.

1.9.3 If system has Application control -> refer to **Trellix section to Change Status > Observe Mode and Update Mode.**

2.0 Dell Server Performance Settings – OTArmor

Note: The adjustment of the server performance settings will require two reboots. It is important to power down the virtual machines and place the server in maintenance mode.

2.1 Set System and Workload Profile

2.1.1 Open the server **iDRAC** -> login with the root user.

2.1.2 On the menu -> select **Configuration – System Settings** -> select the **BIOS Settings** tab.

2.1.3 Expand the **System Profile Settings**.

2.1.3.1 Next to **System Profile** -> click the drop-down and select **Performance**.

2.1.3.2 Next to **Workload Profile** -> click the drop-down and select **Virtualization Optimized Performance Profile**.

2.1.3.3 Click **Apply**.

2.1.3.4 Click **OK** on the Success pop-up.

2.1.3.5 The **Pending Value** should change to **Performance** and **Virtualization Optimized Performance Profile**.

2.1.3.6 On the bottom left -> select **Apply And Reboot**.

2.1.4 After server reboot -> log back into the **iDRAC**.

- 2.1.5 Navigate to *Configuration – System Settings – BIOS Settings – System Profile Settings* -> verify the **System Profile** is now set to **Performance** and **Workload Profile** is now set to **Virtualization Optimized Performance Profile**.

2.2 Enable Dell Controlled Turbo Setting

- 2.2.1 Open the server **iDRAC** -> login with the root user.
- 2.2.2 On the menu -> select **Configuration – System Settings** -> select the **BIOS Settings** tab.
- 2.2.3 Expand the **Processor Settings**.
 - 2.2.3.1 Expand the **Dell Controlled Turbo**.
 - 2.2.3.2 Next to Dell Controlled Turbo setting -> click the drop-down and select **Enabled**.
 - 2.2.3.3 Click **Apply**.
 - 2.2.3.4 Click **OK** on the Success pop-up.
 - 2.2.3.5 The **Pending Value** should change to **Enabled**.
 - 2.2.3.6 On the bottom left -> select **Apply And Reboot**.
- 2.2.4 After server reboot -> log back into the iDRAC.
- 2.2.5 Navigate to *Configuration – System Settings – BIOS Settings – Processor Settings – Dell Controlled Turbo* -> verify the **Dell Controlled Turbo** setting is now **Enabled**.
- 2.2.6 When completed -> remove the server from maintenance mode and power on the VMs.

3.0 NTP Server – OTArmor

3.1 **(Important)** Check System Time and Time Zone

- 3.1.1 **For Host(s):**
- 3.1.2 Verify the following:
 - 3.1.2.1 Login to the Host (e.g., VH1).
 - 3.1.2.2 Select Manage -> click System tab -> select Time & Date.
 - 3.1.2.3 In the Time & Date window -> verify the Current date and time are correct based on the AnswerFile.
Important: The Current date and time will be in UTC which is 6 hours ahead of MST.
 - 3.1.2.4 Repeat for each Host (e.g., VH2).
 - 3.1.2.5 If the Time & Date are incorrect or not configured -> **Contact the Build Team**.
- 3.1.3 **For vCenter:**
- 3.1.4 Verify the following:

- 3.1.4.1 Login to vCenter/VAMI -> **vcsa.hmi.local:5480** (change for non-standard domain name).
- 3.1.4.2 On the left menu -> select Time.
- 3.1.4.3 Verify the Time Zone is correct based on the AnswerFile.
- 3.1.4.4 Verify the Time Synchronization/current Server Time is correct based on the AnswerFile.
- 3.1.4.5 If the Time Zone and Time Synchronization are incorrect or not configured -> **Contact the Build Team.**

3.2 Devices Sync to NTP

Verify domain controller(s) are set to NTP Server.

Note: It is recommended to sync each domain controller to the NTP server. DC1 and DC2 should both be configured to sync to NTP server for redundancy on the domain.

3.2.1 Configure DC1 to sync to NTP:

3.2.1.1 Login to the **HMI** -> remote desktop to **DC1**.

3.2.1.2 Open **PowerShell (Run as Administrator)**.

3.2.1.3 Run the following command:

3.2.1.3.1 **w32tm /config /manualpeerlist:<IP of NTP>
/syncfromflags:manual /reliable:no /update**

Note: make sure to include the spaces.

3.2.1.4 Verify the command completes successfully.

3.2.1.5 Run the following command:

3.2.1.5.1 **w32tm /query /status**

3.2.1.5.1.1 Verify *Source* is <IP of NTP>

3.2.1.5.2 **w32tm /query /peers**

3.2.1.5.2.1 Verify *Peer* is <IP of NTP>

3.2.1.5.3 **W32tm /resync**

3.2.1.5.3.1 Verify command completes successfully.

Note: make sure to include the spaces.

3.2.2 Next, check the OTArmor™ servers are syncing to the DC1.

3.2.2.1 Remote desktop to each OTArmor™ Server (DR1, CA1, LM1, PM1, SP1, etc.) -> open **PowerShell (Run as Administrator)** -> enter the following commands:

3.2.2.1.1 **w32tm /query /status**

3.2.2.1.1.1 Verify *Source* is DC1.HMI.local

3.2.2.1.2 **w32tm /query /peers**

3.2.2.1.2.1 Verify *Peer* is DC1.HMI.local

Note: make sure to include the spaces.

Important: If the servers aren't pointing to DC1. Run the following command to route to DC1 as the NTP address.

3.2.2.2 Desktop -> right-click **Start** -> open **Command Prompt (Admin)** and type:

w32tm/config/manualpeerlist: DC1.HMI.local /syncfromflags: manual /reliable: No /update

w32tm /query /status

3.2.2.2.1 Verify *Source* is DC1.HMI.local

w32tm /query /peers

3.2.2.2.2 Verify *Peer* is DC1.HMI.local

Note: make sure to include the spaces.

3.2.3 (If needed) Configure DC2 to sync to NTP:

3.2.3.1 Login to the **HMI** -> remote desktop to **DC2**.

3.2.3.2 Open **PowerShell (Run as Administrator)**.

3.2.3.3 Run the following command:

3.2.3.3.1 **w32tm /config /manualpeerlist:<IP of NTP> /syncfromflags:manual /reliable:no /update**

Note: make sure to include the spaces.

3.2.3.4 Verify the command completes successfully.

3.2.3.5 Run the following command:

3.2.3.5.1 **w32tm /query /status**

3.2.3.5.1.1 Verify *Source* is <IP of NTP>

3.2.3.5.2 **w32tm /query /peers**

3.2.3.5.2.1 Verify *Peer* is <IP of NTP>

3.2.3.5.3 **W32tm /resync**

3.2.3.5.3.1 Verify command completes successfully.

Note: make sure to include the spaces.

3.2.4 Verify ToolboxST is synchronized to the NTP server.

3.2.4.1 Open the ToolboxST file -> in the **Time Synchronization** (left pane) verify it is pointing to the correct time source (**per network topology**).

3.3 Set NTP logging to LM1(Splunk)

IMPORTANT: The following NTP instructions may vary depending on the time server manufacturer. This example was configured using a Symmetricom Network Time Server.

3.3.1 Login to the HMI. Open an internet browser and type the NTP server IP address into the URL -> hit <enter>.

3.3.2 Login to the NTP server (use NTP manufacturer login).

3.3.3 Navigate to the **Admin** button -> select the **Logs Config** tab.

3.3.4 In **System Log Configuration** -> set parameters based on the 4108 (LM1) and manufacturer specification.

3.4 Change NTP Hostname and Password

IMPORTANT: The following NTP instructions may vary depending on the time server manufacturer. This example was configured using a Symmetricom Network Time Server.

3.4.1 Login to the HMI. Open an internet browser and type the NTP server IP address into the URL -> hit <enter>.

3.4.2 Login to the NTP server (use NTP manufacturer login).

3.4.3 To change the NTP hostname -> navigate to the **System** -> select the **General** tab.

3.4.4 For the **Hostname** type the new hostname -> click **Apply**.

3.4.5 To change the NTP password -> navigate to the **Admin** -> select the **Users** tab.

3.4.6 For the **User** dropdown -> select **admin**.

3.4.7 Enter the **Old Password** -> type in **New Password** and confirm. Logout of the NTP server and login with new credentials to verify the change.

4.0 DC1 and DC2 – OTArmor

4.1 Check Domain Replication (For Redundant Domain Controllers)

If the system has a DC1 and DC2, verify replication.

4.1.1 Open **Command Prompt as Administrator**.

4.1.1.1 Enter the command: **repadmin /replsummary**

4.1.1.1.1 Under Source and Destination DSA -> verify there are no errors listed in the *error* column.

4.1.1.2 Enter the command: **repadmin /showrepl**

4.1.1.2.1 Under Inbound Neighbors -> verify all *attempts* are showing successful.

4.1.2 Open the DC1 Event Viewer -> look for **Event ID: 1104** -> verifying The DFS Replication service successfully restarted replication after a backup or restore operation.

4.2 Verify GE Accounts and Credentials

4.2.1 From the HMI -> remote desktop to **DC1** or **DC2** -> login with **Admin** credentials.

4.2.2 **Desktop** -> click **Start** -> open **Server Manager** (if pop-up, click yes).

4.2.3 Select **Tools** -> open **Active Directory Users and Computers**.

4.2.4 Expand **HMI.local** -> **GEV Controls** -> **GEV Users and Groups**.

4.2.5 Verify the following accounts exist:

4.2.5.1 **GEADMIN**: **See Default Passwords Document*

4.2.5.2 **GEMAIN**: **See Default Passwords Document*

4.2.5.3 **GEOPER**: **See Default Passwords Document*

4.2.6 Expand **HMI.local** -> **Users**.

4.2.6.1 Verify the following accounts exist:

4.2.6.2 **GESWITCH**: **See Default Passwords Document*

4.2.7 Login with the GE Accounts to verify the accounts are active and the passwords are correct.

4.3 Add Network Devices to RADIUS

Note: If the IP Scheme is non-standard, the policy will need to be reviewed and adjusted to allow operation. Example, remove the current IP restriction and/or delete an empty user group.

Please review the Configure RADIUS Policies in the DC1 Virtual Post Configuration document 313T7465P0401BA for more information.

Note: FUTURE: need to discuss resolution for IP constraint.

4.3.1 Verify RADIUS Conditions: Client IPv4 Address

Note: The standard client IPv4 address allows the following networks to authenticate to the RADIUS server. If your network is not in any of these ranges, you need to modify the Client IPv4 Address for your network.

172.16.32.1 to 172.16.47.254

10.10.10.5 to 10.10.10.6

172.16.192.1 to 172.16.207.254

4.3.1.1 Login to the **DC1 Server**.

4.3.1.2 Open the **Network Policy Server**.

4.3.1.3 Expand **Policies** -> select **Network Policies**.

4.3.1.4 In the Network Policies -> double-click the **Switch Admin Access Control**.

4.3.1.4.1 Select the **Conditions Tab** -> select the **Client IPv4 Address**.

4.3.1.4.2 Select **Edit**.

4.3.1.4.3 Example: If your network is 192.168.201.0/24 change the Client IPv4 Address to: **(192\168\201\([1-9]|[1-9][0-9]|1[0-9]{2}|2[0-4][0-9]|25[0-4]))**

4.3.1.4.4 Click **OK**.

4.3.1.4.5 Click **OK**.

4.3.1.4.6 Repeat for the **Switch Operator Access Control**.

This section is only applicable if the system has a **FortiGate Firewall**.

4.3.1.5 In the **Network Policies** -> double-click the **FortiGate Admin Access Control**.

4.3.1.5.1 Select the **Conditions Tab** -> select the **Client IPv4 Address**.

4.3.1.5.2 Select **Edit**.

4.3.1.5.3 The standard configuration is allowing: **10.10.10.5 OR 10.10.10.6**

4.3.1.5.4 If the FortiGate does not match the standard IP Addresses -> replace with the actual FortiGate IP Address.

4.3.1.5.5 Example: **(192.168.201.110)**

Note: The IP Address needs to be in parenthesis.

4.3.1.5.6 Click **OK**.

4.3.1.5.7 Click **OK**.

4.3.1.6 If the system does not have a FortiGate Firewall:

4.3.1.6.1 Right-click the **FortiGate Admin Access Control** -> select **Disable**.

4.3.1.6.2 Right-click the **FortiGate Operator Access Control** -> select **Disable**.

4.3.2 Add Switches to RADIUS

4.3.2.1 From the HMI -> remote desktop to **DC1** -> login with **Admin** credentials.

4.3.2.2 Desktop -> click **Start** -> open **Server Manager** (if pop-up, click yes).

4.3.2.3 Select Tools -> Network Policy Server (NPS) -> open RADIUS Clients and Servers -> RADIUS Clients.

4.3.2.4 Verify the **Templates** are created. If not, refer to drawing 313T7465P0401AA to create it.

- 4.3.2.5 Expand RADIUS Clients and Servers folder -> right-click on **RADIUS Clients** -> select **New**.
- 4.3.2.6 Check/select the box for **Select an Existing Template** -> then Uncheck/deselect it.
- 4.3.2.7 Type the Friendly Name and IP address.
- 4.3.2.8 Select the **Advanced** tab -> under **Vendor Name** select **Cisco**.
- 4.3.2.9 Click OK.

4.3.3 Add Firewalls to RADIUS

- 4.3.3.1 Expand RADIUS Clients and Servers folder -> right-click on **RADIUS Clients** -> select **New**.
- 4.3.3.2 Check/select the box for **Select an Existing Template** -> then Uncheck/deselect it.
- 4.3.3.3 Type the Friendly Name and IP address.
- 4.3.3.4 Select the **Advanced** tab -> under **Vendor Name** select **RADIUS Standard**.
- 4.3.3.5 Under **Additional Options** check/select the box for **Access-Request messages must contain the Message-Authenticator attribute**.
- Note:** This must be done for Fortinet Firmware 7.2.10 and newer.
- 4.3.3.6 Click OK.

4.3.4 Export RADIUS Configuration

- 4.3.4.1 Right-click on **NPS (Local)** -> select **Export Configuration** -> save the file name as **RADIUS DC1** on the **C: Drive**.
- 4.3.4.2 Remote into the **DC2 Server** -> open **Network Policy Server (NPS)** -> right-click on **NPS (Local)** -> select **Import Configuration** -> navigate to **\\DC1\C\$** and select the **RADIUS DC1** file -> click Open.
- 4.3.4.3 Select RADIUS Clients -> verify the Switch Clients have been imported.

4.4 Add Switches/Firewalls to DNS

- 4.4.1 From the HMI -> remote desktop to **DC1** or **DC2** -> login with **Admin** credentials.
- 4.4.2 Desktop -> click **Start** -> open **Server Manager** (if pop-up, click yes).
- 4.4.3 Select **Tools** -> open **DNS**.
- 4.4.4 (Left panel) Expand **DC1** or **DC2** -> **Forward Lookup Zones** -> **HMI.local**.
- 4.4.5 Right-click on **HMI.local** -> select **New Host (A or AAAA)**.
- 4.4.6 Type the **Name** and **IP address** of each Switch/Firewall. Click **Add Host**.
- 4.4.7 Repeat for all Switches/Firewall on the system.

4.4.8 Verify the HMI.local is showing the added hosts.

4.5 **Active Directory – Computer Configuration**

Verify the computers are listed properly inside GEControls/HMI. Create HMIs according to 4108:

4.5.1 From the HMI -> remote desktop to **DC1** or **DC2** -> login with **Admin** credentials.

4.5.2 Desktop -> click **Start** -> open **Server Manager** (if pop-up, select yes).

4.5.3 Select **Tools** -> open **Active Directory Users & Computers**.

4.5.4 To add Windows10 HMIs that are in the control Room and are non-Touchscreen, select:

4.5.5 Expand **HMI.Local** -> **GEV Controls** -> **HMI** -> **Windows 10 HMIs** -> **Control Room**.

4.5.6 Right-click on **Control Room** and select **New** -> **Computer**.

4.5.7 Under **Computer Name** add the name of the HMI and click **OK**.

Note: Add all the HMIs based on functionality (HMI/HST), Windows version (Windows10), Location (Control Room/Non-Control Room).

Note: The Windows 7 HMI folder option has been removed (starting in OTArmor 4.0 Active Directory Users and Computers).

4.5.8 Verify **HMI.local - Computers** folder in Active Directory Users and Computers is empty:

4.5.8.1 From the HMI -> remote desktop to **DC1** or **DC2** -> login with **Admin** credentials.

4.5.8.2 **Desktop** -> click **Start** -> open **Server Manager** (if pop-up, select yes).

4.5.8.3 Select **Tools** -> open **Active Directory Users & Computers**.

4.5.8.4 Expand **HMI.Local – Computers**.

4.5.8.4.1 Verify there are no machines listed. If there is, move the HMIs to the correct folder by drag & drop to the correct folder.

4.6 **Disable Accounts**

Disable user accounts not applicable to the project.

4.6.1 From the HMI -> remote desktop to **DC1** or **DC2** -> login with **Admin** credentials.

4.6.2 **Desktop** -> click **Start** -> open **Server Manager** (if pop-up, select yes).

4.6.3 Select **Tools** -> open **Active Directory Users & Computers**.

4.6.4 Expand **HMI.local** -> **GEV Controls** -> **GEV Users and Groups**.

4.6.5 Review users -> disable the **accounts** not applicable to the project.

4.6.5.1 Sort the User by **Type**.

4.6.5.2 Locate the **User** to be disabled.

- 4.6.5.3 Right-click on the **User** -> select **Disable Account**.
- 4.6.6 Review groups -> delete the **groups** not applicable to the project.
 - 4.6.6.1 Sort the Group by **Type**.
 - 4.6.6.2 Locate the **Group** to be deleted.
 - 4.6.6.3 Right-click on the **Group** -> select **Delete**.
- 4.6.7 Expand **HMI.local** -> **Users**.
- 4.6.8 Review users -> disable the accounts not applicable to the project.
 - 4.6.8.1 Sort the User by **Type**.
 - 4.6.8.2 Locate the **User** to be disabled.
 - 4.6.8.3 Right-click on the User -> select **Disable Account**.
- 4.7 **Modify GEV HMI Computer Policy**

Verify the correct layout name is in the policy.

Note: The shared drive and Cimplicity Startup Screen Shortcut should be matched to the project. It may be different for every project. Review the policy and verify the Target Path is correct.

Note: The policy is linked to the HMI OU and the Virtual HMI OU.

 - 4.7.1 From the HMI -> remote desktop to **DC1** or **DC2** -> login with **Admin** credentials.
 - 4.7.1.1 **Desktop** -> click **Start** -> open **Server Manager** (if pop-up, select yes).
 - 4.7.1.2 Select **Tools** -> open **Group Policy Management**.
 - 4.7.1.3 Expand **HMI.local** -> **Group Policy Objects** -> **GEV HMI Computer Policy**.
 - 4.7.1.4 Right-click **GEV HMI Computer Policy** -> click **Edit**.
 - 4.7.1.5 Expand **User Configuration** -> **Preferences** -> **Windows Settings** -> **Shortcuts**.
 - 4.7.1.6 Right pane -> **Cimplicity** Startup Screen.
 - 4.7.1.7 Double-click -> **Cimplicity** Startup Screen -> Target Path and update the **Name of the CimLayout** to match the CimLayout on the screens for the project.
 - 4.7.1.8 Click **OK**.

Note: Repeat steps to create shortcuts for the software TCW and Alarms.

 - 4.7.1.9 **Desktop** -> right-click **Start** -> open **Command Prompt (Admin)** and type:
gpupdate /force
- 4.8 **Banner**

Add the login banner (if) provided by customer.

 - 4.8.1 From the HMI -> remote desktop to **DC1** or **DC2** -> login with **Admin** credentials.

- 4.8.2 Desktop -> click Start -> open **Server Manager** (if pop-up, select yes).
- 4.8.3 Select **Tools** -> open Group Policy Management.
- 4.8.4 Expand **Domains** -> **HMI.local** -> right-click **GE Domain Policy** -> select **Edit**.
- 4.8.5 Expand:
 - Computer Configuration -> Policies -> Windows Settings -> Security Settings -> Local Policies -> Security Options.**
- 4.8.6 Double-click on **Interactive logon: Message text for users attempting to log on**.
- 4.8.7 Select **Define this policy setting** in the template and insert the Customer provided banner.
- 4.8.8 Click **OK**.
- 4.8.9 In the same path, select -> **Interactive logon: Message title for users attempting to log on**.
- 4.8.10 Select **Define this policy setting**, use the Customer provided title text.
- 4.8.11 Click **OK**
- 4.8.12 On the HMIs/Servers, Desktop -> right-click Start -> open Command Prompt (Admin) and type: **gpupdate /force**.

4.9 **Backup Domain Administrator**

Login to a server with the Backup Domain Administrator (DAdmin/**See Default Passwords Document*) Account and verify it has administrator privileges.

5.0 **Switches – OTArmor**

5.1 **Verify NTP IP Address**

5.1.1 Do not complete this step for test switches. If this is not a test switch, use the following instructions (per network topology):

- 5.1.1.1 **config t**
- 5.1.1.2 **no ntp server <NTP ip address>**
- 5.1.1.3 **ntp server <NTP ip address>**
- 5.1.1.4 **exit**
- 5.1.1.5 **wr mem**

Note: These commands are specific to the Cisco 2960x models.

5.2 **Verify Name and IP Address for PDH VLAN**

5.2.1 Login to the switch and verify the name and IP Address for PDH VLAN is assigned.

5.2.2 If an IP Address isn't set for the VLAN PDH, define one:

5.2.2.1 **Config t**

5.2.2.2 **interface Vlan201** (Or the PDH Vlan defined for the project).

5.2.2.3 **ip address XXX.XXX.XXX.XXX(IP) XXX.XXX.XXX.XXX (mask)<enter>**

5.2.2.4 **Exit**

5.2.2.5 **Wr mem**

5.2.3 If the name isn't set, define one:

5.2.3.1 **Config t**

5.2.3.2 **Hostname <name of the switch based on 4108>**.

5.2.4 **Exit**

5.2.5 **Wr mem**

5.3 **Verify Switch Settings**

Login to the switch and verify the following. If further instructions are needed refer to document: **PRO – 427A7365**.

5.3.1 Make sure the logging server is configured per network topology:

5.3.1.1 Verify **logging source-interface Vlan201** is set and pointing to PDH VLAN.

5.3.1.2 Verify the **Logging host <ip address>** (LM1 IP Address) is set.

5.3.2 Verify DR1 is added in the access list:

5.3.2.1 **access-list 22 permit <ip address>** (DR1 IP Address) is set.

5.3.2.2 If not, use **PRO – 427A7365** to add DR1 to the access list.

5.3.3 Verify Default Gateway is set correctly if needed (Usually when in a DMZ zone)

5.3.3.1 **ip default-gateway <ip address>** (Firewall IP Address) is set.

5.3.4 Verify the **RADIUS Host** is correct and check the following settings:

5.3.4.1 **aaa group server radius RadiusServers**

server name rad1

server name rad2

!

5.3.4.2 **radius server rad1**

address ipv4 <DC1 ip address> auth-port 1812 acct-port 1813

timeout 1

key 7 <string>

!

5.3.4.3 **radius server rad2**

```
address ipv4 <DC2 ip address> auth-port 1812 acct-port 1813
timeout 1
key 7 <string>
!
```

5.3.5 Verify the following for **IP Domain Name**:

```
ip domain-name <Domain name: HMI.local >
ip name-server <DC1 IP Address>
```

5.3.6 Verify all switches are in RADIUS

5.3.6.1 Open **Server Manager** -> select **Tools** -> **Network Policy Server (NPS)** -> **Radius**.

5.3.6.2 Verify all switches listed in the 4108 are listed in RADIUS.

Note: This must be done in both DC1 and DC2, they do not mirror RADIUS.

5.4 Banner

Do not complete this step for test switches. Always check with the customer before adding a banner.

5.4.1 Default example:

```
config t
banner motd ^ <ENTER>
NOTICE TO USERS!
```

This is an official computer system and is the property of the owner. It is for authorized users only. Unauthorized users are prohibited. Users (authorized or unauthorized) have no explicit or implicit expectation of privacy. Any or all uses of this system may be subject to one or more of the following actions: interception, monitoring, recording, auditing, inspection and disclosing to security personnel and law enforcement personnel, as well as authorized officials of other agencies, both domestic and foreign. By using this system, the user consents to these actions. Unauthorized or improper use of this system may result in administrative disciplinary action and civil and criminal penalties. By accessing this system you indicate your awareness of and consent to these terms and conditions of use. Discontinue access immediately if you do not agree to the conditions stated in this notice.

```
^
exit
wr mem
```

5.5 References

Do not complete this step for test switches. If this is not a test switch, use the following instructions (per network topology):

5.5.1 Delete references to stacking switches that won't be used:

```
config t
no switch 3 provision ws-c2960x-24ts-l
no switch 4 provision ws-c2960x-24ts-l
exit
wr mem
```

Note: These commands are specific to the Cisco 2960 models.

5.6 Event Link Status

Do not complete this step for test switches. If this is not a test switch, use the following instructions (per network topology):

5.6.1 Enable the events on each port of the switch:

```
config t
int range gi 1/0/1 - 28, gi 2/0/1 - 28
logging event link-status
exit
exit
wr mem
```

Note: These commands are specific to the Cisco 2960 models.

Note: These commands can also be executed from CatTools to deploy in major quantities.

5.7 Span Ports for NIDS

Note: Typically, this is configured on the root bridge switches. Refer to the network topology for the location of the SPAN port connection.

5.7.1 Login to switch, and enter the following:

```
config t
monitor session 1 source vlan 101<space>,<space>172<space>,<space>201
monitor session 1 destination interface <gi1/0/7> rx (selected port)
```

Note: These commands are specific to Cisco switches.

5.8 MAC Filtering

Configure switch for MAC Address Filtering. Verify customer has either accepted or declined this option during the CWT. If accepted, refer to the following document: **PRO – 427A4442**. Do not complete this step for test switches.

5.9 Disabled Unused Ports

Shutdown the unused interfaces on the switch. Verify customer has either accepted or declined this option during the CWT. If accepted, refer to the following document: **PRO – 427A7365**.

5.10 **Firmware Upgrade**

Upgrading the firmware on the Switch.

To address the Cyber Security vulnerabilities identified in the Cisco IOS operating system, utilize the following procedure (FMI) **PRO – 131M0028**.

5.11 **Test Switch Failover (Stacked Switches)**

Perform a switch failover test. This applies to all switch models in a stack configuration scoped for the project.

6.0 **Firewall and NIDS – OTArmor**

6.1 **Fortinet Firewall Configuration**

For the steps below, refer to **152M3647 on PLM** for the Fortinet Firewall Configuration document.

6.1.1.1 Files required for the configuration:

6.1.1.2 **Firmware Version** – Verify current firmware version on the firewall, check CCB for latest firmware version - Controls Configuration Board (CCB) – **Switches/Firewalls – Upgrade to the latest according to CCB.**

<https://engineeringconfluence.pw.ge.com/dpconfluence/pages/releaseview.action?pageId=24839079>

6.1.1.2.1 Firmware_[xxx]F_v[y.y.y].out

6.1.1.3 To check the current firmware version:

6.1.1.4 Login to the Fortinet Firewall.

6.1.1.5 Select **Dashboard** -> **Status**.

6.1.1.6 Under **System Information** -> check **Firmware**.

6.1.1.7 **FortiManager & FW License** – Verify the Fortinet has a license.

6.1.1.7.1 EntitlementExport (license) File

6.1.1.8 To check if the firewall is licensed:

6.1.1.9 Login to the Fortinet Firewall.

6.1.1.10 Select **System** -> **FortiGuard**.

6.1.1.11 Under **License Information** -> check **License**.

6.1.1.12 **Fortinet Configuration** – Load the base configuration file if not already loaded.

6.1.1.12.1 Config_[xxx]F_v[y.y.y].conf

6.1.1.13 **Fortinet IP Script** – Review text file and update per project requirement.

6.1.1.13.1 IP_Fortinet_Script_[xxx]F.txt

6.1.1.14 **Signature Definitions** (PVP Disk) – Review the project BOM to verify the following are required – per license type.

6.1.1.14.1 apdb_OS6.4.0_[xx.xxx].APDB.pkg

6.1.1.14.2 nids_OS6.4.0_[xx.xxx].NIDS.pkg

6.1.1.14.3 vsigupdate-OS6.4.0_[xx.xxx].ETDB.High.pkg

6.1.1.14.4 isdb_OS6.4.0_xx.00xxx.ISDB.pkg

6.2 Create a Certificate for the Firewall

6.2.1 Login to the **CA1** Server.

6.2.2 Open an **Administrative PowerShell**.

6.2.3 At the prompt -> enter: **cd C:\CertificateSigning**

6.2.4 Run the script -> enter: **./Create-GEV-Cert.ps1**.

6.2.5 Press Enter to continue.

6.2.6 At the prompt -> enter **"2"** to select "request a signed certificate..."

6.2.7 At the prompt -> enter **"y"** to "generate a non-encrypted private key..."

6.2.8 Press Enter to continue.

6.2.9 Enter the "FQDN": (**example**): **GENIDS.HMI.local**

6.2.10 Enter the "Common Name": (**example**): **GENIDS**

6.2.11 Enter the "IP Address": (**example**): **172.16.201.xxx**

6.2.12 It will prompt for a password -> enter the password: **GEV.23cert**

6.2.13 When complete -> the script will create a folder with the certificates (C:\CertificateSigning\<name of firewall>).

Note: Certificates used for the Firewall are: **.crt** and **.key** files.

6.3 Install the Security Certificate

6.3.1 Open internet browser -> type the **firewall IP address** into the URL.

- 6.3.2 Log into the **FortiGate** firewall using the administrator credentials.
- 6.3.3 Navigate to **System** → **Certificates**.
- 6.3.4 On the top menu -> select **Create/Import** -> select **Certificate**.
- 6.3.5 Select **Import Certificate**.
- 6.3.6 Select **Certificate**.
- 6.3.7 Click **Upload** -> select the **.crt** file
- 6.3.8 Click **Upload** -> select the encrypted **enc.key** file
- 6.3.9 Navigate to **System** -> **Settings**.
- 6.3.10 Under **Administration Settings** -> select the **HTTPS server certificate drop-down** -> select the **NIDS certificate**.
- 6.3.11 Enter the password (password used in Create-GEV-Cert.ps1)
- 6.3.12 Enter the confirm password.
- 6.3.13 Click **Create**.

6.4 Test Security Certificate

- 6.4.1 Open an internet browser -> type the firewall IP address (**https://<FW IP Address>**) into the URL.
- 6.4.2 Verify the security warning and certificate error is now gone in the Internet Explorer URL field.

6.4.3 (If necessary) Command Line to Restore Certificate

Note: If an error is encountered during installation of the certificate, it is possible to set the web certificate back to the default certificate.

6.4.3.1 Log into the HMI -> remote desktop to **CA1**.

6.4.3.2 Open **Administrative PowerShell** -> type the following:

```
config system global
unset admin-server-cert
end
```

6.5 Add Customer Rules (If applicable)

- 6.5.1 Please refer to **REF – 136M5796** Policies Template for adding customer rules to the firewall.

6.6 Add Customer Login Banner (If applicable)

- 6.6.1 Login to the **DR1** server.
- 6.6.2 Open internet browser -> type the firewall IP address into the URL.
- 6.6.3 Login to the firewall with user credentials.
- 6.6.4 Go to CLI Console and type the following:

```
config system global
set pre-login-banner enable
end
```
- 6.6.5 Close the CLI Console and go back to the GUI Interface.
- 6.6.6 Go the **System** -> **Replacement Message**.
- 6.6.7 Switch to Extended View.
- 6.6.8 Under **Admin** -> select **Pre-Login Disclaimer Message**.
- 6.6.9 Click **Edit**.

Manage Images	Edit	Search	Simple View Extended View
Name	Description	Modified	
Admin 2			
Post-login Disclaimer Message	Replacement message for post-login disclaimer		
Pre-login Disclaimer Message	Replacement message for pre-login disclaimer		

- 6.6.10 Copy/Paste the customer banner text -> click **Save**.

6.7 Manually Configure/Verify NIDS (If necessary)

Note: Only if the baseline configuration has not been deployed.

- 6.7.1 Please refer to **PRO – 152M3647** Fortinet Firewall Configuration **Section 12.0**.

6.8 Verify RADIUS (If necessary)

Note: RADIUS is configured in the Baseline Configuration (Please refer to **PRO – 152M3547**).

- 6.8.4 Verify/Test the **RADIUS authentication**.

6.8.4.1 Log out of the firewall.

6.8.4.2 Try logging in as a user under **Network Administrators** and verify privileges as **Read-Write**.

6.9 Manually Configure/Verify HA Configuration (If necessary)

- 6.9.1 Please refer to **PRO – 152M3647** Fortinet Firewall Configuration **Section 13.0**.

6.10 Verify NTP Address

- 6.10.1 Login to the **DR1** server.

- 6.10.2 Open internet browser -> type the firewall IP address into the URL.

6.10.3 Login to the firewall with user credentials.

6.10.4 Navigate to **System** -> **Settings**.

6.10.5 Under **System Time** -> verify **NTP** is selected, **Custom** is selected and enter the **NTP IP address**.

System Time

Current system time 2021/10/12 16:57:17

Time zone (GMT-6:00) Central Time (US & Canada)

Set Time **NTP** PTP Manual settings

Select server FortiGuard **Custom** NTP IP Address

Sync interval 60 Minutes (1 - 1440)

Setup device as local NTP server ☐

6.10.6 Scroll down -> click **Apply**.

6.11 Verify LM1 (Splunk) Configuration

6.11.1 Login to the **DR1** server.

6.11.2 Open internet browser -> type the firewall IP address into the URL.

6.11.3 Login to the firewall with user credentials.

6.11.4 Navigate to **Log & Report** -> select **Log settings**.

6.11.5 Configure Logging and Archiving for SIEM server

6.11.5.1 Scroll down to **Send Logs to Syslog**.

6.11.5.2 Verify **Send logs to Syslog** is checked/enabled and the server IP address is the **LM1 IP Address**.

6.11.5.3 Verify the following **Log Settings** are set:

6.11.5.3.1 Event Logging: **All**

6.11.5.3.2 Local Traffic Log: **Customize: Log Allowed Traffic** and **Log Local Out Traffic**

Send logs to syslog ☒

IP Address/FQDN

☐ Cloud Logging Settings

UUIDs in Traffic Log [i](#)

Address ☐

Log Settings

Event Logging ☒ All ☐ Customize

Local Traffic Log ☐ All ☒ Customize

- ☒ Log Allowed Traffic
- ☐ Log Denied Unicast Traffic
- ☒ Log Local Out Traffic
- ☐ Log Denied Broadcast Traffic

Note: Depending on the site, some Log Settings may need to be deselected (tuned) to prevent the Firewall Syslog data from overutilizing the Splunk license.

6.12 Test NIDS

6.12.1 Login to **DR1** Server.

6.12.2 Open File Explorer -> type the path: **\\CA1\Admin_Procedures\OTArmor**

6.12.3 For OTArmor 4.0 or older

IMPORTANT: If you have an OTArmor system that is 4.0 or older and are testing NIDS on a **FortiGate F** model edit the Test-NIDS2.ps1 and change the string name to NC NIDS.

6.12.3.1 Right-click the **Test-NIDS2.ps1** and select Edit.

6.12.3.2 Scroll down and change the **\$str1** = to **NC NIDS**.

```
}
Write-Host "Connected, sending message"

$str1 = "NC NIDS "
$str2 = "TEST ATTACK"

$writer.Write($str1 + $str2)
$writer.Flush()

if ($writer -ne $Null)
```

6.12.3.3 Save the **Test-NIDS2.ps1** file.

6.12.3.4 Right-click the **Test-NIDS2.ps1** and **Run with PowerShell**:

6.12.3.5 Type the **hostname** ex: DC1 (should be a physical to virtual) -> hit **<enter>**

6.12.3.6 Type **Y** (for yes) -> hit **<enter>**

- 6.12.4 Log into the firewall and under **Network -> Interfaces** check the port is configured for NIDS.
- 6.12.5 Also, under **Log & Report -> Intrusion Prevention** -> verify you got the logs for the attack (Test-NIDS2.ps1).
- 6.12.6 If LM1 is a part of the scope.
 - 6.12.6.1 Open **Splunk** -> Login with credentials
 - 6.12.6.2 Navigate to **Search & Reporting** app and type: NIDS.
 - 6.12.6.2.1 Verify time matches with the ones you executed and from the devices they were executed.
- 6.12.7 **For OTArmor 4.0.1 or later**
- 6.12.8 Right-click the **Test-NIDS3.ps1** and **Run with PowerShell**:
 - 6.12.8.1 Type the **hostname** ex: DC1 (should be a physical to virtual) -> hit **<enter>**
 - 6.12.8.2 Type **Y** (for yes) -> hit **<enter>**
- 6.12.9 Log into the firewall and under **Network -> Interfaces** check the port is configured for NIDS.
- 6.12.10 Also, under **Log & Report -> Intrusion Prevention** -> verify you got the logs for the attack (Test-NIDS3.ps1).
- 6.12.11 If LM1 is a part of the scope.
 - 6.12.11.1 Open **Splunk** -> Login with credentials
 - 6.12.11.2 Navigate to **Search & Reporting** app and type: NIDS.
 - 6.12.11.2.1 Verify time matches with the ones you executed and from the devices they were executed.

6.13 Test Lockout

Change lockout to 5 attempts.

- 6.13.1 Login to the **DR1** server.
- 6.13.2 Open internet browser -> type the firewall IP address into the URL.
- 6.13.3 Login to the firewall with user credentials.
- 6.13.4 On the Firewall menu -> click the **CLI Console** button -> type the following:
config system global
set admin-lockout-threshold 5
set admin-lockout-duration 60
end

Note: The threshold and duration numbers may change per customer request.

6.13.5 Logout and enter the username/password incorrectly 5 times -> verify the Too many login failures... message appears.

7.0 **PRO – HA Switch Configuration – OTArmor**

Note: Before beginning the configuration process, check vSAN health in vSphere. Resolve any warnings or errors. Switches will need to reboot as part of firmware upgrade/stack configuration, so redundancy issues could potentially force a server migration.

- 7.1.1 Log into the switch -> start with the **Primary Switch** (first in the stack).
- 7.1.2 Connect a cable from the **PC** to the **OOB port** on switch.
- 7.1.3 **OOB port IP** is set to **172.16.1.254** -> configure PC adapter to an address on that subnet (e.g. 172.16.1.253).
- 7.1.4 Login to the switch -> default credentials are: Username: **cisco** Password: **See Default Passwords Document*
- 7.1.5 When prompted -> configure a new Username and Password -> standard is Username: **Admin** Password: **See Default Passwords Document*
- 7.1.6 At the top of the screen -> select the **Display Mode** dropdown -> select **Advanced** to see all configuration options.

7.1 **HA Switch Certificate (FUTURE)**

7.2 **Upgrade Firmware**

- 7.2.1 On the left Menu -> select **Administration -> File Management -> Firmware Operations**.
- 7.2.2 Select the following options:
 - 7.2.2.1 Operation Type: **Update Firmware**
 - 7.2.2.2 Copy Method: **HTTP/HTTPS**
 - 7.2.2.3 Choose File: **Select latest approved firmware**
 - 7.2.2.4 Currently: **image_c1300_4.1.4.1_official_key.bin**

Note: Must use the current firmware version (image_c1300_4.1.4.1_official_key.bin) to successfully failover.

- 7.2.3 Click **Apply**.

7.3 **Update Switch Configuration with Baseline**

- 7.3.1 Check the project BOM for Switch Model. Download the Baseline Configuration for the Switch Model from PLM and save to your computer.

- 7.3.2 Login to the Switch (SW1) with Admin local credentials.
- 7.3.3 On the left Menu -> select **Administration -> File Management -> File Operations**.
- 7.3.4 Set **File Operations** to **Update File**.
- 7.3.5 Set **Destination File Type** to **Startup Configuration**.
- 7.3.6 Set **Copy Method** to **HTTP/HTTPS**.
- 7.3.7 Browse to the switch configuration file on your computer (e.g., **sw1ota.txt**)
- 7.3.8 Select the file -> click **Open**.
- 7.3.9 Select **Apply**.
- 7.3.10 The configuration file will take some time to apply -> wait for completion.
- 7.3.11 When finished -> navigate to **Administration -> Reboot**.
- 7.3.12 Select **immediate** to restart the switch immediately.
- 7.3.13 Reboot will take a few minutes to complete.

Note: The new configuration file has an ACL and switch login will have to be completed on the DR1 Server only. If RADIUS was configured, use only a RADIUS account to login to the switch.

- 7.3.14 Test switch login from the DR1 Server.
- 7.3.15 Repeat the steps above for the remaining switches (e.g., SW2, secondary).

Important: For the secondary (SW2) switch verify the configuration file is the correct text file for the switch (e.g., **sw2ota.txt**).

Information: Once the switch configuration files have been uploaded to the respective switches, they can now be backed up using CatTools. Please see the CatTools section in this document for steps on backing up the switches.

7.4 Change Management IP Address

- 7.4.1 Once the switch has rebooted -> login (Username: **Admin** Password: **See Default Passwords Document*).
- 7.4.2 Select **IP Configuration -> IPv4 Management and Interfaces -> IPv4 Interface**.

Note: Cannot edit the existing OOB entry.

- 7.4.3 Click **New**.
- 7.4.4 Select the following options:
 - 7.4.4.1 Interface: **Out of Band**
 - 7.4.4.2 IP Address Type: **Static IP Address**
 - 7.4.4.3 IP Address: **Address on your OTArmor PDH network**
 - 7.4.4.4 Network Mask: **Subnet mask of your OTArmor PDH network**

7.4.5 Select **Apply**.

Note: If another window pops up, confirm the change. You will likely lose connection to the switch and will need to log in again using the new IP address.

7.5 vCenter & Host & iDRAC Certificate

7.5.1 VCSA Secure Access – Generate Certificate

7.5.2 Generate certificates using (Create-GEV-Cert.ps1) script

7.5.3 Login to **CA1**.

7.5.4 Open **Windows PowerShell as Admin**.

7.5.5 Navigate to the following path: **cd C:\CertificateSigning**

7.5.6 Enter the following command: **.\Create-GEV-Cert.ps1**

7.5.7 Hit **Enter** to continue.

7.5.8 At the CSR Request prompt -> Enter **"2"**.

7.5.9 At the Generate a non-encrypted private key -> Enter **"y"**.

7.5.10 Hit **Enter**.

7.5.11 Hit **Enter** to continue.

7.5.12 At the FQDN -> enter the **VCSA FQDN** (example: VCSA.HMI.local).

7.5.13 At the Common Name -> enter the **VCSA Hostname** (example: VCSA).

7.5.14 At the Devices IP Address -> enter the **VCSA IP Address** (example: 172.16.201.xxx).

7.5.15 At the Password for private key file -> hit **Enter** (leave blank, no password).

7.5.16 Verify the command completed successfully.

7.5.17 The certificates will be created in the following directory: **C:\CertificateSigning\VCSA**.

7.5.18 Generate the certification chain (.cer) file

7.5.19 From CA1 VM -> open run and type **certlm.msc** to open the **Local Computer Certificate Manager**.

7.5.20 Under **certificate – local computer** -> select (Trusted Root Certificate Authority) -> **Certificates**.

7.5.21 Double-click on **root certificate** (HMI-CA1-CA) -> select **details** tab and click on **copy to file**.

7.5.22 Select **Next**.

7.5.23 Select **Base-64 encoded X.509 (.cer)**.

7.5.24 Select **Next**.

7.5.25 Browse where do you want to export the certificate -> name the file (VCSA.cer).

- 7.5.26 Click **Save**.
- 7.5.27 Login to the **vSphere** client.
- 7.5.28 On the main menu -> select the three lines -> click **Administration**.
- 7.5.29 Under **Certificates** -> select **Certificate Management**.
- 7.5.30 Select the **Trusted Root** tab -> select **ADD TRUSTED ROOT CERTIFICATE**.
- 7.5.31 Browse for the root certificate -> click **Add**.
- 7.5.32 Select the **Machine SSL** tab -> select the new certificate -> click **IMPORT AND REPLACE CERTIFICATE**.
- 7.5.33 Select **Replace with external CA Certificate** (Requires private key).
- 7.5.34 In the **Machine SSL Certificate** box -> select **Browse File** -> navigate to the VCSA folder -> select **VCSA_Cert.crt** file -> click Open.
- 7.5.35 In the **Chain of trusted root certificates** box -> select **Browse File** -> navigate to the **VCSA.cer** file -> select the file -> click Open.
- 7.5.36 In the **Private Key** box -> select **Browse File** -> navigate to the VCSA folder -> select **VCSA_Cert_noenc.key** file -> click Open.
- 7.5.37 Click **Next**.
- 7.5.38 Select/check the **Backup acknowledgment**.
- 7.5.39 Click **Next** -> click **Finish**.
- 7.5.40 **Virtual Host Secure Access – Generate Certificate**
- 7.5.41 Generate certificates using (Create-GEV-Cert.ps1) script
- 7.5.42 Login to **CA1**.
- 7.5.43 Open **Windows PowerShell as Admin**.
- 7.5.44 Navigate to the following path: **cd C:\CertificateSigning**
- 7.5.45 Enter the following command: **.\Create-GEV-Cert.ps1**
- 7.5.46 Hit **Enter** to continue.
- 7.5.47 At the CSR Request prompt -> Enter **"2"**.
- 7.5.48 At the Generate a non-encrypted private key -> Enter **"y"**.
- 7.5.49 Hit **Enter**.
- 7.5.50 Hit **Enter** to continue.
- 7.5.51 At the FQDN -> enter the **VHX FQDN** (example: VH1.HMI.local).
- 7.5.52 At the Common Name -> enter the **VHX Hostname** (example: VH1).
- 7.5.53 At the Devices IP Address -> enter the **VHX IP Address** (example: 172.16.201.xxx).

- 7.5.54 At the Password for private key file -> hit **Enter** (leave blank, no password).
- 7.5.55 Verify the command completed successfully.
- 7.5.56 The certificates will be created in the following directory: **C:\CertificateSigning\VCSA**.
- 7.5.57 Repeat these steps for every host.
- 7.5.58 Login to the **vSphere** client.
- 7.5.59 Click on **VCSA.<Domain Name>**. Click on **Configure** tab->**Settings**->**Advance Settings**.
At the top left, click on **EDIT SETTINGS**. Look for **vpzd.certmgmt.mode** and change **vcsa** for **custom**. Click Save.
- 7.5.60 Under **Inventory** -> select **VHx.HMI.local**.
- 7.5.61 Select the **Configure** tab -> under **System** -> select **Certificate**.
- 7.5.62 Select the **MANAGE WITH EXTERNAL CA** -> click **Refresh CA Certificates** > select **Yes**.
- 7.5.63 Select the **MANAGE WITH EXTERNAL CA** -> click Import and Replace.
- 7.5.64 Select "Replace with external CA certificate where CSR is generated from a Certificate Authority (Requires private key). Click Next.
- 7.5.65 Under Certificate, click Browse and navigate to the folder specific to the host you are doing and Select **VHX_Cert** and click Open.
- 7.5.66 Under Private Key, click Browse **VHX_Cert_nonenc.key** and click Open.
- 7.5.67 Click Next
- 7.5.68 Verify the data shown matches the name of the host you are working on and if everything is OK, click Finish.
- 7.5.69 Open another browser window.
- 7.5.70 Enter: **https://VHx.HMI.local** -> verify the connection is secure and has a valid certificate.
- 7.5.71 Repeat above for all Virtual Hosts in the environment.
- 7.5.72 FUTURE: Virtual Host – General Certificate – No VCSA
- 7.5.73 **iDRAC Secure Access – Generate Certificate**
- 7.5.74 Add the iDRAC to DNS
- 7.5.75 Login to **DC1**.
- 7.5.76 Open the **DNS Manager**.
- 7.5.77 Under **Forward Lookup Zones** -> select the **HMI.local**.
- 7.5.78 Right-click the **HMI.local zone** -> select **New Host (A or AAAA) ...**
- 7.5.79 In the New Host window -> enter the **iDRAC Hostname** -> enter the **IP Address**.
- 7.5.80 Select/check the box for **Create associated pointer (PTR) record**.

7.5.81 Select **Add Host**.

7.5.82 Generate certificates using (Create-GEV-Cert.ps1) script

7.5.83 Login to **CA1**.

7.5.84 Open **Windows PowerShell as Admin**.

7.5.85 Navigate to the following path: **cd C:\CertificateSigning**

7.5.86 Enter the following command: **.\Create-GEV-Cert.ps1**

7.5.87 Hit **Enter** to continue.

7.5.88 At the CSR Request prompt -> Enter **"2"**.

7.5.89 At the Generate a non-encrypted private key -> Enter **"y"**.

7.5.90 Hit **Enter** to continue.

7.5.91 At the FQDN -> enter the **iDRAC FQDN** (example: iDRAC.HMI.local).

7.5.92 At the Common Name -> enter the **iDRAC Hostname** (example: iDRAC).

7.5.93 At the Devices IP Address -> enter the **iDRAC IP Address** (example: 172.16.xxx.xxx).

7.5.94 At the Password for private key file -> hit **Enter** (leave blank, no password).

7.5.95 Verify the command completed successfully.

7.5.96 The certificates will be created in the following directory: **C:\CertificateSigning\iDRAC**.

7.5.97 Login to the **iDRAC**.

7.5.98 Select **iDRAC Settings** -> **Services** tab -> **Web server** -> **SSL/TLS Custom Certificate**.

7.5.99 Click **Upload Certificate**.

7.5.100 Select **Choose File** -> navigate to the iDRAC folder created on the CA1.

7.5.101 Select the **.pfx** file -> click **Open**.

7.5.102 (no password) Click **Upload**.

7.5.103 Click **Continue**.

7.5.104 Click **OK** on the Success window.

7.5.105 Close the browser window -> open a new browser window to the iDRAC.

7.5.106 Verify the Web console has a valid certificate.

7.5.107 Repeat the above steps for each additional iDRAC server (if applicable).

7.6 HA Failover

7.6.1 HA Failover for Hosts

Note: Before performing the HA Failover test, verify VH1, VH2 and VH3 are running and all VM's on each hypervisor are Powered On and operating normally.

Note: Before performing any HA Failover testing, the Trellix SIEM must be properly Shut down. Refer to the Trellix SIEM section of this document.

IMPORTANT: Failure to properly Shutdown or Reboot of the Trellix SIEM will cause a corruption of the database requiring a Trellix support ticket to resolve.

7.6.1.1 Remove the **Power Connections** from one of the Virtual Hosts (VH1 or VH2 or VH3).

7.6.1.2 Login into **VSphere** with admin credentials.

7.6.1.3 Verify all the Virtual Machines on the Virtual Host where power was removed are **migrating** to the remaining Hosts.

Note: The migration will take some time.

7.6.1.4 Reconnect the **Power Connection** on the Virtual Host which is powered off. Wait for the Virtual Host to power on completely.

7.6.1.5 Manually move all the migrated Virtual Machines back to the Virtual Host.

7.6.1.5.1 Right-click the Virtual Machine -> select **Edit Settings** and **Uncheck the CD Drive**.

7.6.1.5.2 Click **Save**.

7.6.1.5.3 Right-click the Virtual Machine -> select **Migrate -> Computer Resources -> Next -> Finish**.

7.6.1.6 Verify the Virtual Machine migration is successful.

7.6.2 Failover for HA Switches

Note: Applies to the C1300 switches.

7.6.2.1 Remove power from **SW1OTA**.

7.6.2.2 Verify the **SW2OTA** switch is still active, and no issues occur in vCenter.

7.6.2.3 Reconnect the power to the **SW1OTA** -> wait for the switch to power-up.

7.6.2.3.1 Login to the reconnected switch -> verify access.

7.6.2.4 Remove power from the **SW2OTA**.

7.6.2.5 Verify the **SW1OTA** is still active, and no issues occur in vCenter.

7.6.2.6 Reconnect the power to **SW2OTA** -> wait for the switch to power-up.

7.6.2.6.1 Login to the reconnected switch -> verify access.

8.0 CatTools – OTArmor

8.1 Change the CatTools Service Log On

8.1.1 Login to the DR1 server.

- 8.1.2 Open Services.
- 8.1.3 Find the CatTools service -> right-click and select the Log On tab.
- 8.1.4 Select This account -> enter the Acronis Agent domain user (AcronisAgent@HMI.local)
- 8.1.5 Enter the Password.
- 8.1.6 Confirm the Password.
- 8.1.7 Click OK.

8.2 Enable the TFTP Server

Add the Windows Firewall port (UDP 69) on DR1. This port is not added by default in the build and will need to be added before enabling the TFTP port.

To open the port on Windows Firewall, open **advanced security (inbound rules and outbound rules)** and add it to both (i/o). Add a new Rule, select UDP, port 69 and allow disconnection. Apply to all, and name: **TFTP**.

- 8.2.1 Login to the HMI -> remote desktop to **DR1**.
- 8.2.2 Open **CatTools** -> under Devices -> select the **TFTP** tab.
- 8.2.3 Verify the server has started. (It will be button on the bottom, next to clear and stop).
- 8.2.4 At the menu, select **Options** -> click **Setup** -> **TFTP Server** tab. Check/select the following:
 - 8.2.4.1 **Automatically start TFTP Server on program start-up**
 - 8.2.4.2 **Allow overwrite of existing files**
 - 8.2.4.3 Download folder in path: **\\NAS\Backup\Switches\Download**
 - 8.2.4.4 Click **OK**.

8.3 Device Entries

Create device entries for the switches and firewalls.

8.3.1 Switch:

- 8.3.1.1 Open **CatTools** -> navigate to the **Devices** window -> right-click -> select **Add**.
- 8.3.1.2 Select the **Device info** tab -> enter the following:
 - 8.3.1.2.1 Device Type: **Cisco.Switch.IOS**
 - 8.3.1.2.2 Name: **Switch Name**
 - 8.3.1.2.3 Host Address: **Switch IP address**
 - 8.3.1.2.4 Model: **Catalyst 2955**
 - 8.3.1.2.5 Connect via: **Direct connect**
 - 8.3.1.2.6 Method: **SSH2**

8.3.1.3 Select the **Passwords** tab -> enter the following:

8.3.1.3.1 Enable Password: **See Default Passwords Document*

8.3.1.3.2 SSH Username: **GESwitch**

8.3.1.3.3 SSH Password: **<CatTools credentials>**

8.3.1.3.4 Click **OK**.

Note: To copy/paste for remaining switches, right-click on switch device and select Copy.

8.3.2 **HA Switch (Small Business C1300):**

8.3.2.1 Open **CatTools** -> navigate to the **Devices** window -> right-click -> select **Add**.

8.3.2.2 Select the **Device info** tab -> enter the following:

8.3.2.2.1 Device Type: **Cisco.SmallBusiness**

8.3.2.2.2 Group: (Type/Enter Manually) **C1300**

8.3.2.2.3 Name: **<Switch Name> (SW1OTA)**

8.3.2.2.4 Host Address: **<Switch IP address>**

8.3.2.2.5 File Name: **SW1OTA**

8.3.2.2.6 Model: **Other**

8.3.2.2.7 Connect via: **Direct connect**

8.3.2.2.8 Method: **SSH2**

8.3.2.2.9 Port: **22**

8.3.2.3 Select the **Passwords** tab -> enter the following:

8.3.2.3.1 SSH Username: **GESwitch**

8.3.2.3.2 SSH Password: **<CatTools credentials>**

8.3.2.3.3 Click **OK**.

Note: To copy/paste for remaining switches, right-click on switch device and select Copy.

8.3.2.4 Select **Variations** tab -> enter the following:

8.3.2.4.1 Click on **Use Variations** button.

8.3.2.4.2 Select **Prompts** Tab -> select/enter the following:

8.3.2.4.3 Under **Mode Prompts** -> select/check **Privileged prompt** -> enter: **#**

8.3.2.4.4 Select **Device Messages** tab -> under **Invalid/Incomplete command**:

8.3.2.4.5 Select/check **Invalid Command** -> change the default to: **%
Unrecognized command**

8.3.2.4.6 Under **Config Header Text** -> select/check **Config Header Text** -> enter the following: **config-file-header**

8.3.2.4.7 Select **Additional Commands** tab -> select **Disable Paging** -> Type **"terminal datadump"**.

8.3.2.4.8 Select/check **Show running config** -> type **"show running-config"**.

8.3.2.4.9 Click **Save**.

8.3.2.5 Click **OK** to close.

8.3.3 **Firewall:**

8.3.3.1 Open **CatTools** -> navigate to the **Devices** window -> right-click -> select **Add**.

8.3.3.2 Select the **Device info** tab -> enter the following:

8.3.3.2.1 Device Type: **Fortinet.FortiOS**

8.3.3.2.2 Name: **Firewall Name**

8.3.3.2.3 Host Address: **Firewall IP address**

8.3.3.2.4 Version: **Other**

8.3.3.2.5 Connect via: **Direct connect**

8.3.3.2.6 Method: **SSH2**

8.3.3.3 Select the **Passwords** tab -> enter the following:

8.3.3.3.1 SSH Username: **GEAdmin**

8.3.3.3.2 SSH Password: **<Firewall credentials>**

8.3.3.4 Select **Variations** tab -> enter the following:

8.3.3.4.1 Click on **Use Variations** button.

8.3.3.4.2 Select Pre/Post Login tab.

8.3.3.4.3 Under **Post-Login Keystroke** -> select/check the box for **"Post-login keys"** -> type in the box: **a**

8.3.3.4.4 Click **Save**.

8.3.3.5 Click **OK** to close.

8.4 **OS Version Activity**

8.4.1 Open **CatTools** -> Select **Add** under **Activities**.

8.4.1.1 Type: **Report.Version table**.

8.4.1.2 Name: **Switch Version Table**

8.4.1.3 Description: **Table of current network device operating system versions for Gather Inventory Power Shell.**

8.4.1.4 Report file: **C:\Program Files (x86)\CatTools3\Reports\GEV_Version_Information.txt**

- 8.4.1.5 Check **Overwrite existing report file**.
- 8.4.1.6 Select **Times Tab**.
 - 8.4.1.6.1 Reoccurring **Never**.
- 8.4.1.7 Select **Devices Tab**.
 - 8.4.1.7.1 **Select All Devices** checkbox.
- 8.4.1.8 Select **E-Mail**.
 - 8.4.1.8.1 Uncheck all boxes on this page.
- 8.4.1.9 Select **Options**.
 - 8.4.1.9.1 Check **Include version table data for stack members (when possible)**.
- 8.4.1.10 Click **Ok**.
- 8.4.1.11 Execute the task to populate the inventory information in the report.

8.5 Backup Activities

Create backup activities for the switches and firewalls.

- 8.5.1 Open **CatTools** -> navigate to the **Activities** window -> right-click -> select **Add**.
- 8.5.2 Select the **Activity** tab -> enter the following:
 - 8.5.2.1 Type: **Device.Backup.Running Config**
 - 8.5.2.2 Name: **Backup Switches**
- 8.5.3 Select the **Time** tab -> enter the following:
 - 8.5.3.1 Days of the week: **Sunday**
 - 8.5.3.2 Reoccurring: **At Midnight**
- 8.5.4 Select the **Devices** tab -> enter the following:
 - 8.5.4.1 Check **Select All Devices**.
 - 8.5.4.2 Select **e-mail**.
 - 8.5.4.3 Uncheck Everything.
- 8.5.5 Select the **Options** tab -> enter the following:
 - 8.5.5.1 CURRENT CONFIG FILE:
\\NAS\Backup\Switches\Configs\Config.Current.Running.%BaseFile%.txt
 - 8.5.5.2 DATED CONFIG FILE:
**\\NAS\Backup\Switches\DatedConfigs\Config.Dated.Running.%BaseFile%.%DateISO
 %-%TimeHHMM%.txt**
 - 8.5.5.3 HTML COMPARE REPORT:

\\NAS\Backup\Switches\Reports\ConfigChanges\Config.Changes.Running.%BaseFile
%.%DateISO%-%TimeHHMM%.htm

8.5.5.4 Uncheck other options.

8.5.5.5 Click **OK**.

8.6 Line Command

Create line command activity.

8.6.1 Open **CatTools** -> navigate to the **Activities** window -> right-click -> select **Add**.

8.6.2 Select the **Activity** tab -> enter the following:

8.6.2.1 Type: **device cli.send commands**

8.6.2.2 Name: **send switch commands**

8.6.2.3 Persistence: **permanent**

8.6.2.4 Report file: **C:\Program Files
(x86)\cattools3\reports\device.cli.send.commands.txt**

8.6.3 Select the **Time** tab -> enter the following:

8.6.3.1 Reoccurring: **Never**

8.6.3.2 Select the **Devices** tab -> enter the following:

8.6.3.3 Check **All Switches** (Not Firewalls(s)).

8.6.4 Select the **Email** tab -> enter the following:

8.6.4.1 Uncheck all the boxes.

8.6.5 Select the **Options** tab -> enter the following:

8.6.5.1 Check all boxes EXCEPT **Password protect Zip file**.

8.6.5.2 Click **OK**

Note: In the **List of commands to be entered on the device**, you can use the following steps, if needed, to push commands to all switches simultaneously. Once the test has been completed, make sure to delete any commands entered in the *List of commands to be entered on the device*.

See example commands below:

config t

int range gi 1/0/1-28, gi 2/0/1-28

logging event link-status

Note: The **Link status** is entered to log the power on/off, by default.

Exit

Exit (leave the command prompt there)

8.6.6 Test the command line configs:

8.6.6.1 Right-click on the task **send switch commands** -> click Run now.

8.6.6.2 Verify the commands were executed successfully in the switch.

8.7 Testing

Test backup activities.

8.7.1 **Backup:**

8.7.1.1 Open **CatTools** -> navigate to the **Activities** window.

8.7.1.2 Select **Device.Backup.Running.config** -> select **Run now** button.

8.7.1.3 Verify in the **CatTools Activity Log** that all the devices are backed up with no errors.

8.7.1.4 In **File Explorer** -> navigate to: **\\NAS\Backup\Switches\Configs** -> verify all the switch configs are listed.

8.8 Logging

Set CatTools logging to SIEM (If no LM1, do not configure).

8.8.1 Open **CatTools** -> navigate to the **Options** window.

8.8.2 Select **Setup** -> **Logging**.

8.8.3 Check **Send logs to Syslog Server**.

8.8.4 Input the LM1 IP address in the **Syslog Server field**.

8.8.5 Click **OK**.

9.0 **HMI Initial Setup – OTArmor**

Note: In Control Server 2.0:

- The HMI *EWS* does not have a Data Drive (the single hard drive has been increased to 150GB).
- The HMI(s) *RDS1* and *RDS2* have a Data Drive (this should be initialized during build).
- The HP Smart Client is installed on *RDS* only.
- If not using RDS, the HP Smart Client will be installed on *EWS*.

9.1 **Remove Local AV & Backup Software**

Remove local Acronis, Trellix and/or Sophos (if installed).

Note: Remove Sophos and Acronis before joining the domain (the Sophos/Trellix software will not uninstall completely if the machine is in the domain).

9.1.1 Login to the HMI as an Administrator.

9.1.2 To remove/uninstall **Acronis**.

9.1.2.1 Desktop -> click **Start** -> expand **Acronis folder**.

9.1.2.2 Select **Uninstall Acronis Cyber Protect** and check/select box to delete logs, etc -> if pop-up select Restart...

9.1.3 Local AV Removal for Windows 10:

9.1.3.1 To remove/uninstall **Sophos**.

9.1.3.1.1 Desktop -> right-click **Start** -> select **Control Panel** -> select Programs -> **Programs and Features**.

9.1.3.1.2 Remove/Uninstall the following:

9.1.3.1.2.1 **sophosautoupdate**

9.1.3.1.2.2 Right-click -> **sophos system protection** -> select Uninstall.

9.1.3.1.2.3 Right-click -> **sophos anti-virus** -> select Uninstall.

9.1.3.1.2.4 Complete this step last: **Uninstall sophos endpoint defense**.

9.1.3.2 To remove/uninstall **Trellix** (for windows 10):

9.1.3.2.1 Desktop -> right-click **Start** -> select **Control Panel** -> select Programs -> **Programs and Features**.

9.1.3.2.2 Remove/Uninstall the following:

9.1.3.2.3 Right-click -> **Trellix Endpoint Security Threat Prevention** -> select Uninstall.

9.1.3.2.4 **Trellix Endpoint Security** (will uninstall with Trellix with Threat Prevention).

9.1.3.2.5 Right-click -> **Trellix Agent** -> select **Uninstall**.

9.1.3.2.6 Desktop -> click **Start** -> select power -> **Restart** the machine.

9.2 DNS and PDH Network

Configure the DNS & DNS suffix for the PDH Network Adapter.

9.2.1 Login to the HMI as an Administrator.

9.2.2 Open the PDH Network Adapter. Desktop -> right-click **Start** -> open **Control Panel**.

9.2.3 Select **Network and Internet** -> **Network and Sharing Center** -> open **Change Adapter Settings**.

9.2.4 Right-click on the PDH network adapter -> select **Properties**.

9.2.5 Double-click on **Internet Protocol Version 4 (TCP/IPv4)** and enter the **Preferred DNS Server** and **Alternate DNS Server** (DC1/DC2 IP Addresses) in the General tab.

9.2.6 Click the **Advanced** button.

Note: If the system has more than two DNS Servers, add the additional DNS server addresses under the DNS Tab, Add.

9.2.7 DNS suffix for this connection: **HMI.local**

9.2.8 Click **OK** to exit Internet Protocol Version 4 (TCP/IPv4) Properties.

9.2.9 Click **OK** to exit the Ethernet Properties.

9.3 HMI into Domain

Add the HMIs to the Domain.

9.3.1 From the HMI, remote desktop to **DC1** or **DC2**, login with **Administrator** credentials.

9.3.2 Desktop-> click **Start** -> open **Server Manager** (if window/pop-up, select yes).

9.3.3 Select Tools -> open **Active Directory Users and Computers**.

9.3.4 Expand **HMI.local** -> **GEV Controls** -> **HMI**.

9.3.5 Select **Windows 10 HMIs**.

9.3.6 Select **Control Room HMIs** OR **Non-Control Room HMIs** (based on 4108).

9.3.7 Create the HMIs you are joining to the domain.

9.3.7.1 Right-click on either **Control Room or Non-Control Room** -> select **New** -> click **Computer**.

9.3.7.2 Type the computer (HMI) name -> click **OK**.

Note: If the name has an underscore _ a dialog box will show with a warning -> click **Yes**.

9.3.7.3 Double check that the new HMIs are created in:

9.3.7.4 **hmi windows 10 hmis** -> **controlroom HMIs OR non-control room HMIs** -> **HMIs**

Navigate back to the HMI.

9.3.7.5 Verify the Date and Time matches the project ubication. If it doesn't, please proceed to change it to match the project.

9.3.7.6 Change HMI system properties:

9.3.7.6.1 Desktop -> right-click Start -> select **System**.

9.3.7.6.2 Click **Change Settings** under Computer name, domain, and workgroup settings.

9.3.7.6.3 Click **Yes** on User Account Control window/pop-up.

9.3.7.6.4 Click the **Change** button.

9.3.7.6.5 Select/check the **Domain** option under Member of.

9.3.7.6.6 Type **HMI.local**

9.3.7.6.7 Click **OK** -> click **Yes** on pop-up.

9.3.7.6.8 In the **Windows Security** -> enter a **Domain Administrator** to authenticate.

9.3.7.6.9 A window/pop-up welcome to the HMI.local domain will appear.

9.3.7.6.10 Click **OK** -> wait for Welcome to the HMI.local domain pop-up -> click **OK**.

9.3.7.6.11 Click **OK** to **Restart HMI** (when prompted).

9.3.7.6.12 Click OK on the Change Settings window -> click Restart Now on pop-up.

9.4 Install Agents

Install Trellix, Trend Micro, Splunk and Sophos agents.

Note: The Trellix agent will be installed in the last step of the HMI Initial Setup section.

Note: Please refer to section Apex for Trend Micro agent installation.

9.4.1 To install **Splunk agent (if applicable):**

9.4.1.1 Desktop -> click Start -> open Command Prompt (Admin).

9.4.1.2 In command prompt -> type: `\\LM1\splunk\install-agent.bat`

9.4.1.3 Enter the Splunk Admin password: **See Default Passwords Document*

IMPORTANT: Verify the following:

1. Open **Services** (in Windows Search bar -> type services -> select the Services app).
2. Scroll to find the **SplunkForwarder** service.
3. Verify the **Log On As** is set to **Local System**.
4. If it's NOT set to **Local System** open the service **Properties**.
 - a. Select the **Log On** tab and select the radio button next to **Local System**.
 - b. Click **Apply** and click **OK**.
 - c. Restart the SplunkForwarder service.

9.5 Disable Local Accounts

Disable Oper and Maint accounts.

Note: Check with the customer before proceeding. It is NOT recommended to disable the local admin account. It is recommended to change the password.

9.5.1 Login to the HMI as an Administrator.

9.5.2 Desktop -> click **Start** -> open **Control Panel**.

9.5.3 Open **System and Security** -> select **Administrative Tools** -> double-click on **Computer Management** (if window/pop-up, select yes).

9.5.4 Expand System Tools -> Local Users and Groups -> click Users.

9.5.5 Right-click on **Maint** account -> select **Properties**.

9.5.6 Select/check Account is disabled box.

9.5.7 Click **Apply** and **OK**.

9.5.8 Right-click on **Oper** account -> select **Properties**.

9.5.9 Select/check Account is disabled box.

9.5.10 Click **Apply** and **OK**.

10.0 Thin Client HMI – OTArmor

10.1 Thin Client HMI System Configuration

IMPORTANT: DO NOT PATCH the Windows Thin Client. The Thin Client configuration requires a lockdown described in the document (GEH-6857). There is no patch validation for Windows Thin Clients.

10.1.1 Please refer to document **GEH-6857 – Control Server HP Windows 10 Thin Client HMI System Support and Maintenance Guide**.

10.1.2 Follow the procedures in GEH-6857 for system overview, initial configuration, USB drive support, remote desktop, dual NIC, system configuration and maintenance.

11.0 NAS – OTArmor

11.1 Generate NAS Certificate

11.1.1 Login to **CA1**.

11.1.2 Open **Windows PowerShell as Admin**.

11.1.3 Navigate to the following path: **cd C:\CertificateSigning**

11.1.4 Enter the following command: **.\Create-GEV-Cert.ps1**

11.1.5 Hit **Enter** to continue.

11.1.6 At the CSR Request prompt -> Enter **"2"**.

11.1.7 At the Generate a non-encrypted private key -> Enter **"y"**.

11.1.8 Hit **Enter**.

11.1.9 Hit **Enter** to continue.

11.1.10 At the FQDN -> enter the **NAS FQDN** (example: NAS.HMI.local).

11.1.11 At the Common Name -> enter the **NAS Hostname** (example: NAS).

11.1.12 At the Devices IP Address -> enter the **NAS IP Address** (example: 172.16.201.120).

11.1.13 At the Password for private key file -> hit **Enter** (leave blank, no password).

11.1.14 Verify the command completed successfully.

11.1.15 The certificates will be created in the following directory:
C:\CertificateSigning\NAS.

11.1.16 Login to the **NAS**.

11.1.17 Open the Control Panel -> in the panel -> select **Security**.

11.1.18 In the Security window -> select the **Certificate tab**.

- 11.1.19 Select the **Add** button.
- 11.1.20 In the Create Certificate window -> select **Add a new certificate** -> click **Next**.
- 11.1.21 Select **Import Certificate** -> click/check box for **Set as default certificate** -> click **Next**.
- 11.1.22 In the Import Certificate Files window -> next to **Private Key** -> click **Browse**.
- 11.1.23 In the files window -> navigate to **\\CA1\C\$\CertificateSigning\NAS**
- 11.1.24 Select the **"NAS_Cert_noenc.key"** -> click **Open**.
- 11.1.25 In the Import Certificate Files window -> next to **Certificate** -> click **Browse**.
- 11.1.26 In the files window -> select the **"NAS_Cert.crt"** (Security Certificate) -> click **Open**.
- 11.1.27 Click **OK**.
- 11.1.28 Verify the NAS now has issued certificates.
- 11.1.29 Select the **Settings** button.
- 11.1.30 Under the **Configure** option -> change all services to the new **NAS.HMI.local** cert.
- 11.1.31 Click **OK** -> click **Yes** on the prompt.
- 11.1.32 Wait for web server to restart.
- 11.1.33 Close the browser window.
- 11.1.34 Re-open a browser.
- 11.1.35 Login to the NAS using the SSL URL (example: **https://NAS.HMI.local:5001**).
- 11.1.36 Verify the connection is now private -> no webpage warning.

12.0 Acronis – OTArmor

12.1 **Verify Acronis Services**

Verify Acronis Services have the correct startup type and logon account.

12.1.1 Login to **DR1**.

12.1.2 Navigate to **Server Manager -> Tools -> Services**.

12.1.3 Verify the following services have the settings below:

12.1.3.1 **Acronis Cyber Protect Management Server**

12.1.3.1.1 Startup Type: Automatic (Delayed Start)

12.1.3.1.2 Log On Account: **hmi\acronisdb**

12.1.3.2 **Acronis Managed Machine Service**

12.1.3.2.1 Startup Type: **Automatic (Delayed Start)**

12.1.3.2.2 Log On Account: **hmi\acronisagent**

12.1.3.3 Acronis Scheduler2 Service

12.1.3.3.1 Startup Type: **Automatic (Delayed Start)**

12.1.3.3.2 Log On Account: **Local System Account**

12.1.3.4 Acronis Service Manager Service

12.1.3.4.1 Startup Type: **Automatic (Delayed Start)**

12.1.3.4.2 Log On Account: **hmi\acronisdb**

Note: If the Acronis Service Manager Service is not set to Automatic (Delayed Start) this can affect the Acronis Cyber Protect Management Console access. If this service starts before the SQL Server (SQLEXPRESS) it will cause access issues between Acronis and the SQL Database.

12.2 Verify Acronis Certificate

Verify Acronis Cyber Protect does not have a certificate error.

12.2.1 Login to **DR1**.

12.2.2 Open the **Acronis Cyber Protect** desktop icon.

12.2.3 Verify the browser authenticates without a certificate error.

12.2.4 If the browser states = **There is a problem with this website's security certificate** -> create the certificate (instructions **313T7465P0404BA**) and re-verify the state.

12.3 Verify Acronis License(s)

Verify Acronis Cyber Protect is licensed.

12.3.1 Login to **DR1**.

12.3.2 Open the **Acronis Cyber Protect** desktop icon.

12.3.3 Click **Sign In**.

12.3.4 On the left navigation panel -> select **Settings** -> **License usage**.

12.3.5 Verify correct number of licenses per Virtual Hosts, Workstations, and Servers.

12.3.6 Add license keys if needed (instructions **313T7465P0404BA – Add and Assign Acronis Licenses**).

Note: Acronis Cyber Protect 15 has an auto-assign feature that will assign the licenses according to the type added. This feature may auto-assign a Virtual Host license to the DR1 Server after assigning to each Virtual Host. If this is the case, leave the Virtual Host license assigned to the DR1 Server, do not unassign. The auto-assign to DR1 Server may affect some settings in the console, verify the Backup Location and the Protection Plans are accessible and available for edit.

12.4 Add HMIs to Acronis Console

Add Physical/Virtual HMIs to Acronis Console

12.4.1 Login to **DR1**.

- 12.4.2 Open the **Acronis Cyber Protect** desktop icon.
- 12.4.3 Click **Sign In**.
- 12.4.4 (In Acronis) On the left navigation pane select **Devices** -> **All Devices** -> click the **Add** button.
- 12.4.5 Select **Windows icon** under **Workstations**.
- 12.4.6 Under **Select the deployment agent** -> select **DR1**.
- 12.4.7 Under **Specify a machine running Windows** -> Enter: < **Machine Name**> or <**IP Address**> (e.g. EWS.HMI.local) under Specify a machine running Windows.
 - 12.4.7.1 Enter the following:
Username: **HMI\Admin**
Password: **See Default Passwords Document*
 - 12.4.7.2 Verify the management server **DR1.HMI.local** is selected.
 - 12.4.7.3 Click **Add**.
- 12.4.8 Close the Activity window once the status shows Succeeded. Once the machine is deployed -> verify it's showing under **All Devices** in the Acronis console.

12.5 Create a Backup Plan for HMIs

Create a backup plan for each Physical HMI

- 12.5.1 Login to **DR1**.
- 12.5.2 Open the **Acronis Cyber Protect** desktop icon.
- 12.5.3 Click **Sign In**.
- 12.5.4 (In Acronis) On the left navigation pane select **Plans** -> **Protection**.
- 12.5.5 In the right menu -> select **Create Plan**.
- 12.5.6 Select the following:
 - 12.5.6.1 Devices: Click on **Add devices**.
 - 12.5.6.1.1 Click on the **All devices** folder.
 - 12.5.6.1.2 Select/check the **Physical HMI**.
 - 12.5.6.1.3 Click **Add** at bottom of screen.
 - 12.5.6.1.4 Click **Done**.
- 12.5.7 Hover next to **New protection plan** -> select the **pencil icon**.
 - 12.5.7.1 Rename the plan to the <**HMI NAME**> **Backup Plan** and click **OK**.
 - 12.5.7.2 What to Back Up: **Entire Machine**.
 - 12.5.7.3 Where to Back Up: Click on **Specify** and select **\\NAS\Backup**
 - 12.5.7.3.1 If needed -> **Add Location**.

12.5.7.3.2 Select Network Folder and type **\\NAS\Backup**(click the green arrow). Enter the following:

Username: **HMI\AcronisAgent**

Password: **See Default Passwords Document*

12.5.7.3.3 Verify the **Backup** folder is selected -> click **Add**.

12.5.7.4 Schedule: Click on the **Monday to Friday at 11:00 PM** -> enter the following:

Note: Schedule each HMI Backup Plan 30 mins apart OR per site requirement. For reference, review the system OTArmor Backup Plan and verify the backup start times do not interfere with each other (e.g. the next HMI Backup Plan should be changed to 2:30 AM, 3:00 AM...etc.).

12.5.7.4.1 Backup Scheme: **Monthly full, Weekly differential, Daily incremental (GFS)**.

12.5.7.4.2 Full Backup: Click on **1 of each month at 11:00 PM** -> change the following:

12.5.7.4.2.1 Select **Monthly**

12.5.7.4.2.2 Months: **All months**

12.5.7.4.2.3 Days of the month: **1**

12.5.7.4.2.4 Start at: **2:00 AM (See Note)**

12.5.7.4.2.5 Click **Done**.

12.5.7.4.3 Differential backup: Click on **Saturday at 11:00 PM** -> change the following:

12.5.7.4.3.1 Verify **Weekly** is selected.

12.5.7.4.3.2 Deselect **Saturday** and select **Sunday** (only).

12.5.7.4.3.3 Start at: **2:00 AM (See Note)**

12.5.7.4.3.4 Select **Done**.

12.5.7.4.4 Incremental Backup: Click on **Monday to Friday at 11:00 PM** -> change the following:

12.5.7.4.4.1 Verify **Daily** is selected.

12.5.7.4.4.2 Select **Run every day**.

12.5.7.4.4.3 Start at: **2:00 AM (See Note)**

12.5.7.4.4.4 Select **Done**.

12.5.7.4.4.5 Click **Done** on the **Schedule** window.

12.5.7.5 How long to keep: Click on **Full: 6 months, Differential...** -> edit the following:

12.5.7.5.1 Full: **6 months**

12.5.7.5.2 Differential: **4 weeks**

12.5.7.5.3 Incremental: **7 days**

12.5.7.5.4 Select **Done**.

12.5.8 Modify Backup Options: Select **Change** next to **Backup options**.

12.5.9 Select **Backup file name** -> delete the entry in the File name template field.

12.5.9.1 Select **[Machine Name]**.

12.5.9.2 Type **_Backup** after [Machine Name] (e.g. **[Machine Name]_Backup**).

12.5.10 Select **Changed block tracking (CBT)** -> select **No** to disable the feature.

12.5.11 Select **File filters** -> under **Do not back up files matching the following criteria**
->

12.5.11.1 First line -> ***.tibx**

12.5.11.2 Add a second line -> ***.tib**

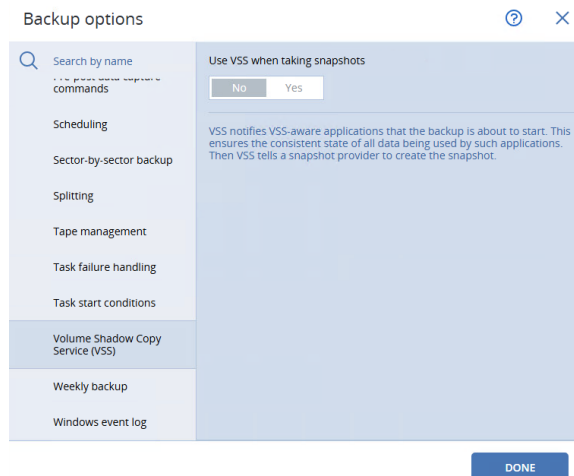
12.5.12 Select **Scheduling** and click **Distribute backup start times within a time window**.

12.5.12.1 Set **Maximum delay** to **4 Hours**.

12.5.13 Select **Windows Event Log** -> click **Yes** to Log event in the Application Event log of Windows.

12.5.13.1 Select **All events**.

Note: FOR Historian (HST) ONLY: select the “**Volume Shadow Copy Service (VSS)**” in the backup options -> Select “**No**” to disable the Volume Shadow Copy Service (VSS).



12.5.14 Click **Done**.

12.5.15 Disable **Active Protection**.

12.5.16 Disable **Vulnerability assessment**.

- 12.5.17 Select **Create**.
- 12.5.18 Wait for the Plan to finish.
- 12.5.19 On the left navigation pane select **Devices -> All devices**.
- 12.5.20 Verify the Physical Machine has a **Scheduled backup** with a local agent.
- 12.5.21 Repeat the above steps to create a backup plan for each Physical HMI.

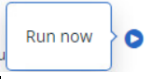
12.6 Verify Backup Plans

Verify ALL Backup Plans have been created and configured properly.

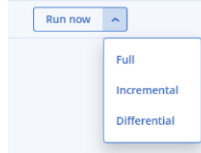
- 12.6.1 Login to **DR1**.
- 12.6.2 Open the **Acronis Cyber Protect** desktop icon.
- 12.6.3 Click **Sign In**.
- 12.6.4 On the left Navigation -> click **Plans** -> select **Protection**.
- 12.6.5 Verify the Backup Plans have been created and are configured properly:
 - **DR1 Backup Plan** – A plan specifically for DR1.HMI.local.
 - **OTArmor Backup Plan (Virtual Machines)** – A plan for Virtual Machines (the Acronis Agent should be deployed to ALL Virtual Machines).
 - **Each Virtual HMI** – A plan for each Virtual HMI.
 - **Each Physical HMI or Server** – A plan for each Physical HMI. A plan for each Physical Server (DC2 physical) (e.g. *EWS1 Backup Plan* or *DC2 Backup Plan*).
 - **(If Applicable) vCenter Backup Plan** – A plan specifically for vCenter Server.
 - **(If Applicable) SplunkArchiveData Backup Plan** – A plan specifically for the SplunkArchiveData folder located on the LM1 Server (D:\SplunkArchiveData).

12.7 Test Backup

Test the Backup Plans.

- 12.7.1 Login to **DR1**.
- 12.7.2 Open the **Acronis Cyber Protect** desktop icon.
- 12.7.3 Click **Sign In**.
- 12.7.4 On the left navigation pane select **Plans -> Protection**.
- 12.7.5 Highlight a backup plan by selecting it.
- 12.7.6 In the right menu -> select **Details**.
- 12.7.7 In the Details window -> hover over **Backup** -> click the **blue play icon**  to **Run now**.

12.7.8 Or expand the arrow on the right to display the **Run now** drop-down options.



12.7.8.1 Select **Go to Activities** -> verify the backup plan is running.

12.7.9 On the left navigation pane select **Devices** -> **All devices** -> verify under **Status** the device(s) are **Backing up**.

12.7.10 **Verify that each backup completes successfully.**

Note: It is important to backup ALL machines before continuing with the Pre-CWT steps.

12.8 Acronis Bootable Media

Verify the Acronis bootable media is present.

12.8.1 Login to **DR1**.

12.8.2 Desktop -> open **File Explorer** -> open the **Acronis Bootable Media** folder.

12.8.3 Verify the following Bootable ISO files are in the folder:

- **WINPE-x64-Recovery-Media.iso**
- **WINPE-x64-RAID-Recovery-Media.iso**
- **WinPE-x64-RAID-Recovery-Media.iso**
- **WinPE-x64-Recovery-Media-Acronis15U6.iso**

Note: The WinPE.iso file is for Non-Standard **RAID1 HMI ONLY or Z4G5 HMI**. The WinPE .iso uses Windows-based drivers which are requested from the respective hardware vendor and embedded into WinPE-based media manually. Please use the WinPE .iso file when restoring a Non-Standard **RAID1 HMI or Z4G5 HMI**.

Acronis Cyber Backup 12.5:

- **WINPE-x64-Recovery-Media.iso** is specific to **MicroSEMI RAID**.
- **WINPE-x64-RAID-Recovery-Media.iso** is specific to **VROC RAID**.

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- **WinPE-x64-RAID-Recovery-Media.iso** is specific to **VROC RAID**.
- **WinPE-x64-Recovery-Media-Acronis15U6.iso** is specific to **Z4G5 HMI**.

12.8.4 Create the **Acronis-Boot-Media.iso** file.

12.8.4.0 Login to **DR1**.

12.8.4.1 In the Windows task bar search -> enter: **Run Acronis Bootable Media Builder**

12.8.4.2 Click **Yes** on the pop-up.

- 12.8.4.3 Click **Next** to continue.
- 12.8.4.4 On the functionality of the created media depends on... window -> select the radio button for "**Create the media without specifying a license key**".
- 12.8.4.5 Click **Next**.
- 12.8.4.6 On the Select the bootable media type to create window -> select the radio button for "**Windows-like representation**".
- 12.8.4.7 Click **Next**.
- 12.8.4.8 On the Advanced Settings window -> **leave the defaults**.
- 12.8.4.9 Click **Next**.
- 12.8.4.10 On the Select the components to place on the bootable media window -> select/check "**Acronis Cyber Protect (64-bit with UEFI support)**".
- 12.8.4.11 Click **Next**.
- 12.8.4.12 On the Review the options and change the settings... window -> select "**Remote logon settings**".
 - 12.8.4.12.1 Check the box for "**Enable remote logon**" -> enter the **domain\admin user and password**.
- 12.8.4.13 On the Review the options and change the settings...window -> select "**Network settings**".
 - 12.8.4.13.1 **Clear all fields** for each Network Interface Card (NIC), including **uncheck** the "Configure the settings automatically" box.
- 12.8.4.14 Click **Next**.
- 12.8.4.15 In the media output window -> select "**ISO image**".
- 12.8.4.16 Click **Next**.
- 12.8.4.17 Navigate to the **Acronis Bootable Media** folder -> save the **.ISO** file as: **Acronis-Boot-Media**
- 12.8.4.18 Click **Next**.
- 12.8.4.19 Confirm the information displayed is accurate in the Review window and begin the media create process.
- 12.8.4.20 Click **Proceed**.
- 12.8.4.21 Click **OK** when completed.
- 12.8.4.22 Verify the Acronis-Boot-Media.iso is now available in the Acronis Bootable Media folder.

12.9 Restore HMI from Backup (Restore Disk)

Restore the HMI from the Acronis Cyber Protect.

12.9.1 Create/burn a **restore disk** from the **Acronis-Boot-Media.iso** (or **WINPE-x64-xxx.iso** for Non-Standard RAID1 HMI).

12.9.2 Insert the disk and boot the HMI from the **restore disk**.

12.9.3 Select **Boot Menu** -> select the correct disk drive -> hit <enter>.

12.9.4 The **Start Acronis Loader...** will appear.

12.9.5 Select **Acronis Cyber Protect** from the **Menu Options** -> hit <enter>.

12.9.6 Wait for the backup to load.

12.9.7 Navigate to **Acronis bootable agent** -> **Network Settings** -> select **configure the network**.

12.9.7.1 Enter the **IP address/subnet mask** of the HMI/Computer.

12.9.7.2 Enter the **DNS Server** -> click **OK**.

12.9.8 Under the **Actions** tab -> select **Start shell**.

12.9.8.1 Ping the IP Address of **DR1** and verify the response.

12.9.9 Click **Recover** -> select **Data**.

12.9.9.1 Browse: **\\NAS\Backup\<yourmachinename>_Backup**

12.9.9.2 Click **OK**.

12.9.10 Data to recover section enter credentials (domain admin) -> click **OK**.

12.9.10.1 Select **Backup** (only one datastore).

12.9.10.2 Select **Disk1** and **Disk2**.

Note: If using the **WINPE-x64-xxx.iso**, Disk1 and Disk2 may need to be swapped due to the RAID1 configuration.

12.9.10.3 Click **OK**.

12.9.11 Review recovery options -> click **OK**.

12.9.12 **My Recover_1 Details** window appears -> wait for the system to restore.

12.10 Restore HMI from Backup (Acronis Management Console)

Note: Cannot restore a Non-Standard RAID1 HMI or Z4G5 HMI using the **Acronis Management Console**. Please use the applicable WinPE.iso restore disk.

Note: If an HMI has teamed NICs or does NOT have redundant PDH/UDH, make sure to unteam or connect all redundant cables for the HMI before recovering from the console.

12.10.1 On the **DR1** server -> open the **Acronis Cyber Protect Console**.

12.10.1.1 Sign-in -> navigate to **Devices** -> **All devices**.

12.10.1.2 Select the HMI -> on the right panel -> select **Recovery**.

12.10.1.3 Select the backup for recovery -> click **Recover...**

Note: Select the latest backup or run a new Full backup of the HMI to prevent the loss of recent data or changes.

12.10.1.3.1 Select **Entire Machine** or **Files/folders**.

12.10.1.3.2 For **Files/folders** -> navigate to the drive with the file or folder and select -> select **Recover**.

12.10.1.3.2.1 (If prompted) Specify the backup location credentials.

12.10.1.3.2.1.1 HMI\AcronisAgent

12.10.1.3.2.1.2 **See Default Passwords Document*

12.10.1.3.2.2 For **Files/folders** -> select **original location** -> **Start Recovery**.

12.10.1.3.2.3 Select **Overwrite existing files** -> click **Proceed**.

12.10.1.3.3 For **Entire Machine** -> Under **Recover To** select either: **Physical Machine** or **Virtual Machine**.

12.10.1.3.3.1 Select **Physical Machine** (for physical HMI).

12.10.1.3.3.2 Verify machine under **Target Machine** (XXX.HMI.LOCAL)

12.10.1.3.3.3 Verify disks under **Disk mapping** (disk 1 -> disk 1, disk 2 -> disk 2, etc.).

12.10.1.3.3.4 (If prompted) Specify the backup location credentials.

12.10.1.3.3.4.1 HMI\AcronisAgent

12.10.1.3.3.4.2 **See Default Passwords Document*

12.10.1.3.3.5 Select **Start Recovery**.

12.10.1.4 On the Activity details window -> verify the recovery progress.

12.10.1.5 Close the Activity details window when the recovery is done.

12.10.1.6 Login to the recovered machine -> verify disks, files and folders have been recovered correctly.

IF RECOVERING VIRTUAL MACHINE from Acronis Management Console:

12.10.1.6.1.1 **(If Virtual Machine)** is selected -> click, **Select Machine** under Target Machine.

12.10.1.6.1.2 Select **Existing Machine**.

12.10.1.6.1.3 Select the Virtual Machine - > click **OK**.

Note: For vCenter/HA systems, the Virtual Machine option will not include the VMs with the agent inside (local agent machines xxx.HMI.local). After recovery this may create a new network adapter. If so, open the Device Manager on the VM, expand Network adapters (select View – show hidden devices) and verify the correct number of

network adapters appear. If the recovery created a new adapter, delete, and verify the IP configuration settings are correct on PDH/UDH.

13.0 **Ivanti Security Controls – OTArmor**

IMPORTANT: If the system has Application control, place all systems in Update Mode before patching.

IMPORTANT: Review all TINs located on the documentation ISO for applicability before proceeding.

13.1 **Patches Update**

Deploy the latest patches.

13.1.1 Login to **PM1**.

13.1.2 Open File Explorer -> select **D: Drive** -> **Shavlik folder**.

13.1.3 Check **PatchList** file/date to verify which Patch Disk to run.

13.1.4 Run the latest cumulative patch disk.

13.1.4.1 Run **Copy_Updates** -> right-click the **.bat** file -> **Run as Administrator**.

13.1.5 Run the latest monthly patch disk.

13.1.5.1 Run **Copy_Updates** -> right-click the **.bat** file -> **Run as Administrator**.

13.2 **Verify all HMIs and servers are in Ivanti.**

Scanning system for devices.

13.2.1 Login to **PM1**.

13.2.2 Double-click the **Ivanti Security Controls** desktop shortcut (if window/pop-up, click yes).

13.2.3 Select **Machine Groups** (top left) -> under My Machine Groups -> double-click **OTArmor Domain**.

13.2.4 Select **Run Operation** and click the **Scan now** button.

13.2.5 Let the scan run -> in the **Operations Monitor** -> verify all machines (HMIs/Servers) are present.

13.3 **Push/Deploy the latest patches using Ivanti (Shavlik).**

Scanning system for missing patches

Note: If the system has **Application control**, place the systems to be patched in **Update Mode**.

13.3.1 Open **Ivanti Security Controls** desktop shortcut -> navigate to **View** -> select **Machines**.

13.3.2 In the top pane or **Machine view** -> select an HMI or Server for patching.

13.3.3 In the pane below -> click on the **Windows Patches** tab.

13.3.4 Select a group of **Patch Missing** products -> right-click -> select **Deploy** -> click **Selected Patches**.

13.3.5 Click **Deploy**.

13.3.6 When service packs are installed -> right-click on the HMI or Server -> select **Deploy all missing patches**.

Note: (if needed) Login to each machine (HMI/Server) and reboot to apply patches.

14.0 Splunk – OTArmor

14.1 Check License

Verify the Splunk Enterprise license is applied and check data volume levels.

INFORMATION: Please refer to **Confluence – Gas Power Engineering Function – Splunk Enterprise – Troubleshooting Guidance OR GEH-6887 – Splunk Troubleshooting** for additional Splunk Licensing information.

14.1.1 Login to **LM1**.

14.1.2 Open **Splunk** (double-click the shortcut on desktop).

14.1.3 Login with **Admin** credentials (admin – **See Default Passwords Document*).

Note: If the web page doesn't load, try deleting the Browsing History.

14.1.4 From the **Splunk console** -> select **Settings** -> **Licensing**.

14.1.5 On the Licensing page -> verify the project license(s).

14.1.5.1 Check **Volume** (1GB, 5GB, etc.), **Expiration** (annual) and **Status** (valid, FROM FUTURE).

Note: The Splunk Enterprise license is an annual renewal. If the project purchased more than 1 year, verify all licenses have been applied (and should state FROM FUTURE).

14.1.6 Licensing Errors:

14.1.6.1 If there is a "**litsearch**" command Error message in the console, this is indicating a problem with the Splunk license.

14.1.6.2 It can be an issue with **license expiration and/or license overutilization**.

14.1.6.3 When this "litsearch" Error occurs, Splunk will continue to index data however the Search functionality pauses until the license is corrected (exception for the *_internal* index for license diagnosis).

14.1.7 **License Expired:** If the license is “*expired*”, apply a valid license to continue using the Splunk application.

Note: This may also require a “*reset*” license (from Splunk) to reenable the Search functionality. **Contact Splunk to acquire the “reset” license if the console is still showing a ‘litsearch’ error.*

14.1.8 **License Overutilization:** If the license is “*overutilized*”, there will be overutilization alerts under Current or Permanent Alerts.

14.1.8.1 If there are several **pool quota overage warnings** reported, this indicates an overutilization of the license (currently and/or historically).

14.1.8.2 The amount of log data collected (daily) is exceeding the daily indexing limit (license - GB).

14.1.8.3 It would be recommended to increase license size to support the actual daily log consumption.

14.1.8.4 Purchase an additional license to prevent license overutilization and support the (actual) daily log consumption.

14.1.8.5 Licenses can be “stacked” to increase the TOTAL daily indexing limit (GB).

Note: this may also require a “*reset*” license (from Splunk) to reenable the Search functionality. **Contact Splunk to acquire the “reset” license if the console is still showing a ‘litsearch’ error.*

Note: If the volume has exceeded the stack size please use the Splunk Enterprise License Calculation Tool guidance table to evaluate the license. This is available in the Quoting and BOM tools. Adjust the Host Quantity per project and calculate the Low Gb and High Gb totals to find an average.

14.1.1 To determine the license size needed -> select the **Usage Report** button on the Licensing page.

14.1.2 Select the **Previous 60 days** tab.

14.1.3 In the Daily License Usage panel -> verify the volume is under the stack size.

14.1.4 The Usage Report window has an option to view previous 30 or 60 days.

Note: if the license has been in overutilization for a prolonged period of time it may affect the results.

Note: to view license usage by endpoint click the “**Split by:**” drop-down and select “**Host**”.

14.1.5 Hover over the volume column for each endpoint to view the individual license consumption.

- 14.1.6 Review the endpoints that are consuming a large amount of the license and evaluate. This may require a log forwarding adjustment to reduce the license overutilization.

14.2 Check Data

Verify log data from all devices.

14.2.1 Login to **LM1**.

14.2.2 Open **Splunk** (double-click the shortcut on desktop).

14.2.3 Login with **Admin** credentials (admin – **See Default Passwords Document*).

Note: If the web page doesn't load, try deleting the Browsing History.

14.2.4 From the **Splunk console** -> select the **OTArmor™ – Visibility** app.

14.2.4.1 On the **Home** page:

14.2.4.2 (If a Controller is connected) Verify the **Controller panels** are populated.

Note: Controller logs will appear (in Splunk) under the IP Address or FQDN of the HMI where the .tcw file was configured. Make sure the master HMI is downloaded as well.

The controller log will include the controller's name in the following format: *device:<name of controller>*. The *device:<name of controller>* can be searched in Splunk to find controller log data as well.

Note: If the controller logs are not appearing in the panels, verify the UDP 514 Windows Firewall port has been added to LM1 and the profile is configured for "Domain, Private, and Public (All)". The controller logs are forwarded via UDP 514 on the UDH network.

14.2.4.2.1 The **Windows Active Directory** panel will alert on the following Windows event codes:

- Event code **4726: User Deleted**
- Event code **4720: User Created**
- Event code **4722: User Enabled**
- Event code **4725: User Disabled**
- Event code **4738: User Changed**
- Event code **4625: Login Failed**
- Event code **4624: Login Success**

14.2.4.2.2 If needed, simulate the event code(s) to verify.

Note: The lookup file = **eventcodes.csv**
(d:\security\etc\apps\security\lookups\)

14.2.4.3 Verify all devices in the network are listed in the following panels:

14.2.4.3.1 **Hosts – Windows**

14.2.4.3.2 **Hosts – Syslog**

14.2.4.3.2.1 **Note: Hosts (Windows)** = Windows devices with a Splunk Agent installed.

14.2.4.3.2.2 **Note: Hosts (Syslog)** = Other devices sending syslog data (e.g. switches, firewalls, virtual hosts, etc.).

Note: The **virtual hosts are not configured to forward log data to Splunk by default**. Due to the frequency of logs, it is recommended to forward only if requested/required by the customer. The Splunk Licensing Tool has been updated to adjust for the virtual host log increase if forwarding is required. Please see section below to configure the virtual hosts to forward to Splunk.

14.2.4.4 (If a Firewall is connected) Verify the Top Denied Sources – **Network Traffic** panel is populated.

14.2.4.5 Select the **Windows and Syslog** drop-down on the OTArmor™ – Visibility dashboard:

14.2.4.6 On the **Windows** page.

14.2.4.6.1 The **Hosts – Windows** shows Windows machines with forwarding Splunk agents.

14.2.4.6.2 The Windows **Login Failure** panel will alert on the following Windows event code:

Event code 4625: Login Failed

14.2.4.6.3 The Windows **Login Success** panel will alert on the following Windows event code:

Event code 4624: Login Success

14.2.4.6.4 If needed, simulate the event code(s) to verify.

14.2.4.6.5 The **Windows Disabled Login** panel will alert on the following Windows event codes:

Event code 4625: Login Failed

Event code 4624: Login Success

Event code name: Account is currently disabled

14.2.4.6.6 If needed, disable an account, and try a login to verify.

14.2.4.6.7 The **Windows Account Changes** panel will alert on the following Windows event codes:

Event code 4726: User Deleted

Event code 4720: User Created

Event code 4722: User Enabled

Event code 4725: User Disabled

Event code 4738: User Changed

Event code 4740: User Locked

Event code 4767: User Unlocked

14.2.4.6.8 If needed, simulate the event code(s) to verify.

Note: The lookup file = **eventcodes.csv**
(d:\security\etc\apps\security\lookups\)

14.2.4.6.9 The **Windows GPO Status Changes** panel will alert on the following Windows event codes:

Event code 5136: A Directory Service Object was Modified

Event code 5137: A Directory Service Object was Created

Event code 5138: A Directory Service Object was Undeleted (Restored)

Event code 5139: A Directory Service Object was Moved

Event code 5141: A Directory Service Object was Deleted

14.2.4.6.10 If needed, simulate the event code(s) to verify.

14.2.4.6.11 The **Windows DFSr Errors** panel will alert on the Domain Controller Distributed File System Replication errors using the following Windows event code levels:

Event code level: 1

Event code level: 2

Event code name: DFS Replication

Note: The lookup file = **dfseventcodes.csv**
(d:\security\etc\apps\security\lookups\)

14.2.4.7 Select the **Windows and Syslog** drop-down on the OTArmor™ – Visibility dashboard:

14.2.4.8 On the **Windows Performance** page:

14.2.4.8.1 The **Physical Disk** panel:

Note: Free Space: Monitors the Microsoft (Windows) Host overall available disk space in a percentage value.

- Includes: Host Name and Value (in %). When the disk space falls below 30% the value will be highlighted Yellow indicating a Warning. When the disk space falls below 10% the value will be

highlighted Red indicating a Critical status. All events are presented in a Statistical Table for organized viewing.

14.2.4.8.2 The **Network Interface** panel:

Note: Utilization: Monitors the Microsoft (Windows) Host Network Interface. The connection between the computer and network.

- Includes: Host, Bytes Received/sec, and Bytes Sent/sec. The end user can view how many bytes have been sent or received on each host. The table is heat mapped based on total number of sent or received bytes. As the number increases or decreases the color transparency will adjust. The opaquer the color, the higher the number. This allows the end user to view Host(s) activity within the network. These numbers will typically fluctuate during processes such as: Data Transfers, Patching, Reboots, Logging, etc. All events are presented in a Statistical Table for organized viewing.

14.2.4.8.3 If needed, simulate bytes received/bytes sent by either running a backup or patching a machine.

14.2.4.8.4 The **Memory** panel:

Note: Committed Bytes in Use: Monitors the Microsoft (Windows) Host committed memory in use based on the paging file threshold, or how much memory is reserved for all processes.

- Includes: Host and Value (in %). When the Committed Bytes in Use rises above 85% the value will be highlighted in Yellow to indicate a Warning. When the Committed Bytes in Use rise above 95% the value will be highlighted in Red to indicate a Critical status. All events are presented in a Statistical Table for organized viewing.

14.2.4.8.5 The **Processor** panel:

Note: Idle Time: Monitors the Microsoft (Windows) Host Processor Idle Time, or time in which the computer processor is not being used by any program.

- Includes: Host and Value (in %). When the Idle Time falls below 30% the value will be highlighted Yellow indicating a Warning. When the Idle Time falls below 10% the value will be highlighted Red indicating a Critical status. All events are presented in a Statistical Table for organized viewing.

14.2.4.8.6 Select the **Host** drop-down to filter per **Host Name**.

14.2.4.9 Select the **Windows and Syslog** drop-down on the OTArmor™ – Visibility dashboard:

14.2.4.10 On the **Syslog** page:

14.2.4.10.1 The **Hosts – Syslog** shows other devices sending syslog data (e.g. switches, firewalls, virtual hosts, etc.).

Note: The **virtual hosts are not configured to forward log data to Splunk by default**. Due to the frequency of logs, it is recommended to forward only if requested/required by the customer. The Splunk Licensing Tool has been updated to adjust for the virtual host log increase if forwarding is required. Please see section below to configure the virtual hosts to forward to Splunk.

14.2.4.10.2 The **Login Failure** panel will alert on the following index and action:

Index: syslog

Action: login failed

14.2.4.10.3 The **Login Success** panel will alert on the following index and action:

Index: syslog

Action: login success

14.2.4.10.4 If needed, simulate the action to verify.

14.2.4.10.5 The **Critical Syslog Events** panel will alert on the following index and log level:

Index: syslog

Log level: critical

14.2.4.10.6 The **Firewall Interface Status** panel will alert on the following index and action:

Index: syslog

Action: interface-stat-change

14.2.4.10.7 If needed, simulate the action on a FW to verify.

14.2.4.10.8 The **Switch Interface Status** panel will alert on the following index and action:

Index: syslog

Action: changed state

14.2.4.10.9 If needed, simulate the action on a SW to verify.

14.2.4.10.10 The **Switch Command Status** panel will alert on the following index and action:

Index: syslog

Action: CatTools AND switch OR sw

14.2.4.10.11 If needed, simulate the action in CatTools to verify.

14.2.4.10.12 The **Total Syslog Events** panel counts total number of syslog events per host over a period of 7 days.

14.2.4.11 Select the **Backup and Patching** drop-down on the OTArmor™ – Visibility dashboard:

14.2.4.12 On the **Backup** page:

14.2.4.12.1 Verify the **Acronis Cyber Protect Console** button links to **<https://dr1:9877>** -> sign in and verify functionality (admin/**See Default Passwords Document*).

Note: The backup panels gather log data from Windows Event Log using the source names: '**Acronis Backup and Recovery**' and '**Microsoft-Windows-WindowsUpdateClient**'.

14.2.4.12.2 Click the **Machine Backup Success** and **Machine Backup Failure** panels and verify the Backup Event Details -> if needed, simulate a backup to verify.

14.2.4.13 Select the **Backup and Patching** drop-down on the OTArmor™ – Visibility dashboard:

14.2.4.14 On the **Patching** page:

14.2.4.14.1 The **Installed Patches** panel populates total patches installed on the system, including all Servers and/or Workstations.

14.2.4.14.2 The **Last Patching Date** panel populates based on the last patch date of a Server and/or Workstation per WinEventLog data.

14.2.4.14.3 The **Patch Installs Over Time** panel will populate after a patch or patches have been installed -> if needed, simulate a patch install to verify or adjust the time picker.

14.2.4.15 Select **Endpoint Protection** on the OTArmor™ – Visibility dashboard:

14.2.4.16 On the **Endpoint Protection** (IF Trellix) page:

14.2.4.16.1 The **DLP Events** panel:

Note: This panel uses the **sourcetype=epo:csv**. This sourcetype is a log vehicle for the input file in the deployment-app **epo_inputs\default\input.conf**. The input.conf monitors a report generated by Trellix ePO saved on **SP1 c:\reports*.csv** (This report is configured in the **SP1 Build Document – 128M6050 - Create DLP Query Report**). To verify configuration, review the **Splunk DLP** section below).

14.2.4.16.2 **Test DLP** and Verify the **Time** and **Computer name** match the event details. If the event(s) are not populating in Splunk, open the DLP Block Events.csv file on SP1 C:\reports. Verify the .csv file contains the DLP events.

14.2.4.16.3 The **Antivirus Events** panel:

Note: This panel gathers log data from the Windows Event Log using the source name: '**Trellix Endpoint Security**'.

14.2.4.16.4 If needed, simulate Antivirus events (test virus) to verify.

14.2.4.16.5 The **Application Control Events** panel:

Note: This panel gathers log data from the Windows Event Log using the source name: '**Trellix Solidifier**'. It also uses the lookup file: **epo_events.csv** to translate the WEL code event ID to the Trellix event name.

14.2.4.17 On the **Endpoint Protection** (IF **Trend Micro**) page:

14.2.4.17.1 The **Antivirus Events** panel:

14.2.4.17.1.1 This panel is populated using the forwarding Syslog data from Apex One/Apex Central, logs tagged with AV*.

14.2.4.17.2 The **Device Control Events** panel:

14.2.4.17.2.1 This panel is populated using the forwarding Syslog data from Apex One/Apex Central, logs tagged with Device Access Control.

14.2.4.17.3 The **Application Control Events** panel:

14.2.4.17.3.1 This panel is populated using the forwarding Syslog data from Apex One/Apex Central, logs tagged with *Application Control*.

14.2.4.17.4 If needed, simulate antivirus, device control, and/or application control events to verify panel population.

14.2.4.18 Select **Network Traffic** on the OTArmor™ – Visibility dashboard:

14.2.4.19 On the **Network Traffic** page:

14.2.4.19.1 The **Traffic by Status, Top Applications, and Top Denied Sources** panels, verify the panels are populated, click the chart to drill-down to the details panel.

Note: These panels use the Splunk index=syslog to ingest data from the forwarding network traffic (e.g. switches, firewalls).

14.2.4.20 Test **LDAP Users** Login.

14.2.4.20.1 Login as **GEOper** (**See Default Passwords Document*) -> verify functionality (this account has lowest privilege) -> logout.

14.2.4.20.2 **(If necessary) Splunk Domain Integration (LDAP)** – See OTArmor Log Management (Splunk) Maintenance Guide – **GEH-6887**.

Note: If a domain has more than one Domain Controller, add all to the LDAP Strategies by cloning the DC1 entry and edit accordingly. Make sure to enter the Bind DN Password/Confirm Password (**See Default Passwords Document*) before saving.

14.2.4.21 **Search & Reporting** app.

14.2.4.21.1 From the Splunk Console -> select **Search & Reporting**.

14.2.4.21.2 Select the **Data Summary** button -> review connected hosts/devices.

14.2.4.21.3 **Example Search commands** -> in the **Search** field -> enter a search command.

Note: Adjust the Time Picker to find events older than the 24-Hour default.

14.2.4.21.4 **Syslog Event**, e.g. Firewall, Switch, Controller, etc.:

host= 172.16.201.110

host= 172.16.201.110 NIDS

14.2.4.21.5 **Backup Event**, e.g. Acronis Cyber Protect:

Acronis Backup and Recovery successfully

Acronis Backup and Recovery failed

14.2.4.21.6 **Antivirus Event**, e.g. Trellix:

14.2.4.21.7 **Windows Active Directory Event**, e.g. windows account created, windows account deleted, etc.:

host= DC1

windows_account_created

windows_account_deleted

14.3 Check Resource Utilization

Check the LM1 Server resource utilization using Splunk.

14.3.1 Login to **LM1**.

14.3.2 Open **Splunk** (double-click the shortcut on desktop).

14.3.3 Login with **Admin** credentials (admin – **See Default Passwords Document*).

Note: If the web page doesn't load, try deleting the Browsing History.

14.3.4 From the **Splunk console** -> select **Settings** -> **Monitoring Console**.

14.3.4.1 Select the **Resource Usage** drop-down -> click **Resource Usage: Machine**.

14.3.4.2 Review the panels on the **Resource Usage: Machine** window.

14.3.4.3 Under **Snapshots** -> review the **CPU Usage** and **Physical Memory Usage** panels.

14.3.4.4 Under **Historical Charts** -> review the **Median CPU Usage** and **Median Physical Memory Usage** panels.

14.3.4.5 Verify the percentages are not at capacity/exceeding the configured limit.

14.4 Removing Dashboard View

If a page/view on the OTArmor Visibility Dashboard is not required, perform the following steps to hide the page/view.

14.4.1 Login to **LM1**.

14.4.2 Open **Splunk** (double-click the shortcut on desktop).

14.4.3 Login with **Admin** credentials (admin – **See Default Passwords Document*).

14.4.4 On the Splunk Menu -> select **Settings – User Interface – Navigation menus**.

- 14.4.5 On the Navigation menus page -> click the **App** drop-down -> select **OTArmor – Visibility (security)** app.
- 14.4.6 If the system has *Trellix (McAfee)* -> click/select the **default** navigation menu.
- 14.4.7 If the system has *Trend Micro* -> click/select the **trend_default** navigation menu.
- 14.4.8 In the **Navigation menu XML** -> locate the view/page for removal -> *example*: network for the Network Traffic page OR endpoint_protection for the (Trellix) Endpoint Protection page, etc.
- 14.4.9 Highlight the **<view name=network/>** -> delete the line.
- 14.4.10 Click **Save**.
- 14.4.11 Open the **OTArmor – Visibility** dashboard -> verify the Navigation Menu option is no longer visible.

14.5 (If requested) Forward ESXi Logs to LM1(Splunk)

IMPORTANT: ONLY configure the ESXi Syslog forwarding to LM1(Splunk) if requested by the customer. The virtual hosts are not configured to forward log data to Splunk by default. Due to the frequency of logs, it is recommended to forward only if requested/required by the customer. The Splunk Licensing Tool has been updated to adjust for the virtual host log increase if forwarding is required.

14.5.1 For Host(s) ESXi:

- 14.5.2 Login to the Host (e.g., VH1).
- 14.5.3 Select **Manage** -> click **System tab** -> select **Advanced Settings**.
- 14.5.4 In the **Advanced Settings** window -> type **Syslog** into the search bar.
- 14.5.5 Find the Key: **Syslog.global.logHost** -> select and click **Edit Option**.
- 14.5.6 In the Edit Option window -> enter the following (string): **udp://<LM1 IP Address>:514**
- Note:** This can be set UDP or TCP, UDP is default.
- 14.5.7 Click **Save**.
- 14.5.8 Find the Key: **Syslog.global.logLevel** -> select **Edit Option**.
- 14.5.9 In the Edit Option window -> enter the following (string): **error**

Note: It is recommended to filter the log level to “error”. If the customer prefers “warning and error” log messages, set the (string) to “warning”. This is a *quality over quantity* log filter and helps reduce the number of logs being forwarded to Splunk.

14.5.10 (For vSphere Systems) Enable Syslog in the vSphere Firewall:

- 14.5.10.1 When the Key values have been changed -> navigate to **Host – Configure** -> scroll to **System** -> select the **Firewall**.

14.5.10.2 Click **Edit** – in the search bar/filter, type **Syslog**.

14.5.10.3 Check/select the box for **Syslog** -> click **OK**.

14.5.11 **(For vSphere Systems) Restart the Syslog Service in vSphere:**

14.5.11.1 Navigate to **Host – Configure** -> scroll to **System** -> select the **Services**.

14.5.11.2 Find the **Syslog Server** service.

14.5.11.3 Select the service -> click **Restart**.

14.5.11.4 Repeat the above steps for VH2 and VH3 if necessary.

14.5.11.5 Verify the logs are populating in Splunk.

14.5.12 **(For ESXi Systems) Restart the Syslog Service in ESXi:**

14.5.12.1 When the Key values have been changed -> navigate to **Host – Manage** -> select the **Services tab**.

14.5.12.2 Find the vmsyslogd – Syslog Server service.

14.5.12.3 Select the service -> click **Restart**.

14.5.12.4 Repeat the above steps for VH2 and VH3 if necessary.

14.5.12.5 Verify the logs are populating in Splunk.

14.6 **(If applicable) Forward Splunk Data to Third-Party SIEM**

Configure the outputs.conf file on the LM1 Server to forward Splunk (raw syslog) Data to the Third-Party SIEM. This file can be configured ahead of site deployment; however, the filename should be saved as a backup file until connected to the Third-party SIEM.

Note: Please read below (configuration settings) for more information on **maxEventSize** and **priority**. These are optional settings.

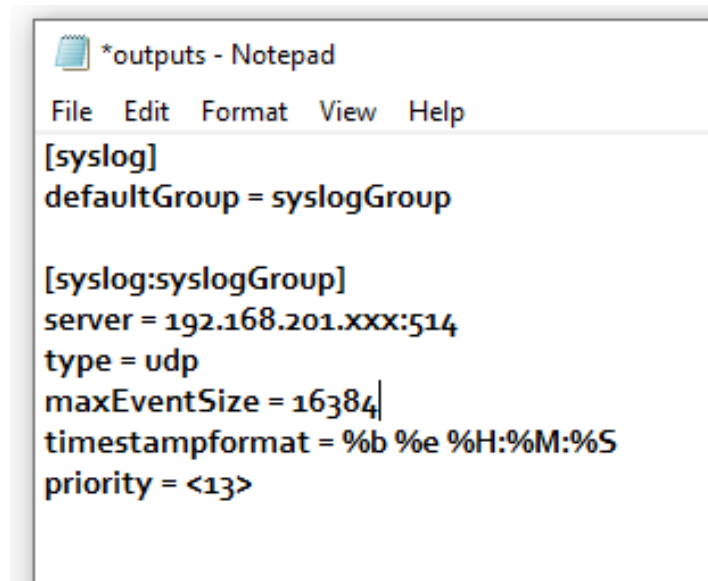
14.6.1 On the LM1 Server, open Notepad as Administrator.

14.6.1.1 Enter the following:

```
[syslog]
defaultGroup = syslogGroup

[syslog:syslogGroup]
server = <Third-Party SIEM IP ADDRESS>:514
type = udp
maxEventSize = 16384
timestampformat = %b %e %H:%M:%S
priority = <13>
```

14.6.1.2 Example **outputs.conf**:



```
*outputs - Notepad
File Edit Format View Help
[syslog]
defaultGroup = syslogGroup

[syslog:syslogGroup]
server = 192.168.201.xxx:514
type = udp
maxEventSize = 16384
timestampformat = %b %e %H:%M:%S
priority = <13>
```

14.6.1.3 Click File -> Save As...

14.6.1.4 Navigate to: D:\security\etc\system\local\

14.6.1.5 Save the file as: **outputs.conf.bak**

Important: Name the outputs.conf file as a backup file until the system is deployed at site and connected to the Third-party system. If this file is active before connecting to the Third-party system, the data will begin forwarding (to an unknown address) and cause issues in Splunk.

14.6.2 Close the file.

14.6.3 **AT SITE** -> rename the file to "**outputs.conf**" -> the data should begin forwarding without rebooting LM1 or restarting the *splunkd* service.

14.6.3.1 This may take a few minutes, be sure to check the receiving end (third-party system) to verify.

14.6.3.2 If needed, restart the *splunkd* service.

Note: Once the Splunk (raw syslog) Data is received by the Third-Party SIEM, it is the responsibility of the Third-Party SIEM owner to configure data parsing (if required).

14.6.3.3 Additional Information.

14.6.3.4 Configuration Settings for *outputs.conf*

<https://docs.splunk.com/Documentation/Splunk/9.2.2/Admin/Outputsconf>

```
[syslog]
defaultGroup = syslogGroup
```

[syslog:syslogGroup]

server = <IP Address of Third-Party SIEM>:514

Set IP Address and Port. Default port is 514.

type = udp (default)

Do not recommend setting type = TCP, some logs are stripped of a priority, breaking into multiple lines, and cannot be parsed correctly on the receiving end. Example, Windows performance logs.

(optional) maxEventSize = 1024 (default)

Maximum size of an event in bytes, set higher for log truncation issues. All events exceeding the configured size are truncated. Some Windows events may exceed the default, requiring an increased maxEventSize. This setting should be increased gradually, not to exceed site bandwidth limitations.

timestampformat = %b %e %H:%M:%S

The syslog output header will not be RFC3164 compliant without this setting. There are multiple specifiers that can be used. Please see the Splunk documentation (link above) for more specifier information.

- *| %b - Abbreviated month name (Jan, Feb, ...)
- * %e - Day of month
- * %H - Hour
- * %M - Minute
- * %s - Second

(optional) priority = <13> (default user and notice)

The priority value included at the beginning of each syslog message, this ranges from 0-191 made up of Facility Value and Severity Value.

- The table of facility and severity (and their values) is located in RFC3164. For example, <http://www.ietf.org/rfc/rfc3164.txt> section 4.1.1

15.0 **Splunk SIEM – OTArmor**

15.1 **Check SIEM1 Resource Utilization**

Check the SIEM1 Server resource utilization using AlmaLinux Cockpit.

15.1.1 Login to a Security Server or HMI as Domain Admin.

15.1.2 Open a **Web Browser**.

15.1.3 Enter the **SIEM1 AlmaLinux Cockpit** URL: **https:\\siem1:9090** OR **https:\\<SIEM1 IP ADDRESS>:9090**

- 15.1.4 The *AlmaLinux* login page will appear.
- 15.1.5 Login with **Admin** credentials (siemadmin – **See Default Passwords Document*).
- 15.1.6 Under System -> select **Overview**.
- 15.1.7 Review the **Usage** panel.
- 15.1.8 Verify the percentages/GiB are stable, not at capacity/exceeding the configured limit (16 CPU, 32GB RAM).
 - 15.1.8.1 Click **View metrics and history** link to review in detail.
 - 15.1.8.2 The **Splunkd service** is expected to be the highest consumer of CPU and Memory.

15.2 Check Splunk License

Verify the Splunk Enterprise license is applied and check data volume levels.

INFORMATION: Please refer to **Confluence – Gas Power Engineering Function – Splunk Enterprise – Troubleshooting Guidance OR GEH-6887 – Splunk Troubleshooting** for additional Splunk Licensing information.

- 15.2.1 Login to a Security Server or HMI as Domain Admin.
 - 15.2.2 Open a **Web Browser**.
 - 15.2.3 Enter the **SIEM1 Splunk URL**: **https:\\siem1:8000** OR **https:\\<SIEM1 IP ADDRESS>:8000**
 - 15.2.4 The *splunk>enterprise* login page will appear.
 - 15.2.5 Login with **Splunk Admin** credentials (splunkadmin – **See Default Passwords Document*).
 - 15.2.6 From the **Splunk console** -> select **Settings -> Licensing**.
 - 15.2.7 On the Licensing page -> verify the project license(s).
 - 15.2.7.1 Check **Volume** (1GB, 5GB, etc.), **Expiration** (annual) and **Status** (valid, FROM FUTURE).
- Note:** The Splunk Enterprise license is an annual renewal. If the project purchased more than 1 year, verify all licenses have been applied (and should state FROM FUTURE).
- 15.2.8 **Licensing Errors:**
 - 15.2.8.1 If there is a **“litsearch” command** Error message in the console, this is indicating a problem with the Splunk license.
 - 15.2.8.2 It can be an issue with license expiration and/or license overutilization.

15.2.8.3 When this “litsearch” Error occurs, Splunk will continue to index data however the Search functionality pauses until the license is corrected (exception for the *_internal* index for license diagnosis).

15.2.9 **License Expired:** If the license is “*expired*”, apply a valid license to continue using the Splunk application.

Note: This may also require a “reset” license (from Splunk) to reenable the Search functionality. **Contact Splunk to acquire the “reset” license if the console is still showing a ‘litsearch’ error.*

15.2.10 **License Overutilization:** If the license is “*overutilized*”, there will be overutilization alerts under Current or Permanent Alerts.

15.2.10.1 If there are several **pool quota overage warnings** reported, this indicates an overutilization of the license (currently and/or historically).

15.2.10.2 The amount of log data collected (daily) is exceeding the daily indexing limit (license - GB).

15.2.10.3 It would be recommended to increase license size to support the actual daily log consumption.

15.2.10.4 Purchase an additional license to prevent license overutilization and support the (actual) daily log consumption.

15.2.10.5 Licenses can be “stacked” to increase the TOTAL daily indexing limit (GB).

Note: this may also require a “reset” license (from Splunk) to reenable the Search functionality. **Contact Splunk to acquire the “reset” license if the console is still showing a ‘litsearch’ error.*

Note: If the volume has exceeded the stack size please use the Splunk Enterprise License Calculation Tool guidance table to evaluate the license. This is available in the Quoting and BOM tools. Adjust the Host Quantity per project and calculate the Low Gb and High Gb totals to find an average.

15.2.11 To determine the license size needed -> select the **Usage Report** button on the Licensing page.

15.2.12 Select the **Previous 60 days** tab.

15.2.13 In the Daily License Usage panel -> verify the volume is under the stack size.

15.2.14 The Usage Report window has an option to view previous 30 or 60 days.

Note: if the license has been in overutilization for a prolonged period of time it may affect the results.

Note: to view license usage by endpoint click the “**Split by:**” drop-down and select “**Host**”.

15.2.15 Hover over the volume column for each endpoint to view the individual license consumption.

15.2.16 Review the endpoints that are consuming a large amount of the license and evaluate. This may require a log forwarding adjustment to reduce the license overutilization.

15.3 (If requested) Hide/Disable Unused Splunk Apps

Splunk Enterprise + Splunk Enterprise Security (SIEM1) has a standard set of supporting applications. If requested, these applications can be disabled/hidden in Splunk if not used. Example applications:

- Splunk Add-on for Unix and Linux
- Fortinet FortiGate App for Splunk
- Ivanti Security Controls App for Splunk
- InfoSec
- Nozomi Networks Universal Add-on

15.3.1 Login to a Security Server or HMI as Domain Admin.

15.3.2 Open a **Web Browser**.

15.3.3 Enter the **SIEM1 Splunk URL**: **https:\\siem1:8000** OR **https:\\<SIEM1 IP ADDRESS>:8000**

15.3.4 The *splunk>enterprise* login page will appear.

15.3.5 Login with **Splunk Admin** credentials (splunkadmin – **See Default Passwords Document*).

15.3.6 On the Splunk home page -> select the **Apps** drop-down -> select **Manage Apps**.

15.3.7 In the filter by app name search box -> enter the App name requested.

15.3.7.1 To Disable: Under the Application **Status** -> select **Disable**.

15.3.7.2 To Hide: Under the Application **Actions** -> select **Edit properties** -> next to **Visible** -> select **No** -> click **Save**.

15.3.8 After Disabling or Hiding an application -> select **Settings – Server controls**.

15.3.9 Click **Restart Splunk**.

15.3.10 Wait for the Splunk application to restart -> login and verify the applications are no longer visible from the home page (Apps).

15.4 (If requested) Forward ESXi Logs to SIEM1 (Splunk)

IMPORTANT: ONLY configure the ESXi Syslog forwarding to SIEM1 (Splunk) if requested by the customer. The virtual hosts are not configured to forward log data to Splunk by default. Due to the frequency of logs, it is recommended to forward only

if requested/required by the customer. The Splunk Licensing Tool has been updated to adjust for the virtual host log increase if forwarding is required.

15.4.1 For Host(s) ESXi:

15.4.2 Login to the Host (e.g., VH1).

15.4.3 Select **Manage** -> click **System tab** -> select **Advanced Settings**.

15.4.4 In the **Advanced Settings** window -> type **Syslog** into the search bar.

15.4.5 Find the Key: **Syslog.global.logHost** -> select and click **Edit Option**.

15.4.6 In the Edit Option window -> enter the following (string): **udp://<SIEM1 IP Address>:514**

Note: This can be set UDP or TCP, UDP is default.

15.4.7 Click **Save**.

15.4.8 Find the Key: **Syslog.global.logLevel** -> select **Edit Option**.

15.4.9 In the Edit Option window -> enter the following (string): **error**

Note: It is recommended to filter the log level to “error”. If the customer prefers “warning and error” log messages, set the (string) to “warning”. This is a *quality over quantity* log filter and helps reduce the number of logs being forwarded to Splunk.

15.4.10 (For vSphere Systems) Enable Syslog in the vSphere Firewall:

15.4.10.1 When the Key values have been changed -> navigate to **Host – Configure** -> scroll to **System** -> select the **Firewall**.

15.4.10.2 Click **Edit** – in the search bar/filter, type **Syslog**.

15.4.10.3 Check/select the box for **Syslog** -> click **OK**.

15.4.11 (For vSphere Systems) Restart the Syslog Service in vSphere:

15.4.11.1 Navigate to **Host – Configure** -> scroll to **System** -> select the **Services**.

15.4.11.2 Find the **Syslog Server** service.

15.4.11.3 Select the service -> click **Restart**.

15.4.11.4 Repeat the above steps for VH2 and VH3 if necessary.

15.4.11.5 Verify the logs are populating in Splunk.

15.4.12 (For ESXi Systems) Restart the Syslog Service in ESXi:

15.4.12.1 When the Key values have been changed -> navigate to **Host – Manage** -> select the **Services tab**.

15.4.12.2 Find the vmsyslogd – Syslog Server service.

15.4.12.3 Select the service -> click **Restart**.

15.4.12.4 Repeat the above steps for VH2 and VH3 if necessary.

15.4.12.5 Verify the logs are populating in Splunk.

15.5 **(If applicable) Forward Splunk Data to Third-Party SIEM**

Configure the outputs.conf.bak file on the SIEM1 Server to forward Splunk (raw syslog) Data to the Third-Party SIEM. This file can be configured ahead of site deployment; however, the filename should be saved as a backup file until connected to the Third-party SIEM.

Note: Please read below (configuration settings) for more information on **maxEventSize** and **priority**. These are optional settings.

15.5.1 Login to a Security Server or HMI as Domain Admin.

15.5.2 Open a **Web Browser**.

15.5.3 Enter the **SIEM1 AlmaLinux Cockpit** URL: **https:\\siem1:9090** OR **https:\\<SIEM1 IP ADDRESS>:9090**

15.5.4 The *AlmaLinux* login page will appear.

15.5.5 Login with **Admin** credentials (siemadmin – **See Default Passwords Document*).

15.5.6 Under Tools -> select **File browser**.

15.5.7 Click the **Filesystem** icon -> navigate to the following directory:
/opt/splunk/etc/system/local.

15.5.8 At the root of the local folder -> select the **three dots** (far right next to Upload) -> select **Create file**.

15.5.8.1 Enter the following:

```
[syslog]
defaultGroup = syslogGroup

[syslog:syslogGroup]
server = <Third-Party SIEM IP ADDRESS>:514
type = udp
maxEventSize = 16384
timestampformat = %b %e %H:%M:%S
priority = <13>
```

15.5.8.2 Example:

Create file

File name:

```
[syslog]
defaultGroup = syslogGroup

[syslog:syslogGroup]
server = <Third-Party SIEM IP ADDRESS>:514
type = udp
maxEventSize = 16384
timestampformat = %b %e %H:%M:%S
priority = <13>
```

Owner:

Create **Cancel**

15.5.8.3 File name: **outputs.conf.bak**

15.5.8.4 Owner: **splunk:splunk**

15.5.8.5 Click **Create**.

Important: Name the outputs.conf file as a backup file until the system is deployed at site and connected to the Third-party system. If this file is active before connecting to the Third-party system, the data will begin forwarding (to an unknown address) and cause issues in Splunk.

15.5.9 **AT SITE** -> rename the file to **"outputs.conf"** -> the data should begin forwarding without rebooting SIEM1 or restarting the *splunkd* service.

15.5.9.1 This may take a few minutes, be sure to check the receiving end (third-party system) to verify.

15.5.9.2 If needed, restart the *splunkd* service.

Note: Once the Splunk (raw syslog) Data is received by the Third-Party SIEM, it is the responsibility of the Third-Party SIEM owner to configure data parsing (if required).

15.5.9.3 Additional Information.

15.5.9.4 Configuration Settings for *outputs.conf*

<https://help.splunk.com/en/splunk-enterprise/administer/admin-manual/10.0/configuration-file-reference/10.0.0-configuration-file-reference/outputs.conf>

```
[syslog]
defaultGroup = syslogGroup
```

```
[syslog:syslogGroup]
server = <IP Address of Third-Party SIEM>:514
```

Set IP Address and Port. Default port is 514.

type = udp (default)

Do not recommend setting type = TCP, some logs are stripped of a priority, breaking into multiple lines, and cannot be parsed correctly on the receiving end. Example, Windows performance logs.

(optional) maxEventSize = 1024 (default)

Maximum size of an event in bytes, set higher for log truncation issues. All events exceeding the configured size are truncated. Some Windows events may exceed the default, requiring an increased maxEventSize. This setting should be increased gradually, not to exceed site bandwidth limitations.

timestampformat = %b %e %H:%M:%S

The syslog output header will not be RFC3164 compliant without this setting. There are multiple specifiers that can be used. Please see the Splunk documentation (link above) for more specifier information.

- *| %b - Abbreviated month name (Jan, Feb, ...)
- * %e - Day of month
- * %H - Hour
- * %M - Minute
- * %s - Second

(optional) priority = <13> (default user and notice)

The priority value included at the beginning of each syslog message, this ranges from 0-191 made up of Facility Value and Severity Value.

- The table of facility and severity (and their values) is located in RFC3164. For example, <http://www.ietf.org/rfc/rfc3164.txt> section 4.1.1

16.0 Certificate Authority – OTArmor

16.1 **Verify the MKVle Controller goes to Secure State**

Note: Make sure you can ping UDH and PDH to verify network connectivity.

16.1.1 Login to the HMI with **GEMaint** credentials.

16.1.2 Open the software *.TCW file located on the **(E:) drive** -> folder **Master**.

16.1.3 In the left pane -> locate **Security Servers** (lower left pane) and change the following:

16.1.3.1 **CA Server Full Name:**

16.1.3.1.1 Login to **CA1** with Administrator credentials.

16.1.3.1.2 Open Server Manager.

16.1.3.1.3 Select **Tools** -> open **Certification Authority** and select CA name.

16.1.3.1.4 Navigate back to the HMI.

16.1.3.1.5 Open **ToolboxST** -> locate **Security Servers** -> paste the CA server name in the CA Server Full Name section (click and type in the box).

Example: **HMI-CA1-CA.hmi** (per network topology).

16.1.3.2 **CA Server IP Address:**

16.1.3.2.1 Enter the UDH IP Address of CA1.

16.1.3.3 **CA Server Thumb Print:**

16.1.3.3.1 Login to **CA1** with Administrator credentials.

16.1.3.3.2 Open Server Manager.

16.1.3.3.3 Select **Tools** -> open **Certification Authority** and right-click on the CA name.

16.1.3.3.4 Select Properties -> General Tab -> View Certificate -> Details Tab -> Thumb Print.

16.1.3.3.5 Click on **Thumb Print** and copy the text.

16.1.3.3.6 Navigate back to the HMI.

16.1.3.3.7 Paste the text into **Thumb Print** on **TCW**.

16.1.3.4 **System Log Server IP Address:**

16.1.3.4.1 Enter the IP Address of the **Log Server** LM1.

16.1.3.4.2 If no Log Server -> use the IP Address of CA1.

16.1.3.5 **Gateway IP Address (only when working with Firewall):**

16.1.3.5.1 Set the correct Gateway IP Address for the PDH and UDH on TCW.

16.1.4 **Download the controller:**

16.1.4.1 In the ***.TCW file** (ToolboxST file). Select from menu -> **File** -> click **Save**.

16.1.4.2 Open the **Controller**.

16.1.4.3 Select from menu -> **File** -> click **Save** controller.

16.1.4.4 Expand: **Device** -> **Download** -> **Download Wizard**.

16.1.4.5 Click **Next** until the download starts.

16.1.4.6 Wait for download to complete.

16.1.4.7 Click **Close**.

16.1.5 **Controller in Controlling state:**

16.1.5.1 Expand: **Device** -> **Online**.

16.1.5.2 Select the **Status Tab** (bottom pane).

16.1.5.3 Verify Control State = Controlling.

16.1.6 Test Secure mode on Controller:

16.1.6.1 Expand: **Device** -> **Security State** -> **Set Security** (pop-up).

16.1.6.2 Click **Go to Secure**.

16.1.6.3 Wait for response = **Security State Secure** (Controller R/S/T is Secure).

16.1.6.4 **(If applicable)** Login to the LM1 Server – verify Controller logs.

16.1.6.5 Open to the **Splunk** application -> select the **OTArmor Visibility dashboard**.

16.1.6.6 On the **Home** page -> verify logs are populated on the **Controller Events** panel and the **Controller Events – Download Completed** panel.

Note: Controller logs will appear (in Splunk) under the IP Address or FQDN of the HMI where the .tcw file was configured. The controller log will include the controller's name in the following format: *device:<name of controller>*. The device:<name of controller> can be searched in Splunk to find controller log data as well.

Note: If the controller logs are not appearing in the panels, verify the UDP 514 Windows Firewall port has been added to LM1 and the profile is configured for "Domain, Private, and Public (All)". The controller logs are forwarded via UDP 514 on the UDH network.

17.0 Trellix AV and Application Control – OTArmor

Attention: If the system has **Application Control**, make sure to put the system in **Enable Mode** before completing the test steps.

17.1 Deploy Trellix Agent to Endpoint

Note: Refer to **GEH-6889 Section 3.1** to verify deployment of Trellix Agent to Endpoint.

17.1.1 From the HMI -> remote desktop to **SP1**.

17.1.2 Desktop -> open the **Launch Trellix ePolicy Orchestrator Console** icon and login with admin credentials.

17.1.3 In the **ePO console** -> click on **System Tree** (in menu).

17.1.4 Select **Workstations** -> open the **Group Details** tab.

17.1.5 On **Synchronization type** -> click **Edit**.

17.1.6 Under **Active Directory Domain** -> select **Use Domain**.

17.1.7 Type the **Domain Name: HMI.local**.

17.1.8 Under **Active Directory Credentials**.

17.1.8.1 Type in Domain **HMI**.

17.1.8.2 Enter Username and Password (use domain administrator account).

17.1.8.3 Scroll to bottom and click **Synchronize Now**.

17.1.8.4 Click **Save**.

17.1.9 Verify the **Synchronize Now** command.

17.1.9.1 Expand **Menu** -> **Automation** -> **Server Task Log** -> confirm **Completed** status under **Active Directory/NT Domain Synchronization**.

17.1.10 Click on **System Tree** -> **Workstations** -> **Systems tab** -> select **This Group and all Subgroups** in the presets dropdown (verify all computers are in the folder).

17.1.11 Select the computer for agent deployment -> scroll to the bottom pane and expand/select **Actions** -> **Agents** -> **Deploy Agents**.

17.1.12 In Deploy Trellix Agent window complete the following:

17.1.12.1 Under Credentials for agent installation.

17.1.12.2 enter the Domain: **HMI**.

17.1.12.3 Username: **Admin**.

17.1.12.4 Password use credentials.

17.1.12.5 Click **OK**.

17.2 Run MsgBusCertsUpdater

Note: The MsgBusCertsUpdater must be run before deploying the rest of the agents.

17.2.1 Navigate to the system in the **System Tree**.

17.2.2 If necessary, select **This Group and All Subgroups** in the **Preset** drop-down menu.

17.2.3 Select the Workstation or Server the update will be deployed to.

17.2.4 Select **Actions** -> **Agent** -> **Run Client Task Now**.

17.2.5 Select the following:

17.2.5.1 Product: **Trellix Agent**

17.2.5.2 Task Type: **Product Update**

17.2.5.3 Task Name: **Create New Task**

17.2.5.3.1 Select the following settings:

17.2.5.3.2 Package selection: **Selected packages**

17.2.5.3.3 Package types: **MsgBus Cert Updater 5.8.1.313**

17.2.5.4 Select **Run Task Now**.

17.2.5.5 Verify that the task completes successfully.

17.2.5.6 Select **Close** on the Running Client Task Status window.

17.3 Deploy Trellix Endpoint Security Platform 10.7.0.6809.

17.3.1 Navigate to **System Tree** -> Workstations or Servers.

17.3.2 If necessary, select **This Group and All Subgroups** in the **Preset** drop-down menu.

17.3.3 Select the Workstation or Server the Trellix Agent will be deployed to.

17.3.4 Select **Actions** -> **Agent** -> **Run Client Task Now**.

17.3.5 Select the following:

17.3.5.1 Product: **Trellix Agent**

17.3.5.2 Task Type: **Product Deployment**

17.3.5.3 Task Name: **Create New Task**

17.3.6 Select the following in the **Run Client Task Now** window:

17.3.6.1 Target Platform: **Windows**

17.3.6.2 Products and components: **Endpoint Security Platform 10.7.0.6809**

17.3.6.2.1 Actions: **Install**

17.3.6.2.2 Branch: **Current**

17.3.6.3 Select **Run Task Now**.

17.3.7 Verify the task completes successfully.

17.3.8 Select **Close** on the Running Client Task Status window.

17.3.9 Reboot the Workstation or Server before proceeding.

17.4 **Deploy Trellix Endpoint Security Threat Prevention 10.7.0.6711**

17.4.1 Navigate to **System Tree** -> Workstations or Servers.

17.4.2 If necessary, select **This Group and All Subgroups** in the **Preset** drop-down menu.

17.4.3 Select the Workstation or Server the Trellix Agent will be deployed to.

17.4.4 Select **Actions** -> **Agent** -> **Run Client Task Now**.

17.4.5 Select the following:

17.4.5.1 Product: **Trellix Agent**

17.4.5.2 Task Type: **Product Deployment**

17.4.5.3 Task Name: **Create New Task**

17.4.6 Select the following in the **Run Client Task Now** window:

17.4.6.1 Target Platform: **Windows**

17.4.6.2 Products and components: **Endpoint Security Threat Prevention
10.7.0.6711**

17.4.6.2.1 Actions: **Install**

17.4.6.2.2 Branch: **Current**

17.4.6.3 Select **Run Task Now**.

- 17.4.7 Verify the task completes successfully.
- 17.4.8 Select **Close** on the Running Client Task Status window.

17.5 **Deploy Trellix Data Loss Prevention 11.11.2.1172.**

- 17.5.1 Navigate to **System Tree** -> Workstations or Servers.
- 17.5.2 If necessary, select **This Group and All Subgroups** in the **Preset** drop-down menu.
- 17.5.3 Select the Workstation or Server the Trellix Agent will be deployed to.
- 17.5.4 Select **Actions** -> **Agent** -> **Run Client Task Now**.
- 17.5.5 Select the following:
 - 17.5.5.1 Product: **Trellix Agent**
 - 17.5.5.2 Task Type: **Product Deployment**
 - 17.5.5.3 Task Name: **Create New Task**
- 17.5.6 Select the following in the **Run Client Task Now** window:
 - 17.5.6.1 Target Platform: **Windows**
 - 17.5.6.2 Products and components: **Trellix Data Loss Prevention 11.11.2.1172**
 - 17.5.6.2.1 Actions: **Install**
 - 17.5.6.2.2 Branch: **Current**
 - 17.5.6.3 Select **Run Task Now**.
- 17.5.7 Verify the task completes successfully.
- 17.5.8 Select **Close** on the Running Client Task Status window.

17.6 **Deploy Trellix Solidcore Client for Windows 8.4.2.411.**

IMPORTANT: Verify the system has a full backup before proceeding.

- 17.6.1 Navigate to **System Tree** -> Workstations or Servers.
- 17.6.2 If necessary, select **This Group and All Subgroups** in the **Preset** drop-down menu.
- 17.6.3 Select the Workstation or Server the Trellix Solidcore product will be deployed to.
- 17.6.4 Select **Actions** -> **Agent** -> **Run Client Task Now**.
- 17.6.5 Select the following:
 - 17.6.5.1 Product: **Trellix Agent**
 - 17.6.5.2 Task Type: **Product Deployment**
 - 17.6.5.3 **Create New Task** -> select **Solidcore Client 8.4.2.411**
 - 17.6.5.4 Select **Run Task Now**.

17.6.6 Verify the task completes successfully.

17.6.7 Select **Close** on the Running Client Task Status window.

Note: Solidification will not begin until the 07-Enable-Application-Control task is run. On the endpoint, the Trellix Solidifier will display (no license) until the following task is run.

17.6.8 Navigate to **System Tree** -> Workstations or Servers.

17.6.9 If necessary, select **This Group and All Subgroups** in the **Preset** drop-down menu.

17.6.10 Select the Workstation or Server with the Solidcore Agent.

17.6.11 Select **Actions** -> **Agent** -> **Run Client Task Now**.

17.6.12 Select the following:

17.6.12.1 Product: **Solidcore 8.4.2.411**

17.6.12.2 Task Type: **SC:Enable**

17.6.12.3 Task Name: **07-Enable-Application-Control**

17.6.12.4 Select **Run Task Now**.

17.6.13 Verify the task completes successfully.

17.6.14 On the server or workstation -> right-click on the Trellix icon in the task bar.

17.6.14.1 Select **Quick Settings – Application Control Solidification Status**.

17.6.14.1.1 The status will indicate if the **solidification** process is in progress and when the **process is complete**.

Note: After running the 07-Enable-Application-Control the Solidification will begin. After deploying the Trellix Application Control agent, it may take 2-4 hours for solidification (Application Control) to complete.

IMPORTANT: DO NOT REBOOT

It is important to run the following **‘Disable Memory Protection’** steps before rebooting the machine.

17.6.15 **Disable Memory Protection**

IMPORTANT

Failure to complete this section may result in issues with Virtual Machines, Servers, and Workstations after a reboot.

17.6.15.1.1 Navigate to **System Tree** -> select **My Organization**.

17.6.15.1.2 If necessary, select **This Group and All Subgroups** in the **Preset** drop-down menu.

17.6.15.1.3 Select a Workstation or Server for the task.

17.6.15.1.4 Select **Actions** -> **Agent** -> **Run Client Task Now**.

17.6.15.1.5 In the Product list -> select **Solidcore 8.x.x**.

17.6.15.1.6 In the Task Type -> select **SC:Run Commands**.

17.6.15.1.7 Select **08-Disable MP**

17.6.15.1.8 Click **Run Task Now**.

17.6.15.1.9 Wait for the task to complete.

IMPORTANT: Verify this has been completed on ALL workstations and servers.

17.6.15.2 After the Disable MP task has been completed -> navigate to the machine and verify the status.

17.6.15.3 Login and open a **Windows PowerShell (Admin)**.

17.6.15.3.1 Enter: **sadmin recover** (to recover CLI lockdown).

17.6.15.3.2 Password: ***See Default Passwords Document**

17.6.15.3.3 Enter: **sadmin features**

17.6.15.3.4 Verify the three tasks have been **Disabled** (mp, deny-read, deny-write).

17.6.15.3.5 If the features appear **Disabled on Reboot** -> complete the following before rebooting the machine.

17.6.15.3.6 **Verify the solidification status -> complete steps below.**

17.7 Check Application Control Solidification Status

17.7.1 On the machine.

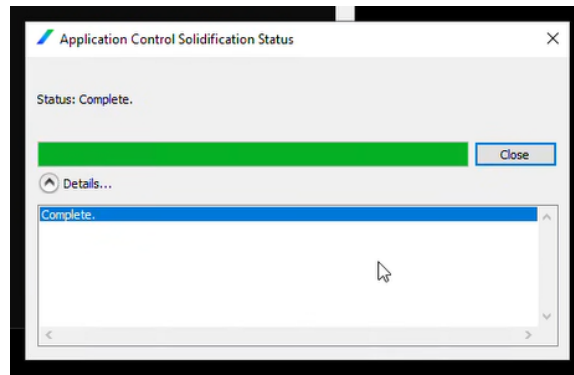
17.7.2 Right-Click on the Trellix icon in the task bar.

17.7.3 Select **Quick Settings – Application Control Solidification Status**.

17.7.3.1 The status will indicate if the **solidification** process is in progress and when the **process is complete**.

WAIT FOR SOLIDIFICATION TO COMPLETE

Rebooting the machine before solidification is complete will result in issues with Virtual Machines, Servers, and Workstations after a reboot.



17.7.4 Once **solidification** is complete (above screenshot) -> **reboot** the endpoint.

17.7.5 Once the server or workstation has rebooted -> re-check the **Disable MP status**.

17.7.6 Open a **Windows PowerShell (Admin)**

17.7.6.1 Enter: **sadmin recover** (to recover CLI lockdown).

17.7.6.2 Password: ***See Default Passwords Document**

17.7.6.3 Enter: **sadmin features**

17.7.6.4 Verify the three tasks have been **Disabled** (mp, deny-read, deny-write).

17.7.6.5 Enter **sadmin lockdown** (to re-enable CLI lockdown).

17.7.7 **Verify all HMI's are in Trellix:**

17.7.7.1 Open **Trellix ePO** (ePolicy Orchestrator) desktop icon.

17.7.7.2 Login and click on **System Tree** open **My Organization** -> **System** Tab.

17.7.7.3 Preset: **select This Group and All Subgroups**.

17.7.7.4 Under List of all the Systems:

17.7.7.4.1 Verify HMI's are in the **Managed State** column, as **Managed**.

17.7.8 **Verify the HMI versions match the Console:**

17.7.8.1 Login to the HMI.

17.7.8.2 Right-click **Trellix Icon** (shield in the taskbar) -> click **About**.

17.7.8.3 Check Trellix **DLP Endpoint**.

17.7.8.4 Check Trellix **Endpoint Security** (if applicable).

17.7.8.5 Check Trellix **Solidcore** (Application control, if applicable).

17.7.8.6 Open **Trellix ePO** (ePolicy Orchestrator) desktop icon.

17.7.8.7 Login and select **Menu -> Software -> Master Repository**.

17.7.8.8 Compare the list to what was deployed and verify the following:

17.7.8.8.1 **DLP Endpoint.**

17.7.8.8.2 **Endpoint Security.**

17.7.8.8.3 **Solidcore.**

17.7.8.8.4 **Trellix for Windows.**

17.7.8.9 Version on the HMIs should match the ones on the ePO.

17.7.9 **Deploy test/fake virus:**

In some situations, the Trellix icon (shield) may not be present in the system tray (taskbar). This can happen during a remote connection, or some other issue may cause the icon not to appear. The Trellix icon can be manually started by doing the following procedure:

17.7.9.1 From any HMI or Computer.

17.7.9.2 Verify that the Trellix icon is now present in the system tray.

17.7.9.3 If the icon is **not** in the tray:

17.7.9.3.1 Navigate to: **C:\Program Files\Trellix\Agent\x86** on the physical or virtual machine.

17.7.9.3.2 Double-click the **UpdaterUI.exe**

17.7.9.4 Place the HMI or Computer in **Observe mode**.

17.7.9.5 Login to SP1. Open **Trellix ePO** (ePolicy Orchestrator) desktop icon.

17.7.9.6 Login -> click on **System Tree** -> open **Servers**.

17.7.9.7 Click the **Systems** tab -> open the **Presets** dropdown -> select **This group and all subgroups**.

17.7.9.8 Select the machine -> click on **Actions** -> **Agent** -> select **Run Client Task Now**.

17.7.9.9 In the **Run Client Task Now** -> select **Solidcore 8.x.x** -> SC Observe Mode -> Begin Observer Mode -> click Create Task Now -> select Run Task Now.

17.7.9.10 Verify the solidifier is in Observe Mode -> open **Command Prompt (Admin)** and run the following: **sadmin status**

17.7.9.11 Verify:

17.7.9.12 Trellix Solidifier: **Observe**

17.7.9.13 Trellix Solidifier on reboot: **Observe**

17.7.9.14 Or in the ePO -> the About section -> verify the solidifier is in Observe Mode.

- 17.7.9.15 Open File Explorer -> type the following path:
\\CA1\Admin_Procedures\OTArmor
- 17.7.9.16 Look for the file **Test-AntiVirus.ps1**
- 17.7.9.17 Right-click and copy file onto **C:**
- 17.7.9.18 Right-click the **Test-AntiVirus.ps1** and select **Run with PowerShell**.
- 17.7.9.19 The PowerShell window will show a "Set-ExecutionPolicy" message in red, this can be ignored.
- 17.7.9.20 Type **Y** and hit enter.
- 17.7.9.21 A pop-up window should appear **Trellix Endpoint Security -> On Access Scan**.
- 17.7.9.22 Verify the test virus (EICAR test file) was **Detected (action=Delete)**.
- 17.7.9.23 Close the Trellix pop-up window.
- 17.7.9.24 Open Windows "Event Viewer".
- 17.7.9.25 Expand the **Windows Logs** -> click on **Application** -> look for **Trellix Endpoint Security (Error)** -> verify the test virus was detected and deleted in the General tab below.
- 17.7.9.26 Close Windows "Event Viewer".
 - 17.7.9.26.1 Login to **SP1**.
 - 17.7.9.26.2 Open **Trellix ePO** (ePolicy Orchestrator) desktop icon.
 - 17.7.9.26.3 Verify the event was logged.
 - 17.7.9.26.4 Select **Dashboards** -> select Dashboards **Dropdown** -> select Endpoint Security: **Detection Status**.
 - 17.7.9.26.5 In the endpoint Security Summary of Threats Detected in the last 24 Hours.
 - 17.7.9.26.6 Select/click the Threat Name -> EICAR test file.
 - 17.7.9.26.7 Verify the threat name was logged, and it comes from the HMI you deployed it from.
 - 17.7.9.26.8 Close the Trellix ePO console.
- 17.7.9.27 If you have LM1:
 - 17.7.9.27.1 Login to **LM1**.
 - 17.7.9.27.2 Open the **Splunk Login** desktop icon -> login with **Splunk admin credentials**.
 - 17.7.9.27.3 Open the **Search & Reporting** app.
 - 17.7.9.27.3.1 In the New Search, type: **EICAR** and hit <enter>.

- 17.7.9.27.3.2 There should be an event in the search list. Verify the event is from the HMI you deployed the virus on.
- 17.7.9.27.4 Open the **OTArmor – Visibility** app -> select Apps – OTArmor Visibility.
 - 17.7.9.27.4.1 Select the Trellix - **Endpoint Protection** page.
 - 17.7.9.27.4.2 Verify the EICAR event appears on the **Antivirus Events** panel. Click the number on the panel to review the details.
- 17.7.9.28 Delete the copied **Test-AntiVirus.ps1** from c:\ drive on the test machine.
- 17.7.9.29 Place the HMI or Computer back in **Enabled mode**.
- 17.7.9.30 Login to SP1. Open **Trellix ePO** (ePolicy Orchestrator) desktop icon.
- 17.7.9.31 Login -> click on **System Tree** -> open **Servers**.
- 17.7.9.32 Click the **Systems** tab -> open the **Presets** dropdown -> select **This group and all subgroups**.
- 17.7.9.33 Select the machine -> click on **Actions** -> **Agent** -> select **Run Client Task Now**.
- 17.7.9.34 In the **Run Client Task Now** -> select **Solidcore 8.x.x** -> SC Enable -> 07-Enable Application Control -> select Run Task Now.
- 17.7.9.35 Verify the solidifier is in Enabled Mode -> open **Command Prompt (Admin)** and run the following: **sadmin status**
- 17.7.9.36 Verify:
- 17.7.9.37 Trellix Solidifier: **Enabled**
- 17.7.9.38 Trellix Solidifier on reboot: **Enabled**
- 17.7.9.39 Or in the ePO -> the About section -> verify the solidifier is in Enabled Mode.
- 17.7.9.40 Close the Trellix ePO console.

17.8 Update Virus Definitions

Trellix ePO With AntiVirus/DLP - Provide Initial Threat Prevention Definitions

This section will outline providing the initial Threat Prevention Definitions (known as AMCore Content) to the Master Repository. These definitions should be obtained from the latest Update disc.

17.8.1 Run the batch file **Copy_McAfee_AV.bat** located on the **OTArmor McAfee XX XXXX AV Definitions Package**.

17.8.2 Login to **SP1** -> login to the **ePO**.

17.8.3 Click **Menu**- > **Software** -> **Main Repository** and click on **Check in Package**.

17.8.4 Leave the default Package Type: **Product or Update (ZIP)** selected -> click **Choose file**.

17.8.4.1 Navigate to **C:\McAfee Definitions** -> select **epoVx_xxxx.0dat.zip**.

17.8.4.2 Click **Open**.

17.8.4.3 Click **Next** on Package screen.

17.8.5 On the **Package Options** screen -> leave the **defaults**.

17.8.6 Click **Save** on the Package Information screen.

17.9 On-Access Scan Policy Exceptions

17.9.1 Verify ToolboxST and Cimplicity policy exceptions.

17.9.1.1 Open **Trellix ePO (ePolicy Orchestrator)** desktop icon and login as admin.

17.9.1.2 Navigate to **Menu** -> under **Policy** select **Policy Catalog**.

17.9.1.3 In the Products menu -> select Endpoint Security Threat Prevention.

17.9.1.4 Expand the On-Access Scan.

17.9.1.5 Select the GEV On-Access Scan policy -> click Edit.

17.9.1.6 Click Show Advanced -> in the Process Settings section -> select Use Standard settings for all processes.

17.9.1.7 Scroll down to **Process Types** -> verify the Standard tab is selected.

17.9.1.8 In the Exclusions box -> click Add.

17.9.1.8.1 **CimView.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.2 **ToolboxST.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.3 **LicenseST.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.4 **LicenseSTMonitor.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.5 **LicenseSTworkerservice.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.6 **LicenseSTMonitorservice.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.7 **PTOPC.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.8 **GeCssOPCServer.exe** When to exclude: On read and On Write -> click Save.

17.9.1.8.9 **GeCssOPCUAServer.exe** P When to exclude: On read and On Write -> click Save.

- 17.9.1.8.10 **GeCssControlSystemHealth.exe** When to exclude: On read and On Write -> click Save.
- 17.9.1.8.11 **Trender.exe** When to exclude: On read and On Write -> click Save.
- 17.9.1.8.12 **Indexer.exe** When to exclude: On read and On Write -> click Save.
- 17.9.1.8.13 C:\ProgramData\GE Energy\LicenseST\ Select Also exclude subfolders -> select When to exclude On read and On Write -> click Save.
- 17.9.1.8.14 C:\ProgramData\GE Vernova\LicenseST\ Select Also exclude subfolders -> select When to exclude On read and On Write -> click Save.
- 17.9.1.8.15 Select **Save**.

17.10 DLP

Make and verify open/closed USB groups using Trellix Data Loss Prevention policy.

- 17.10.1 Open Trellix ePO (ePolicy Orchestrator) desktop icon.
- 17.10.2 Login -> select **Menu** -> **Data Protection** -> **DLP Policy Manager** -> **Rule Sets** Tab.
- 17.10.3 Click on **GEV Removable Storage Device Rule**.
- 17.10.4 Click **Device Control** tab.
- 17.10.5 Select the **GEV Block Removable Device Rule**.
 - 17.10.5.1 Click on **Exceptions** Tab -> select **Excluded Users**.
 - 17.10.5.2 In the **End User** row -> click the ellipsis button **[...]**
 - 17.10.5.3 Select **Removable Media Access Allowed** -> click **Edit**.
Note: Make sure the Users were added in Active Directory before this step.
 - 17.10.5.4 Click **Add Users** button (bottom left).
 - 17.10.5.5 Select **Container and Children** in the **Preset** field.
 - 17.10.5.6 Browse for your **User** in the OU Structure -> click **OK**.
Note: Users Accounts: Check with customer for additional User Accounts.
 - 17.10.5.7 Click **Save**.
 - 17.10.5.8 In the Choose from existing values -> click **OK**.
 - 17.10.5.9 In the **DLP Rule set** -> click **Save**.
 - 17.10.5.10 Click **Close** in the DLP Policy Manager.
- 17.10.6 Click **Policy Catalog** -> select **GEV Block All Devices** policy -> click **Edit**.
- 17.10.7 Click **Apply Policy** (bottom right).

- 17.10.8 Click **OK** in the Policy Selection pop-up.
- 17.10.9 **Wake Up Trellix Agent to Endpoint.**
 - 17.10.9.1 Click on the **System Tree** icon in the menu & icon area at the top of the page.
 - 17.10.9.2 Select the **Workstation or Server** in the tree view -> click the **Systems** tab.
 - 17.10.9.3 Change the preset: select **This group and all subgroups**.
 - 17.10.9.4 Select the endpoint you wish to send the **Wake-up Agent**.
 - 17.10.9.5 Click on **Wake-up Agents**.
 - 17.10.9.6 Select **Force complete policy and task update**.
 - 17.10.9.7 Click **OK**.
- 17.10.10 **Test User Accounts.**
 - 17.10.10.1 Login as a User to verify User access (added in the previous section).
 - 17.10.10.2 Plug in a USB device and verify USB access.
- 17.10.11 **Allow USB Drive Access:**
 - 17.10.11.1 Plug the USB Drive into a computer that has the DLP installed on it. Verify that an **Access is denied** message appears on computer.
 - 17.10.11.2 Right-click on the **Trellix icon (shield)** in the system tray (taskbar) -> select **Managed Features** and **DLP Endpoint Console**.
 - 17.10.11.3 Verify that there is entry showing the USB Drive was blocked.
 - 17.10.11.4 Log into **SP1** using a domain administrator account.
 - 17.10.11.5 Log into the Trellix ePO (ePolicy Orchestrator).
 - 17.10.11.6 Select Dashboards and **DLP: Incident Summary**.
 - 17.10.11.7 Click on the Pie chart in the **DLP: Number of Incidents per type**.
 - 17.10.11.8 Double-click on the Incident (under Incident Type) that you created in step 13.2.11.3.
 - 17.10.11.9 Click on the Incident created and copy the USB serial number under the Additional Information section.
 - 17.10.11.10 Navigate to **Menu -> Data Protection -> DLP Policy Manager**.
 - 17.10.11.11 Select the **GEV Removable Storage Device Rule**.
 - 17.10.11.12 Select the **Device Control** tab.
 - 17.10.11.13 Select the **GEV Block Removable Device Rule** under the **Device Control** tab.
 - 17.10.11.13.1 Select the **Exceptions** tab.
 - 17.10.11.13.2 Select the Excluded Serial Number & User Pairs.

17.10.11.13.3 Set the following settings:

- 17.10.11.13.3.1 Description: **Allowed USB Devices**
- 17.10.11.13.3.2 Set State to **Enabled**.
- 17.10.11.13.3.3 Select the browse in the Serial Number & User Pairs.
- 17.10.11.13.3.4 Check the **GEV_Validation_USB** and click **Edit**
- 17.10.11.13.3.5 In the **Device Serial Number** field -> paste the USB serial number was copied earlier.
- 17.10.11.13.3.6 Select **Add**. The device will appear in the Entries box.
- 17.10.11.13.3.7 Select **Save** in lower right.
- 17.10.11.13.3.8 Verify the **GEV_Validation_USB** checkbox is checked and select **OK**.

17.10.11.13.4 Select the **Save** button on the lower right on the **Exceptions** tab.

17.10.11.14 Select the **GEV Block Plug and Play** under the **Device Control** tab.

17.10.11.14.1 Select the **Exceptions** tab.

17.10.11.14.2 Select the Excluded Serial Number & User Pairs.

17.10.11.14.3 Set the following settings:

- 17.10.11.14.3.1 Description: **Allowed USB Devices**
- 17.10.11.14.3.2 Set State to **Enabled**.
- 17.10.11.14.3.3 Select the browse in the Serial Number & User Pairs.
- 17.10.11.14.3.4 Check the **GEV_Validation_USB** and select **OK**.

17.10.11.14.4 Select **Save**.

17.10.11.14.5 Select the **Save** button on the lower right on the **Exceptions** tab.
Click **Close**.

17.10.11.15 Navigate to **Menu** -> **Policy Catalog**.

17.10.11.16 Select the **Data Loss Prevention 11.x** from the **Product** drop-down menu.

17.10.11.17 Select **GEV Block All Devices** policy and click **Edit**.

17.10.11.18 Verify that Pending Changes states **Yes** for the Device Blocking Rule Set.

17.10.11.19 Select **Apply policy** on lower right.

17.10.11.20 Verify that policy was applied Successfully -> Select **OK**.

17.10.11.21 Wait for policy to be applied to computer or use **Wake up agent** -> check the box for **Force complete policy and task update** in system tree to force policy.

17.10.11.22 Test on local computer that USB device is now allowed.

17.11 Verify DLP Functionality

17.11.1 Verify Device Control Policy

17.11.1.1 This task is to verify a policy exists to control USB Devices in Trellix ePO in SP1 Server:

17.11.1.2 Log into the SP1 server using an administrator account and launch the ePO web management interface shortcut on the desktop.

17.11.1.3 Log in using the ePO administrator account.

17.11.1.4 Go to **Menu -> Policy Catalog**. Verify the Policy **GEV Block All Devices** exists.

17.11.2 Verify USB Device Access

17.11.2.1 This task is to verify you can access USB devices in HMI/Server with the correct user account:

17.11.2.2 Log into an HMI/Server using the **Admin** account.

17.11.2.3 Insert any encrypted USB to be tested.

17.11.2.4 Verify under this account you can access the USB and its contents.

17.11.3 Verify USB Device Restrictions

17.11.3.1 This task is to verify access to USB devices is restricted for users without authorization:

17.11.3.2 Log into an HMI using the **GEMaint** account. Insert an encrypted USB to be tested.

17.11.3.3 Under this account, a pop up should appear explaining *DLP has blocked the device*.

17.11.3.4 Attempt to access files on the device and verify an **Access Denied** message is received when trying to get the USB's contents.

17.11.3.5 Log into the SP1 server using an administrator account.

17.11.3.6 Launch the ePO web management interface shortcut on the desktop.

17.11.3.7 Log in using the ePO administrator account.

17.11.3.8 Go to **Menu -> Policy Catalog -> Data Loss Prevention 11.X -> GEV Block All Devices**.

17.11.3.9 Select **Edit -> GEV Removable Storage Device Rule -> Device Control -> GEV Block Removable Device Rule**.

17.11.3.10 Select **Exceptions** tab and choose **Excluded Users**.

17.11.3.11 Verify that the **GEMaint** account does not have a USB exception.

17.11.4 **Verify USB Device Restrictions Event on LM1 (Splunk)**

- 17.11.4.1 Log into the SP1 server using an administrator account and launch the ePO web management interface shortcut on the desktop.
- 17.11.4.2 Log in using the ePO administrator account.
- 17.11.4.3 Go to **Menu -> Automation -> Server Tasks**.
- 17.11.4.4 Select **Run** on the **DLP Block Events** task.
- 17.11.4.5 Verify the **DLP Block Events** task status is **Completed**.
- 17.11.4.6 Log into the LM1 server using an administrator account and launch the Splunk web management interface shortcut on the desktop.
- 17.11.4.7 Select **OTArmor – Visibility** in the menu.
- 17.11.4.8 Select **Endpoint Protection**.
- 17.11.4.9 Verify that the **DLP** counter increments.
- 17.11.4.10 Click on the **DLP Events** number and verify the **DLP Event Detail** displays the USB block event that was generated in the previous section.

17.12 **Application Control**

17.12.1 **Import Solidcore Rules**

- 17.12.1.1 Login into **SP1**.
- 17.12.1.2 Open **Trellix ePO (ePolicy Orchestrator)** desktop icon and login as admin.
- 17.12.1.3 Navigate to the Menu – “**Configuration**” – “**Solidcore Rules**”.
- 17.12.1.4 Select “**Import**” (bottom left).
- 17.12.1.5 Select “**Choose File**” on the “**Import Rule Groups**” window.
- 17.12.1.6 Browse to the location of the Solidcore rules and select the following rule group:
- 17.12.1.7 Select the “**GEV_Custom_Rules_A.xml**” and select “**Open**”.
- 17.12.1.8 Select “**Overwrite duplicate groups**”.
- 17.12.1.9 Select “**OK**” on the “**Import Rule Groups**” window.
- 17.12.1.10 Verify that rule group imported successfully and select “**OK**”.
- 17.12.1.11 Verify that **.GEV-Custom-Rules** is listed.
- 17.12.1.12 Select the **.GEV-Custom-Rules** and click “**Edit**”.
- 17.12.1.13 Select the “**Certificates**” tab -> verify there are two certificates listed for “**GEV_Cimply**” and “**GEV_Toolbox/Workstation**”.

17.12.1.14 Select “**Cancel**”.

17.12.2 After deploying the agent. Verify the **admin** status: Enabled Solidified and Attached state on each PC.

17.12.2.1 Login into SP1.

17.12.2.2 Open **Trellix ePO (ePolicy Orchestrator)** desktop icon and login as admin.

17.12.2.3 Click on **System Tree** -> select an endpoint.

17.12.2.4 In the **Products** tab -> click on **Solidcore**.

17.12.2.5 Scroll down to **Solidification Status** -> verify its **Solidified**.

Note: If status is Unsolidified that means there are no files on the drive.

17.12.2.6 Scroll up to **General** -> verify **Memory Protection** is **Disabled**.

17.12.2.7 Repeat steps above for every endpoint.

17.12.2.8 If the Solidification Status is **not Solidified** -> enable Solidcore.

17.12.3 **Test the following Solidcore Tasks**

17.12.3.1 Open **Trellix ePO (ePolicy Orchestrator)** desktop icon and login as admin.

17.12.3.2 Navigate to **System Tree**-> select/check a **server or workstation**.

17.12.3.3 Select **Actions** -> **Agent** -> **Run Client Task Now**.

17.12.3.4 In the Product list -> select **Solidcore 8.x.x**.

17.12.3.5 In the Task Type -> select **SC:Run Commands**.

17.12.3.6 In the Task Name -> select **Disable Deny-Write**.

17.12.3.6.1 Click **Run Task Now**.

17.12.3.6.2 Wait for the task to complete.

17.12.4 **(If necessary) (Re)Solidifying**

Note: It's recommended to Re-Solidify at the end of the day.
The Re-Solidify task may take a while to run on each machine.

17.12.4.1 Open **Trellix ePO (ePolicy Orchestrator)** desktop icon and login as admin.

17.12.4.2 Navigate to **System Tree**-> select/check all **servers and workstations**.

17.12.4.3 Select **Actions** -> **Agent** -> **Run Client Task Now**.

17.12.4.3.1 In the Product list -> select **Solidcore 8.x.x**.

17.12.4.3.2 In the Task Type -> select **Run Commands**.

17.12.4.3.3 In the Task Name -> click **Create New Task**.

17.12.4.3.4 In Run Commands -> enter **sadmin so**

17.12.4.3.5 Click **Run Task Now**.

17.12.4.4 To check the Solidification Status on the machine -> login and right-click the **Trellix tray icon**.

17.12.4.4.1 Select **Quick Settings** -> **Application Control Solidification Status**.

17.12.5 **Review HMI Programs blocked by Application Control**

IMPORTANT

Verify the list of programs blocked in every HMI and OTArmor Server. The TCW, Cimplicity, CatTools, etc. **SHOULD NOT BE BLOCKED**.

Also, verify screen and TCW functionality after enabling Application Control.

17.12.5.1 **Add CatTools Service to Solidcore**

17.12.5.1.1 Login to **SP1** -> open **Windows PowerShell (Admin)**.

17.12.5.1.2 Enter the following: **Get-FileHash \\DR1\C\$\Program Files (x86)\CatTools3\CatTools_Service.exe**

17.12.5.1.3 Hit **Enter**.

17.12.5.1.4 The **SHA-256 hash** will be listed.

17.12.5.1.5 Highlight the number -> click **ctrl+C to copy**.

17.12.5.1.6 Open **Trellix ePO (ePolicy Orchestrator)** desktop icon and login as admin.

17.12.5.1.7 Open **Menu** -> **Configuration** -> **Solidcore Rules**.

17.12.5.1.8 Select/check the **.GEV-Customer-Server-Rules** -> click **Edit**.

17.12.5.1.9 Select the **Updater Processes** tab -> click **Add**.

17.12.5.1.10 Select radio button for **Updater By File SHA-256**.

17.12.5.1.11 In the **File SHA-256** -> **Paste** the copied hash number.

17.12.5.1.12 Enter the Updater Label as: **CatTools_Service**

17.12.5.1.13 Click **OK**.

17.12.5.1.14 Click **Save Rule Group**.

17.12.5.1.15 Click the **System Tree** -> in the **Preset** -> select **This Group and All Subgroups**.

17.12.5.1.16 > Select/check the **DR1 Server**.

17.12.5.1.17 Click **Wake Up Agents**.

17.12.5.1.18 Select/check **Force complete policy and task update**.

17.12.5.1.19 Click **OK**.

17.12.5.1.20 Login to **DR1** -> open **CatTools**.

17.12.5.1.21 Run a **backup task** and verify the service is **not blocked** by Solidcore.

17.12.5.2 **Add PowerShell Service to Solidcore**

17.12.5.2.1 Open **Trellix ePO (ePolicy Orchestrator)** desktop icon and login as admin.

17.12.5.2.2 Open **Menu -> Configuration -> Solidcore Rules**.

17.12.5.2.3 Select/check the **.GEV-Custom-Rules** -> click **Edit**.

17.12.5.2.4 Select the **Filters** tab -> expand **Policy Discovery & Events**.

17.12.5.2.5 Click **Add Rule**.

17.12.5.2.6 Enter the following information:

17.12.5.2.7 File – ends with - **.ps1**

17.12.5.2.8 Click + button.

17.12.5.2.9 File – contains – **Windows\SystemTemp**

17.12.5.2.10 Click + button.

17.12.5.2.11 Program – ends with – **powershell.exe**

17.12.5.2.12 Select/check the box **Apply rule to events also**.

17.12.5.2.13 Click **Save Rule Group**.

17.12.5.2.14 Open **Menu -> Configuration -> Solidcore Rules**.

17.12.5.2.15 Select/check the **.GEV-Custom-Server-Rules** -> click **Edit**.

17.12.5.2.16 Select the **Filters** tab -> expand **Policy Discovery & Events**.

17.12.5.2.17 Click **Add Rule**.

17.12.5.2.18 Enter the following information:

17.12.5.2.19 File – ends with - **.ps1**

17.12.5.2.20 Click + button.

17.12.5.2.21 File – contains – **Windows\SystemTemp**

17.12.5.2.22 Click + button.

17.12.5.2.23 Program – ends with – **powershell.exe**

17.12.5.2.24 Select/check the box **Apply rule to events also**.

17.12.5.2.25 Click **Save Rule Group**.

17.12.5.2.26 Click the **System Tree** -> in the **Preset** -> select **This Group and All Subgroups**.

17.12.5.2.27 Select/check **ALL Servers**.

17.12.5.2.28 Click **Wake Up Agents**.

17.12.5.2.29 Select/check **Force complete policy and task update**.

17.12.5.2.30 Click **OK**.

17.12.5.2.31 Verify the pop-up block for Application Control Events does not occur for C:\Windows\SystemTemp files.

17.12.5.3 Adding Cimplicity and ToolboxST to Solidcore

17.12.5.3.1 In the HMI, right-click on the **Trellix icon** (shield) in the system tray (taskbar) -> select **Quick Settings** -> **Application Control events**.

17.12.5.3.2 Review the files being blocked, verify which files need to be added to the allow list for exception (ex: CIMPLICITY\EXE\examst.exe.) (looking at local machine).

17.12.5.3.3 Login to **SP1**. Open Trellix ePO (ePolicy Orchestrator).

17.12.5.3.4 Navigate to **Menu -> Reporting -> Solidcore Events -> Workstations** -> select the event corresponding to the machine.

17.12.5.3.5 In the **Solidcore Events** page select -> **Create policy** located at the bottom of the page -> select **Role Group** -> Choose **existing** -> **Site custom rules** -> click **Save**.

17.12.5.3.6 Under **Menu -> Configuration -> Solidcore Rules -> .GEV-Custom-Rules** -> select **Edit** -> Verify the above policy was created.

17.12.5.3.7 Click on the **System Tree** icon in the menu & icon area at the top of the page.

17.12.5.3.8 Select the **Workstation** in the tree view -> click the **Systems** tab.

17.12.5.3.9 Change the preset: **select This group and all subgroups**.

17.12.5.3.10 Select the endpoint you wish to send the **Wake-up Agent** command.

17.12.5.3.11 Click on **Wake-up Agents**.

17.12.5.3.12 Select **Force complete policy and task update**.

17.12.5.3.13 Click **OK**

- 17.12.5.3.14 Navigate to **Menu -> Automation -> Server Task Log** -> verify wake up agent completed.
- 17.12.5.3.15 On the target HMI -> close all windows.
- 17.12.5.3.16 Logout of the HMI and login again using the **GEMaint** credentials.
- 17.12.5.3.17 Open ***.TCW** ToolboxST.
- 17.12.5.3.18 Select **File -> Open HMI workstation -> Build -> Download**.
- 17.12.5.3.19 Under Trellix Icon Tray-> **Application Control Events** -> verify no events are listed for ToolboxST.
- 17.12.5.3.20 Open **Site -> Screens -> CimLayout Configuration** (double-click).
- 17.12.5.3.21 Under Trellix Icon Tray-> **Application Control Events** -> verify no events are listed for Cimplicity.
- 17.12.5.3.22 Check application control events on every machine.

17.12.6 **Change Status:**

17.12.6.1 Make sure you can change status for Solidcore:

17.12.6.2 **Observe:**

- 17.12.6.2.1 Login to SP1 server and open Trellix ePO (ePolicy Orchestrator) -> login as Admin.
- 17.12.6.2.2 Select System Tree -> My Organization -> on the System tab -> select Preset drop-down -> select Computers and Subgroups.
- 17.12.6.2.3 In the list of Computers -> select/check each Computer -> navigate to **Actions-> Agent -> Run client task now**.
- 17.12.6.2.4 Select **Solidcore 8.x.x -> SC: Observer mode -> Begin observer mode -> click Run task now**.
- 17.12.6.2.5 At the HMI, open **Command Prompt (Admin)** or **Trellix About** to verify the change into Observe Mode.
 - 17.12.6.2.5.1 **sadmin status**
 - 17.12.6.2.5.2 **Trellix solidifier: Observe**
 - 17.12.6.2.5.3 **Trellix solidifier on reboot: Observe**
- 17.12.6.2.6 After verifying Observe Mode -> remove it from solidifier state.
- 17.12.6.2.7 Navigate back to SP1->ePO Console->Actions -> Agent -> **Run client task now -> Solidcore 8.x.x -> SC: Observer mode -> select End observe mode -> click Run task now**.
- 17.12.6.2.8 Check sadmin status to verify the state ended.

17.12.6.3 **Update:**

- 17.12.6.3.1 Login to SP1 server and open Trellix ePO (ePolicy Orchestrator) -> login as Admin.
- 17.12.6.3.2 Select System Tree -> My Organization -> on the System tab -> select Preset drop-down -> select Computers and Subgroups.
- 17.12.6.3.3 In the list of Computers -> select/check each Computer -> navigate to **Actions-> Agent → Run client task now.**
- 17.12.6.3.4 Select **Solidcore 8.x.x → SC: Begin Update mode → Create new task -> click Run task now.**
- 17.12.6.3.5 Open **Command Prompt (Admin)** or **Trellix About** to verify the change into **Update Mode.**
 - 17.12.6.3.5.1 **sadmin status**
 - 17.12.6.3.5.2 **Trellix solidifier: Update**
 - 17.12.6.3.5.3 **Trellix solidifier on reboot: Update**
- 17.12.6.3.6 After verifying Update Mode -> remove it from solidifier state.
- 17.12.6.3.7 Navigate back to SP1->ePO Console-> Actions -> Agent -> **Run client task now -> Solidcore 8.x.x → SC: Update mode -> select end update mode -> click Run test now.**
- 17.12.6.3.8 Check sadmin status to verify the state ended.

17.12.6.4 **Disable/Enable:**

- 17.12.6.4.1 Login to SP1 server and open Trellix ePO (ePolicy Orchestrator) -> login as Admin.
- 17.12.6.4.2 Select System Tree -> My Organization -> on the System tab -> select Preset drop-down -> select Computers and Subgroups.
- Note:** Testing the “Disable” action will require a reboot of the HMI.
- 17.12.6.4.3 In the list of Computers -> select/check ONE Computer (that can be rebooted).
- 17.12.6.4.4 Navigate to **Actions-> Agent → Run client task now.**
- 17.12.6.4.5 Select **Solidcore 8.x.x → SC: Disable → 04-Disable application control -> click Run task now.**
- 17.12.6.4.6 After verifying Disable mode -> run client task to Enable.
- 17.12.6.4.7 Navigate back to SP1-> ePO Console -> Actions -> Agent → **Run client task now → Solidcore 8.x.x → SC: Enable → 02-Enable application control -> click Run task now.**
- 17.12.6.4.8 Reboot after enabling.

17.12.6.5 **Lockdown Mode:**

- 17.12.6.5.1 Login to SP1 server and open Trellix ePO (ePolicy Orchestrator) -> login as Admin.
- 17.12.6.5.2 Select System Tree -> My Organization -> on the System tab -> select Preset drop-down -> select Computers and Subgroups.
- 17.12.6.5.3 In the list of Computers -> select/check each Computer -> navigate to **Actions-> Agent -> Run client task now.**
- 17.12.6.5.4 Select **Solidcore 8.x.x -> SC: Change local CLI access -> Create new task -> Allow -> click Run task now.**
- 17.12.6.5.5 At the HMI, open **Command Prompt (Admin)** and type: **admin status** (check the result):
 - 17.12.6.5.5.1 Local CLI access: **Recover**
Important: Make sure to put the system back in **LOCKDOWN** mode.

17.13 Configure Trellix Web Certificate

Note: Refer to **GEH-6889 Section 3.8** to Renew or Configure the Trellix Web Certificate.

18.0 Trellix SIEM – OTArmor

18.1 (FUTURE) Generate Trellix SIEM Certificate

18.1.1 Add the Trellix SIEM to DNS

18.1.2 Login to **DC1**.

18.1.3 Open the **DNS Manager**.

18.1.4 Under **Forward Lookup Zones** -> select the **HMI.local**.

18.1.5 Right-click the **HMI.local zone** -> select **New Host (A or AAAA) ...**

18.1.6 In the New Host window -> enter the **Trellix SIEM Hostname** -> enter the **IP Address**.

18.1.7 Select/check the box for **Create associated pointer (PTR) record**.

18.1.8 Select **Add Host**.

18.1.9 Generate certificates using (Create-GEV-Cert.ps1) script

18.1.10 Login to **CA1**.

18.1.11 Open **Windows PowerShell as Admin**.

18.1.12 Navigate to the following path: **cd C:\CertificateSigning**

18.1.13 Enter the following command: **.\Create-GEV-Cert.ps1**

18.1.14 Hit **Enter** to continue.

18.1.15 At the CSR Request prompt -> Enter **"2"**.

- 18.1.16 At the Generate a non-encrypted private key -> Enter “y”.
- 18.1.17 Hit **Enter** to continue.
- 18.1.18 At the FQDN -> enter the **Trellix SIEM FQDN** (example: SIEMx.HMI.local).
- 18.1.19 At the Common Name -> enter the **Trellix SIEM Hostname** (example: SIEMx).
- 18.1.20 At the Devices IP Address -> enter the **Trellix SIEM IP Address** (example: 172.16.xxx.xxx).
- 18.1.21 At the Password for private key file -> hit **Enter** (leave blank, no password).
- 18.1.22 Verify the command completed successfully.
- 18.1.23 The certificates will be created in the following directory:
C:\CertificateSigning\SIEMx.
- 18.1.24 Login to the **Trellix SIEM** web console.
- 18.1.25 Click the **Menu** -> select **System Properties** -> select **ESM Management**.
- 18.1.26 Select the **Key Management** tab -> click **Certificate**.
- 18.1.27 Click **Upload** -> click **OK** on the pop-up.
- 18.1.28 Upload the .crt file -> browse to the CA1 - **C:\CertificateSigning\SIEMx folder** -> select the .crt file -> click **Open**.
- 18.1.29 Upload the .key file -> browse to the CA1 - **C:\CertificateSigning\SIEMx folder** -> select the .key file -> click **Open**.
- 18.1.30 Optional: Chain files > select **No**.
- 18.1.31 Click **Close**.

18.2 (If needed) How to Shut down or Reboot the Trellix SIEM

IMPORTANT: Failure to properly Shutdown or Reboot of the Trellix SIEM will cause a corruption of the database requiring a Trellix support ticket to resolve.

18.2.1 Follow the steps below to properly shut down the Trellix SIEM.

- 18.2.1.1 Login to vSphere -> select the “**Trellix SIEM**” VM.
- 18.2.1.2 Click to open the Guest OS console for Trellix SIEM VM.
- 18.2.1.3 Login to the Trellix SIEM VM (Username: **root**, Password: **See Default Passwords Document*).
- 18.2.1.4 At the prompt enter the following:
- 18.2.1.5 To Shutdown -> “Enter Choice>” **4**
- 18.2.1.6 Hit Enter.
- 18.2.1.7 OR

- 18.2.1.8 To Reboot -> "Enter Choice>" **3**
- 18.2.1.9 Hit Enter.
- 18.2.1.10 Wait 5-10 minutes for the Shutdown or Reboot to complete.

19.0 Trend Micro Apex One and Apex Central – OTArmor

19.1 Create List Device Folder

- 19.1.1 Open the SP2 Server.
- 19.1.2 Navigate to the **C: Drive** -> create a new folder called **ListDeviceInfo**.
- 19.1.3 Navigate to: **SP3\C\$\Program Files (x86)\Trend Micro\Apex One\PCCSRV\Admin\Utility>ListDeviceinfo**
- 19.1.4 Copy the **listDeviceInfo.exe**.
- 19.1.5 Navigate back to the **C:\ListDeviceInfo** folder and paste the .exe file.

19.2 Apex Central Dashboard

- 19.2.1 Open the SP2 Server.
- 19.2.2 Login to the **Apex Central** Console (Username: **admin**, Password: **See Default Passwords Document* or log in using domain credentials).
- 19.2.3 On the **Dashboard** tab -> select **Security Posture** -> click the ellipses icon (far right) select **Widget Settings**.
 - 19.2.3.1 In the **Active Directory grouping** drop-down -> select **Site**.
 - 19.2.3.2 Click **Save**.
- 19.2.4 Navigate to **Administration** -> **Settings** -> **Active Directory and Compliance Settings**.
 - 19.2.4.1 Select the **Sites** tab -> click **Add Custom**.
 - 19.2.4.1.1 In the **Display name**: enter **Customer Site Name**.
 - 19.2.4.1.2 In the **IP Address range**: enter Network Topology IP range (e.g. 172.16.201.0/24 (PDH ONLY)) or enter each IP address separately (line by line).
 - 19.2.4.2 Click **Save**.
- 19.2.5 Navigate back to **Dashboard** tab -> verify the Customer Site Name (dot) is now showing in the **Security Posture** dashboard.

19.3 Change Owner on Apex Central Policies

- 19.3.1 Open the SP2 Server.
- 19.3.2 Login to the **Apex Central** Console (Username: **admin**, Password: **See Default Passwords Document*).
- 19.3.3 Select **Policies** -> **Policy Management**.

19.3.4 Select all policies and then select **Change Owner**.

19.3.5 In the **Change Policy Owner** dropdown, select **HMI\Trend Micro Administrators** and then select **Save**.

19.4 Install Security Agent on Workstation (HMI)

19.4.1 Log into a Workstation.

19.4.2 Navigate to the SP2 Network Share: **\\SP2\TM_Agent**

Note: the **\\SP2** server name may vary based on customer site.

19.4.3 In the **TM_Agent** folder -> right-click the **agent_cloud_x64** Windows installer.

19.4.3.1 Click **Install**.

19.4.3.2 Select **Run** on the pop-up window.

19.4.3.3 On the **Trend Micro Security Agent** window -> click **Next**.

19.4.3.4 The **Installing Trend Micro Security Agent** status window will appear.

19.4.3.4.1 Click **Yes** on the UAC prompt.

19.4.3.4.2 Click **Yes** on the second UAC prompt.

19.4.3.4.3 Click **Yes** on the allow app UAC prompt.

19.4.3.5 The installation will begin.

19.4.3.6 When the installation is complete -> click **Finish**.

19.4.3.7 A **Trend Micro Apex One** prompt will appear to **Restart Computer**.

19.4.3.7.1 Click **Restart**.

19.4.4 Log back into the machine.

Note: The agent (tray icon) may disappear and reappear during updates or as it makes connection to the SP2 server. DO NOT uninstall or reconfigure the agent, wait a few minutes for the agent to return online.

19.4.4.1 In the menu tray -> click the arrow to open all tray icons.

19.4.4.1.1 Right-click the **Trend Micro Security Agent** icon.

19.4.4.1.2 Select **Open Security Agent Console**.

19.4.4.1.3 Click the connection link (green dot bottom right) and verify the Agent is connected to the **Apex One Server** (SP3.HMI.local).

19.5 Verify Workstation Security Agent Status – See Trend Micro GEH.

19.6 Add Splunk Application Control Rules

19.6.1 Open the **SP2 Server**.

19.6.2 Login to the **Apex Central** Console (Username: **admin**, Password: **See Default Passwords Document* or log in using domain credentials).

19.6.3 Select **Policies -> Policy Resources -> Application Control Criteria**.

19.6.4 Add Microsoft Rule.

19.6.4.1 Select **Add Criteria -> Allow**.

19.6.4.1.1 Enter Name: **Allow Microsoft**

19.6.4.1.2 Trust Permission: **Application can execute other processes**

19.6.4.1.3 Match Method: **Certificates**

19.6.4.1.4 Under Specify certificate type: -> select **Trusted (valid or expired)**.

19.6.4.1.5 Under Certificate properties: -> select **Issuer Organization (O)** and **Microsoft Corporation**.

19.6.4.2 Click **Save**.

19.6.4.3 Minimize the SP2 Server window.

19.6.5 Login to the **LM1 Server** -> open **Administrative PowerShell**.

19.6.5.1 Enter: **Get-FileHash D:\security\bin\splunkd.exe | Out-File -FilePath C:\splunk.txt -Append**

19.6.5.2 Enter: **Get-FileHash D:\security\bin\splunk-powershell.exe | Out-File -FilePath C:\splunk.txt -Append**

19.6.5.3 Copy/paste the splunk.txt file to **\\SP2\C\$**.

19.6.5.4 Close the LM1 Server window.

19.6.6 Open the **SP2 Server**.

19.6.7 In the Apex Central Console navigate to **Policies -> Policy Resources -> Application Control Criteria**.

19.6.8 Add Splunk PS Rule.

19.6.8.1 Select **Add Criteria -> Allow**.

19.6.8.1.1 Enter Name: **Allow Splunk PS**

19.6.8.1.2 Trust Permission: **Inheritable execution rights (not recommended)**

19.6.8.1.3 Match Method: **Hash values**

19.6.8.1.4 Under Input method: -> select **Manual**.

19.6.8.1.5 Select **SHA-256**.

19.6.8.1.6 Open the saved hash file on the C: drive and copy the hash values.

19.6.8.1.7 Paste the first SHA-256 hash value for **splunkd.exe** -> enter the name splunkd.exe.

19.6.8.1.8 Click the + button.

19.6.8.1.9 Paste the second SHA-256 hash value for **splunk-powershell.exe** -> enter the name splunk-powershell.exe.

19.6.8.1.10 Click **Save**.

19.6.8.2 Add Block Gray Software Rule.

19.6.8.3 Select **Add Criteria** -> **Block**.

19.6.8.3.1 Enter Name: **Block Gray Software**

19.6.8.3.2 Uncheck Enable assessment mode.

19.6.8.3.3 Match Method: **Gray Software List**

19.6.8.3.4 Select **Save**.

19.7 Add the Rules to both the Security and Workstation Agent Policy

19.7.1 Server Security Agent Policy -> expand the **Application Control Settings**.

19.7.1.1 Click on **All user Accounts**.

19.7.1.2 Available Criteria -> move the following rules to **Selected Criteria** window.

19.7.1.2.1 Allow Microsoft

19.7.1.2.2 Allow Splunk PS

19.7.1.2.3 Block Gray Software

19.7.1.3 Click **OK**.

19.7.1.4 Scroll down and click **Deploy**.

19.7.1.5 Repeat steps for the Workstation Security Agent Policy.

19.8 Deploy the Workstation Security Agent Policy to Workstation(s).

19.8.1 Open the **SP2 Server**.

19.8.2 Login to the **Apex Central** Console (Username: **admin**, Password: **See Default Passwords Document*).

19.8.3 Select **Policies** -> **Policy Management**.

19.8.4 Click on the **Workstation Security Agent Policy**.

19.8.4.1 Click **Manage Targets**.

19.8.4.2 Select the **Browse** tab -> select dropdown and select: **Product Directory** OR **Active Directory**.

19.8.4.2.1 In the **Product Directory**:

19.8.4.2.1.1 Expand the **SP2 folder**.

- 19.8.4.2.1.2 Expand the **Local Folder**.
- 19.8.4.2.1.3 Expand the **Apex One** -> **Apex One manager** -> **Domain name folder**.
- 19.8.4.2.1.4 Select/check the Workstations showing Policy Status = **Without policy**.
- 19.8.4.2.1.5 Verify the machines selected are Workstations (HMI's) NOT servers.
- 19.8.4.2.1.6 Click **Add Selected Targets** -> click **OK**.
- 19.8.4.3 (Scroll down) Click **Deploy** (on the edit policy window).
- Note:** Refresh the policy management window to verify the policy has been deployed. It may take a few minutes for the policy to deploy.

19.9 Verify the Agent Policy Matches the Apex Central Policy

- 19.9.1 Login to a workstation with the Apex Security Agent.
- 19.9.2 Right-click the Security Agent tray icon (bottom right) -> select **Component Versions**.
- 19.9.3 In the top panel -> scroll down and verify the policy name matches the policy deployed from Apex Central (e.g. Workstation Security Agent Policy)
- 19.9.4 Click the information icon (i) next to the name and verify the policy number matches the policy deployed from Apex Central.
- 19.9.5 Repeat these steps for every workstation.

Note: The agent (tray icon) may disappear and reappear during updates or as it makes connection to the SP2 server. DO NOT uninstall or reconfigure the agent, wait a few minutes for the agent to return online.

19.10 Deploy the EICAR Test Virus

- 19.10.1 Log into a Workstation (with the Apex Security Agent) using the admin account.
- 19.10.2 Create the **EICAR test file**.
- 19.10.3 Open Notepad as an Administrator.
- 19.10.4 Type or paste in the following text:
X5O!P%@AP[4\PZX54(P^)7CC)7}\$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!\$H+H*
- 19.10.5 Save the file as **eicar.com**.
- 19.10.6 Verify the Workstation displays a **Threat/Violations Found** message.
- 19.10.7 Click on detection count and verify that the threat has been quarantined.
- 19.10.8 Login to the **SP2 Server**.
- 19.10.9 Double-click the **Trend Micro Apex Central** desktop shortcut.

- 19.10.9.1 Login to the console -> select **Log On with Domain Credentials** (HMI\Admin, **See Default Passwords Document*).
- 19.10.9.2 Navigate to **Detections -> Logs -> Log Query**.
- 19.10.9.3 Select **Virus/Malware detections**
- 19.10.9.4 Select **All products**
- 19.10.9.5 Select **Last 24 hours**
- 19.10.9.6 Select **Search** to display results.

Note: It may take some time for the Virus/Malware indication to display in the log.

- 19.10.10 Login to the **LM1 Server**.
- 19.10.11 Double-click the **Splunk** desktop shortcut.
- 19.10.12 Login to the console as Admin (**See Default Passwords Document*).
- 19.10.13 Select the **Search & Reporting** app.

Note: The default syslog setting in Apex Central is set to forward logs every 1 hour. For testing purposes, the schedule can be changed to 2 minutes to verify the log is received by Splunk. If changed, verify the syslog setting schedule is set back to the configured time of 1 hour.

- 19.10.14 Enter the following in the search box: **eicar** or **Eicar_test_file**
- 19.10.15 Click **Search**.
- 19.10.16 Verify the EICAR test file has logged to Splunk.

19.11 Update Antivirus Patterns – See Trend Micro GEH.

19.12 Test Application Control

This task is to verify Application control is working properly.

- 19.12.1 Log into a server or workstation (with the Apex Security Agent) using the admin account.
- 19.12.2 Record the name of the server under **Comments & Observations**. Make sure the HMI/Server is not under any Patching Policy.
- 19.12.3 Navigate to **C:\Staples USB** on B-DR1.
- 19.12.4 Copy it to the C: Drive of the HMI/Server.
- 19.12.5 Double-click the **Staples USB.exe** -> verify the application is blocked (Policy Violation pop-up should appear).
- 19.12.6 Login to the **SP2 Server**.

- 19.12.7 Double-click the **Trend Micro Apex Central** desktop shortcut.
- 19.12.8 Login to the console -> select **Log On with Domain Credentials**.
- 19.12.9 Click on the **Detections** tab -> select **Log – Log Query**.
- 19.12.10 In the drop-down -> select **Application Control violations** -> click **OK** -> click **Search**.
- 19.12.11 Verify the event details.
 - Note:** Application Control events may take some time to appear in the log. Wait and recheck the query until the event(s) appear.
 - 19.12.11.1 Export the application control violations to .csv -> select **Export to CSV**.
 - 19.12.11.2 Open the exported file and copy the **SHA-256** hash value.
- 19.12.12 Create a **PuTTY Policy**.
 - 19.12.12.1 Navigate to **Policies -> Policy Resources -> Application Control Criteria**.
 - 19.12.12.2 Select **+ Add Criteria -> Allow**.
 - 19.12.12.3 Allow Criteria Settings
 - 19.12.12.3.1 Name: **Allow PuTTY**
 - 19.12.12.3.2 Trust Permission: **Application cannot execute external processes**
 - 19.12.12.3.3 Match Method: **Hash Values**
 - 19.12.12.3.4 Input Method: **Manual**
 - 19.12.12.3.5 Select **SHA-256**.
 - 19.12.12.3.6 In the hash value field -> paste the hash value from the csv export.
 - 19.12.12.3.7 In the description field -> enter **PuTTY**.
 - 19.12.12.3.8 Select **Save**.
 - 19.12.12.4 Navigate to **Policies -> Policy Management**.
 - 19.12.12.4.1 Select the either the **Server Security Agent Policy** or **Workstation Security Agent Policy**.
 - Note:** Select the policy that matches the system the event was generated on.
 - 19.12.12.4.2 Expand the **Application Control Settings**.
 - 19.12.12.4.3 Select **All user accounts**.
 - 19.12.12.4.4 Assign Rule Settings.
 - 19.12.12.4.5 Select **Allow PuTTY** and verify it moves to the **Selected criteria** window.
 - 19.12.12.4.6 Select **OK**.
 - 19.12.12.4.7 Select **Deploy** to deploy policy to endpoints and wait for policy to be deployed.

19.12.12.4.8 Note the policy version of the policy that was deployed under **Observation & Comments** below.

19.12.12.5 Log into the endpoint where the application violation was generated.

19.12.12.5.1 Check policy version.

19.12.12.5.2 Right-click on the Trend Micro icon in the task bar and select **Component Versions**.

19.12.12.5.3 Verify that the Apex Central policy version matches the policy number noted in Apex Central.

Note: Hover over the I at the end of the policy name to get version.

19.12.12.6 Verify that the blocked application now can run.

19.13 Application Control Log Lookup – See Trend Micro GEH.

19.14 Application Control Splunk Log

19.14.1 Login to the **LM1 Server**.

19.14.2 Double-click the **Splunk** desktop shortcut.

19.14.3 Login to the console as Admin (**See Default Passwords Document*).

19.14.4 Select the **Search & Reporting** app.

Note: The default syslog setting in Apex Central is set to forward logs every 1 hour. For testing purposes, the schedule can be changed to 2 minutes to verify the log is received by Splunk. If changed, verify the syslog setting schedule is set back to the configured time of 1 hour.

19.14.5 Enter the following in the search box: **application control, putty**

19.14.6 Click **Search**.

19.14.7 Verify the putty.exe test file has logged to Splunk with action Blocked.

19.15 Test Device Control – See Trend Micro GEH to configure Device Control Policy.

19.16 Test Patching – See Trend Micro GEH.

For additional SP2 Trend Micro Apex One and Apex Central configuration, please refer to the following document: **Trend Micro GEH OTArmor™ Cybersecurity Platform – Site Protection Maintenance Guide – Trend Micro Apex One & Apex Central**.

20.0 WhatsUp Gold – OTArmor

20.1 Verify Credentials – See **NCM10015**.

- 20.2 **Create a New Scan** – See **NCM10015.**
- 20.3 **Scan the Network** – See **NCM10015.**
- 20.4 **Verify Network Monitoring** – See **NCM10015.**

- 20.5 **Verify Switch Disconnection**
 - 20.5.1 Select a switch to disconnect.
 - 20.5.2 Verify the system triggers an alert for the disconnected device.
 - 20.5.3 Click in **Live Activity** -> review the results.
 - 20.5.4 Reestablish the connection to a different port.
 - 20.5.5 Proceed to rescan the system.
 - 20.5.6 The old connection should still show as down.
 - 20.5.7 Proceed to erase the connection -> rescan the system to update the asset.

- 20.6 **Verify Connection to SSH for Network Devices**
 - 20.6.1 Choose a switch under **My Network**.
 - 20.6.2 Right-click and choose **Tools** -> **SSH**.
 - 20.6.3 Enter Radius credentials to log into the switch.
 - 20.6.4 Make sure you can see the switch configuration.
 - 20.6.5 Exit the putty connection.

- 20.7 **Verify HMI Disconnection**
 - 20.7.1 Choose an HMI or OTArmor server -> proceed to shut it down.
 - 20.7.2 Verify the system triggers an alert for the device.
 - 20.7.3 Click **Live Activity** -> review the results.
 - 20.7.4 Reboot the device -> verify the system status returns.

- 20.8 **Verify Dashboard**
 - 20.8.1 Log into the NM1 server using an administrator account.
 - 20.8.2 Launch the web management interface by clicking the desktop shortcut.
 - 20.8.3 Log in using the admin account.
 - 20.8.4 Click on **Analyze->Dashboards** -> click on each option and review.
 - 20.8.5 Click on **Analyze->Performance** -> click on each option and review.
 - 20.8.6 Click on **Analyze->Network** -> click on each option and review.
 - 20.8.7 Click on **Analyze->Device** -> click on each option and review.
 - 20.8.8 Click on **Analyze->Alerts & Actions** -> click on each option and review.

20.8.9 Click on **Analyze->Inventory** -> click on each option and review.

20.8.10 See **WhatsUp Gold GEH** for additional Dashboard information.

21.0 **Nozomi Guardian - OTArmor**

21.1 **SPAN Ports for Nozomi**

Note: Typically, this is configured on the root bridge switches. Refer to the network topology for the location of the SPAN port connection.

21.1.1 Login to switch, and enter the following:

config t

monitor session 1 source vlan 101<space>,<space>172<space>,<space>201

monitor session 1 destination interface <gi1/0/7> rx (selected port)

Note: These commands are specific to Cisco switches.

21.2 **Update Definitions**

Note: To download Threat Intelligence files, please visit:

<https://nozominetworks.my.site.com/support/login?ec=302&startURL=%2Fsupport%2Fs%2F>

Steps to get the required files are:

21.2.1 **Threat Intelligence:**

21.2.1.1 Log into the **Nozomi Support Portal** (above).

21.2.1.2 Click **Product Releases**.

21.2.1.3 Click the **Threat Intelligence Update Package** at the top.

21.2.1.4 Select the correct **Threat Intelligence** bundle based on the Nozomi Guardian version.

21.2.1.5 Download the bundle.

21.2.1.6 Log in to Nozomi user interface from browser.

21.2.1.7 Go to **Administration -> Updates & Licenses -> Updates service configuration -> Manual contents upload**.

21.2.1.8 Browse to and select the Threat Intelligence update package to upload.

21.2.2 **Asset Intelligence:**

21.2.2.1 Log into the **Nozomi Support Portal** (above).

21.2.2.2 Click **Product Releases**.

21.2.2.3 Click the **Asset Intelligence Update Package** at the top.

21.2.2.4 Select the correct **Asset Intelligence** bundle based on the version of Nozomi Guardian.

- 21.2.2.5 Download the bundle.
- 21.2.2.6 Log in to Nozomi user interface from browser.
- 21.2.2.7 Go to **Administration -> Updates & Licenses -> Updates service configuration -> Manual contents upload**.
- 21.2.2.8 Browse to and select the asset intelligence update package to upload.

21.3 Malicious Signature Detection

Note: The signatures must be updated for this test to function correctly.

- 21.3.1 On the test machine -> open **Trellix Endpoint Security** from the tray icon.
- 21.3.2 Do the Administrator Log in and enter the password.
- 21.3.3 Settings -> **Threat Prevention** and uncheck **Enable On-Access Scan** and select **Apply**.
- 21.3.4 Create the **EICAR test file**.
- 21.3.5 Open Notepad as an Administrator.
- 21.3.6 Type or paste in the following text:
X5O!P%@AP[4\PZX54(P^)7CC)7}\$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!\$H+H*
- 21.3.7 Save the file as **eicar.txt**.
- Note:** The traffic must cross the root bridge for the Nozomi to detect.
- 21.3.8 Copy Antivirus file and proceed to copy through the network to another asset.
- 21.3.9 Log into the Nozomi device and select **Alerts** and search on **EICAR** in the description field of the alerts page.

22.0 Xona Critical Systems Gateway (CSG) – OTArmor

22.1 Configure Xona – See **PRO-153M6895**.

- 22.1.1 Please refer to the OTArmor Xona CSG Setup and Configuration for Xona CSG.

23.0 Yubikey – OTArmor

Note: Only deploy on physical HMIs and if a firewall is configured, open the ports as necessary.

Yubikey FIPS

Each OTArmor™ product being sold will now come with one Yubikey FIPS. For our customers to fully build in a Two factor authentication solution, they will need to purchase a Yubikey FIPS for each user. The Yubikey FIPS is a cryptographic device that meets FIPS 140-2 standards for Smart Card devices.

Yubikey Certificate Template

Each OTArmor™ Certificate Authority server should now contain a Smart Card template called Yubikey. The automated build process creates this certificate as of the fall 2019 release. The certificate template

along with the Yubikey Policy group policy, allow end users to enroll their Yubikey automatically. The template is configured for Auto Enrollment for domain authenticated users.

Yubikey Policy (Group Policy)

The Yubikey Policy (group policy) is designed to enable certificate enrollment for end users. This allows users who login without a YubiKey to enroll a new Yubikey device. If a user logs into an HMI, or OTArmor™ Server they will be prompted in the Windows system try to launch the auto enrollment for the Yubikey certificate. This process is enabled for authenticated domain users only and does not work for local windows accounts. Please refer to document **313T7465P0420AA** to edit the Yubikey GPO.

Note: DO NOT import the Yubikey GPO from a different domain.

23.1 Configuring HMIs to use Yubikey

23.1.1 Log into an HMI that has been joined to the domain using the Domain Admin account.

23.1.2 Launch **PowerShell as Administrator**.

23.1.2.1 Enter the following command to install the Yubikey mini driver:

\\DC1\Yubikey-Driver\Install-Yubi-Driver.ps1

23.1.2.2 Enter the following command to install the Yubikey Manager software:

\\DC1\Yubikey-Driver\Install-Yubikey-Manager.ps1

23.2 Configuring a Yubikey User

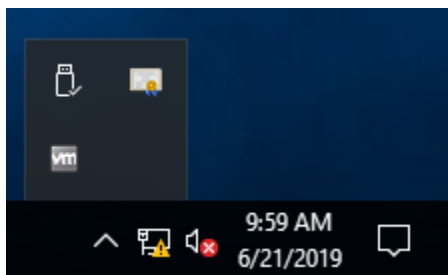
Note: To prevent certificate enrollment error, do not use Remote Desktop (RDP) to access the HMI. You must be physically present at the HMI station and logged in with YubiKey to successfully complete this step.

23.2.1 Login to the HMI as the **YubiAdmin** user with password: **See Default Passwords Document*

23.2.1.1 **Insert the Yubikey** into the USB port on the HMI for configuration.

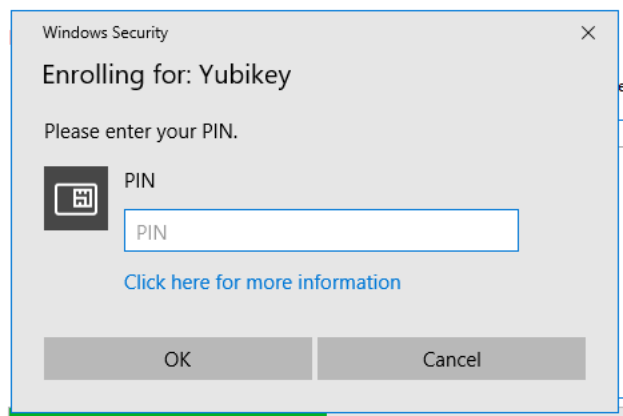
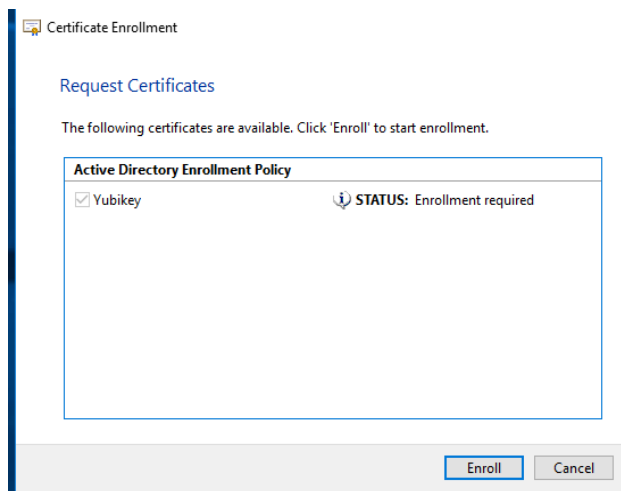
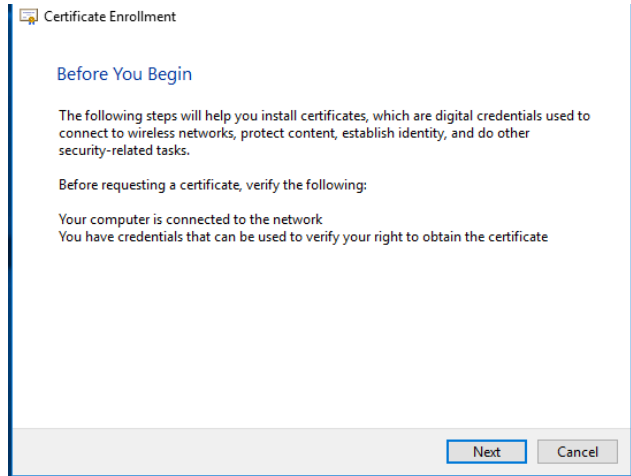
23.2.1.2 After a few seconds you should receive an **auto-enrollment prompt** in the Windows system tray (located near the clock on the desktop). Launch the auto-enrollment wizard.

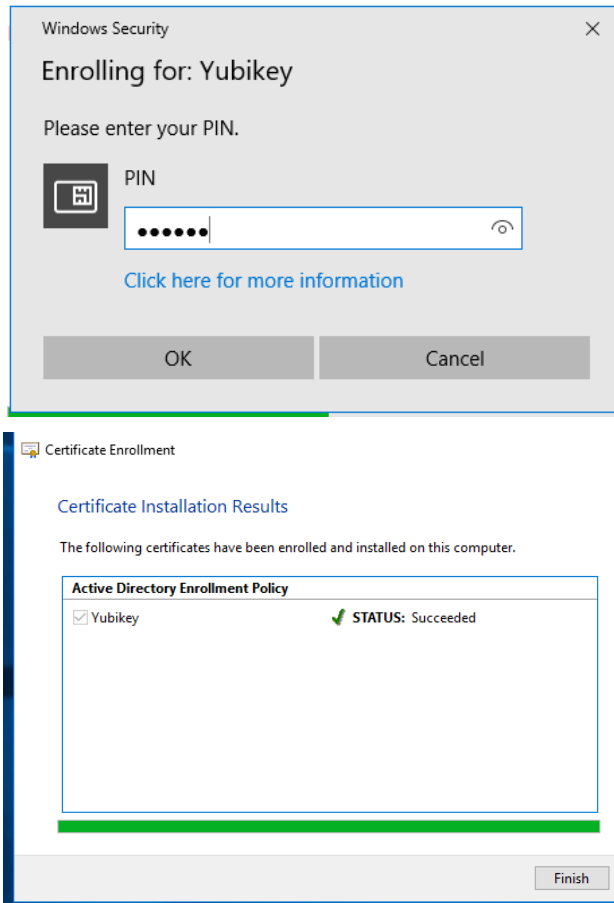
Note: If you missed the pop up in the system tray, click on the ^ symbol and look for the certificate icon:



23.2.1.3 Click the **certificate icon** to begin the self-enrollment process.

23.2.1.4 When prompted for the **Yubikey PIN** enter the default of: **123456**



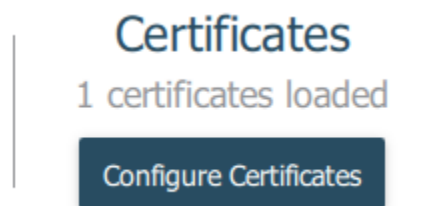


23.2.2 To verify the user has been issued a **Yubikey certificate** -> launch the **Yubikey Manager** tool.

23.2.2.1 Click **Start -> Yubico -> YubiKey Manager**.

23.2.2.2 When the application opens -> select **Applications -> PIV**.

23.2.2.3 You should see that the Yubikey now contains a certificate for the logged in user:



23.3 Enforce Smart Card Login for the YubiAdmin User

23.3.1 Log into the domain controller as the domain Admin user.

23.3.2 Launch the **Active Directory Users and Computers** management window.

23.3.3 Expand the domain (HMI.local for example) -> then click on the **Users** folder.

23.3.3.1 Locate the **YubiAdmin** user in the right pane -> right-click and choose **properties**.

23.3.4 On the **YubiAdmin Properties** window -> click on the **Account** tab.

23.3.4.1 Under **Account Options** -> check the box for **Smart card is required for interactive logon** -> then click **OK**.

23.3.4.2 The **YubiAdmin** user must now logon with the Yubikey and can no longer use the username and password authentication method.

23.4 Configure the YubiMaint Account for Yubikey

23.4.1 Repeat the YubiAdmin steps (19.2) and its sub-steps for the **YubiMaint** account.

The Yubikey is a CCID device (Chip Card Interface Device) and it is read using the CCID protocol for USB. In ESXi the CCID protocol is not available to VM's running on the ESXi host by default because it is reserved for ESXi console login. To allow this protocol an advanced setting must be enabled on each VM.

Note: This setting will eventually be added to the base OVAs for the Security ST automated builds (for now it will be configured manually on all builds) but, may need to be configured manually if a VM is built without an OVA. Currently the setting will need to be enabled for all builds. This document will be updated when the OVAs have been updated and the setting enabled.

23.5 Enable CCID protocol for Virtual Machines

23.5.1 Login to **VMware ESXi** web console -> User: **root** Password: **See Default Passwords Document*

23.5.2 Expand the **Virtual Machines** view -> right-click on the **first VM** in the list and select **Edit Settings**.

23.5.3 On the **Settings tab** -> click on the **VM Options** at the top -> then expand the **Advanced tab** on the left

23.5.3.1 Scroll down to **Configuration Parameters**.

23.5.3.2 Click the **Edit Configuration** button.

23.5.3.3 In the **Configuration Parameters** window -> click the **Add parameter** button at the top.

23.5.3.3.1 You should see a new line in the table below.

23.5.3.3.2 In the **Key** column enter the following: **usb.generic.allowCCID**

23.5.3.3.3 In the **Value** column enter the following:

TRUE

23.5.3.3.4 Click **OK**.

23.5.3.3.5 Repeat these steps for each VM on the ESXi Host.

23.5.4 You may now map a Yubikey inserted into a USB port on the ESXi host to a VM.

Note: If the Yubikey device still fails to attach to the VM you may need to power down the VM and attach the Yubikey with the VM powered off this process is referred to as a cold add.

24.0 Scans – OTArmor

Scan PDH, UDH, or MDH based on network topology.

Note: You will need a GE Vernova supplied laptop with **nmap-zenmap** software installed (cannot be connected to the internet).

24.1 **NMAP-Zenmap**

24.1.1 Before scanning, check the network address on the local computer and check for network connectivity.

24.1.2 Configure the network settings on local computer.

24.1.2.1 Open **Control Panel** -> select **Network and Internet** -> **Network Connections**.

24.1.2.2 Right-click the connected network adapter -> select **Properties**.

24.1.2.3 Double-click **Internet Protocol Version 4 (TCP/IPv4)**.

24.1.2.4 Enter the IP Address and Subnet Mask for the network: PDH, UDH or MDH. (Choose the IP address according to the network you are scanning).

24.1.2.5 Open **Command Prompt (Admin)** -> ping a device in the network to verify connectivity (use network topology if needed).

24.1.3 Double-click on the **nmap-zenmap** software icon located on the desktop.

Note: This scan should be performed while the system is idle/inactive.

24.1.4 **PDH Network**

24.1.4.1 Enter the following in the Zenmap GUI:

24.1.4.2 In the **Target:** <IP Address/Netmask> (e.g., 192.168.1.0/24)

24.1.4.3 In the **Profile:** **LEAVE EMPTY**

24.1.4.4 In the **Command:** Enter: **nmap -sS -sU -T4 -v nmap 172.16.201.0/20**

24.1.4.5 Click **Scan**.

24.1.4.6 Wait for the scan to complete.

24.1.4.7 On the C: Drive -> there should be a folder called **Scans** -> create a folder in **Scans** for the project.

24.1.4.8 Save the scan as **PDH.xml** in the folder created for the project.

24.1.4.9 Open the PDH.xml and save-as **PDH.mht** file.

24.1.4.10 Copy the **PDH.mht** file to \\CA1\C\$\Users\Admin\Documents\Scans.

Note: If the Scans folder is not created, create the folder and copy the file.

24.1.5 UDH Network

24.1.5.1 In the Command -> type/select **nmap -sS -sU -T4 -v nmap 172.16.101.0/20**

24.1.5.2 Click **Scan**.

24.1.5.3 Wait for the scan to complete.

24.1.5.4 On the C: Drive -> there should be a folder called **Scans** -> create a folder in **Scans** for the project.

24.1.5.5 Save the scan as **UDH.xml** in the folder created for the project.

24.1.5.6 Open the UDH.xml and save-as **UDH.mht** file.

24.1.5.7 Copy the **UDH.mht** file to \\CA1\C\$\Users\Admin\Documents\Scans.

Note: If the Scans folder is not created, create the folder and copy the file.

24.2 Nessus Scan

FUTURE WORK INSTRUCTION.

IMPORTANT:

- The Pre-CWT Checklist should be completed at this point – refer to **REF – 136M5795**.
- Next step – Complete CWT Plan – refer to **REF – 136M5793**.

25.0 Before Shipment – OTArmor

25.1 Remove Secure Mode

25.1.1 Login to the HMI with **GEMaint** credentials.

25.1.2 Open **TCW ToolboxST** (located on) the **(E:) drive** -> folder **Master**.

25.1.2.1 On the Menu bar -> select **Device** -> **Secure State**.

25.1.2.2 Check the **Secure State**.

25.1.2.3 If the controller is showing: **Secure State Secure** -> click the **Go to Secure** button.

Note: This switches the controller back to **Security State Open** removing Secure Mode.

25.1.2.4 Verify the controller has been remove from Secure Mode.

25.2 Observe Mode

25.2.1 Login to **SP1**.

- 25.2.2 Open the ePO (ePolicy Orchestrator) desktop icon.
- 25.2.3 Login -> click on **System Tree** -> open **Workstations**.
- 25.2.4 Click the **Systems** tab -> open the **Presets** dropdown -> select **This group and all subgroups**.
- 25.2.5 Select a workstation -> click on **Actions** -> **Agent** -> **Run Client Task Now**.
- 25.2.6 In the **Run Client Task Now** -> select **Solidcore 8.x.x** -> **SC Observer Mode** -> **Begin Observer Mode** -> click **Create Task Now** -> select **Run Task Now**.
- 25.2.7 Verify the solidifier is in Observer Mode -> open **Command Prompt (Admin)** and run the following: **sadmin status**
- 25.2.8 Verify:
 - 25.2.8.1 Trellix Solidifier: **Observe**
 - 25.2.8.2 Trellix Solidifier on reboot: **Observe**
- 25.2.9 Or in the ePO -> the About section -> verify the solidifier is in Observe Mode.
- 25.2.10 Open **Trellix ePO** (ePolicy Orchestrator) desktop icon.
- 25.2.11 Login -> click on **System Tree** -> open **Servers**.
- 25.2.12 Click the **Systems** tab -> open the **Presets** dropdown -> select **This group and all subgroups**.
- 25.2.13 Select a server -> click on **Actions** -> **Agent** -> select **Run Client Task Now**.
- 25.2.14 In the **Run Client Task Now** -> select **Solidcore 8.x.x** -> **SC Observer Mode** -> **Begin Observer Mode** -> click **Create Task Now** -> select **Run Task Now**.
- 25.2.15 Verify the solidifier is in Observer Mode -> open **Command Prompt (Admin)** and run the following: **sadmin status**
- 25.2.16 Verify:
 - 25.2.16.1 Trellix Solidifier: **Observe**
 - 25.2.16.2 Trellix Solidifier on reboot: **Observe**
- 25.2.17 Or in the ePO -> the About section -> verify the solidifier is in Observe Mode.

25.3 **Services & Switch Configurations**

Download and save the services and switch configurations for archive.

25.3.1 **Services & MAC Addresses:**

- 25.3.1.1 Remote desktop to **DC1**.
- 25.3.1.2 Navigate to the following CA1 Share:
\\CA1\Admin_Procedures\OTArmor.
- 25.3.1.3 Copy the **GEV-Inventory-Gather.ps1** file into the **(C:)Drive** of DC1.
- 25.3.1.4 Open **Windows PowerShell ISE as Administrator** -> open the **GEV-Inventory-Gather.ps1**.

- 25.3.1.5 Run the script -> enter the information at the prompts.
- 25.3.1.6 Open the newly created **ArchiveInfo** folder
(C:\Users\Admin\Documents\) -> verify the **.txt** and **.csv** files are in the folder.
- 25.3.1.7 Copy the **ArchiveInfo** folder -> save it on a **Removable Drive** for project archive.
- 25.3.1.8 Delete the **GEVGE-Inventory-Gather.ps1** script from the DC1 Server.

25.3.2 Switch Configurations (CatTools):

- 25.3.2.1 Login to **DR1**
- 25.3.3 Open **File Explorer** -> click on the **(D:)Drive** -> select the **Backups** folder.
- 25.3.4 Copy the **Switches** folder -> save it on a **Removeable Drive** for project archive.

25.4 **Clean System**

- 25.4.1 Review servers for any files that don't belong.
- 25.4.2 Remove all the temporary files created during configuration.
- 25.4.3 Empty the Recycled Bin.
- 25.4.4 Once the servers have been **Cleaned** -> shut down the system.
- 25.4.5 If **OTArmor™ Pro (HA)** – use the Shutdown Procedure (**Startup-Shutdown VSAN Cluster.docx**).